

Continuous Hidden Markov Models for Equity Returns: Spectral Rank, Heavy-Tail Emission Families, and Regime-Conditional Value-at-Risk

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Abstract

The classical verdict that hidden Markov models with $K = 2$ –3 Gaussian states cannot reproduce the slow autocorrelation of absolute equity returns [1] has shaped two decades of regime-switching work toward semi-Markov sojourn extensions. We provide empirical evidence that this verdict is a marginal-mixture limit, not a temporal one: applying the textbook bilinear identity for the absolute-return ACF over the non-unit eigenvalues of the transition matrix and recasting the failure as a rank statement on $\mathbf{T} - \mathbf{1}\bar{\pi}^\top$, an effective-rank diagnostic on the fitted $\hat{\mathbf{T}}$ shows the implied $K - 1$ rank bound is non-binding at $K \geq 3$ at the cross-ticker median.

A unified ECM scaffold supports four emission families (Gaussian, Student- t , Laplace, Generalised Error Distribution) that share identical forward-backward recursions and quantile-based initialisation; the M-step is the only architectural difference. The empirical study covers SPY 2014–2026, a sector-balanced 30-ticker panel, a CRSP cross-decade slice (1994–2006), and a six-asset US-equity copula extension. Headline operating result: a regime-conditional Value-at-Risk that propagates the one-step-ahead state forecast through the predictive mixture passes the Christoffersen joint conditional-coverage test cleanly across emission families on the headline OoS window, and on 19/24 rows of a six-fold rolling-origin walk-forward, with rejections concentrated on out-of-distribution stress folds.

Positioning is honest: the i.i.d. bootstrap and an ML hidden semi-Markov model both exceed the CHMM on raw single-window OoS Kolmogorov–Smirnov; the CHMM’s value-add is the structural use cases that those alternatives cannot serve (regime-conditional VaR, multi-asset copula composition, parametric synthetic-data privacy), and a unified emission-family ablation that the literature lacks. The empirical scope is the three symmetric Cont (2001) stylized facts on daily US-equity returns under stationary OoS; a non-equity stress test (GLD, SLV) collapses under static fitting, and periodic refit is the deployment recipe.

Keywords: Continuous Hidden Markov Model, Baum-Welch, Student- t Emissions, Stylized Facts, Volatility Clustering, Value-at-Risk, Synthetic-Data Generator.

1 Introduction

Synthetic generators of equity-return time series support stress testing, regulatory backtesting, scenario design, and the training of downstream machine-learning models on data that cannot be

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released for privacy or licensing reasons [2–4]. The bar for a useful generator is that it reproduce, simultaneously, the canonical stylized facts of daily returns: heavy-tailed marginals, negligible linear autocorrelation, and slow decay of the autocorrelation function (ACF) of absolute returns [5, 6]. Generalized autoregressive conditional heteroskedasticity (GARCH) models [7, 8] capture volatility clustering but not the multi-regime structure of the marginal; i.i.d. generators match the marginal but destroy temporal persistence; deep generators [9–11] reproduce some stylized facts at the cost of interpretability and reproducibility.

An influential negative result has shaped two decades of work on the regime-switching route. Rydén et al. [1] fitted Gaussian-emission hidden Markov models (HMMs) at $K = 2$ –3 states and concluded that the geometric sojourn-time distribution of a standard Markov chain produces exponential ACF decay too fast to match the observed slow decay; Bulla and Bulla [12] restored ACF fidelity by replacing the Markov chain with a hidden semi-Markov chain at the cost of substantially increased complexity.

We identify the binding axis. The empirical Rydén et al. low- K failure compounds two constraints: a temporal one (the absolute-return ACF must admit slow-decay modes) and a distributional one (the marginal mixture must carry enough components to match observed kurtosis). The closed-form mixture-of-eigenvalues identity $\rho_{|G|}(\tau) = \sum_{k \geq 2} c_k \lambda_k^\tau / \sigma_{|G|}^2$ over the non-unit eigenvalues of $\mathbf{T}$ [13–15] (restated in Section 3.8) gives an algebraic upper bound of $K - 1$ on the number of decay modes through the rank of $\mathbf{T} - \mathbf{1}\bar{\pi}^\top$. An empirical effective-rank diagnostic on the fitted $\hat{\mathbf{T}}$ (Appendix A.5.4) shows this bound is non-binding at $K \geq 3$ at the cross-ticker median on equity-return data: across a 30-ticker panel the dominant-mode share has median 75.6% with interquartile range [66%, 86%] and minimum 32.6% (NEM); the SPY value at $K = 18$ is 93.6%, in the right tail of this distribution. The binding axis at $K \geq 3$ is therefore the marginal mixture, whose representational capacity must match the empirical kurtosis target.

We instantiate this on ten years of daily SPY ETF data with a 2.5-year held-out window, using a continuous HMM trained by Baum-Welch [16, 17] under quantile-based initialization in four emission variants (Gaussian, per-state Student- t in the tradition of 18, 19, Laplace, and GED) that share identical forward-backward recursions. The empirical scope is daily US equities under stationary OoS: a non-equity stress test on GLD and SLV ETFs (Appendix A.7.6) collapses to 0% OoS KS under static IS-fitted CHMM, a documented scope limit rather than a deferred follow-up.

The body headline is $K^* = 3$, selected by held-out per-observation log-likelihood and held-out KS on the strictly pre-2020 slice under both single-fold and four-fold / six-fold rolling-origin CV (Appendix A.8). At this operating point CHMM-N attains 89.7% IS / 80.5% OoS KS at $|G_t|$ ACF-MAE 0.0460 (Table 1), and the penalised CHMM-t at $\lambda = 20$ gives the cleanest IS / OoS heavy-tail match at 14.91/8.50 against observed 7.68/5.29. The single-window OoS sits at the upper end of a six-fold rolling-origin walk-forward whose median CHMM-N OoS KS is 62.1% at $K^* = 3$ and 67.7% at $K = 18$ (Table 32); the walk-forward distribution, not the single-window pair, is the operationally informative summary for production deployment. VaR and Expected-Shortfall envelopes bracket observed values at 1% and 5% on both windows.

Multi-asset synthesis on a US-equity universe composes through a rank-based Student- t copula on the CHMM marginals; profile MLE selects $\nu^* = 6$ at IS off-diagonal correlation MAE 0.027, but on OoS the Gaussian and Student- t copulas are statistically indistinguishable, so the dependence-family choice is an IS-calibration distinction. On the sector-balanced 30-ticker panel the IS distribution is tight (median KS 99.5%) and 11/30 tickers fall below 60% OoS KS, concentrated on single-name regime introductions; this reflects the well-known stationarity-scope limit on financial time series [20–22] and motivates periodic refit at deployment [23].

Our contributions are empirical and architectural. (i) We provide empirical evidence that the

algebraic $K - 1$ rank bound on $\mathbf{T} - \mathbf{1}\bar{\pi}^\top$ from the bilinear identity is non-binding at $K \geq 3$ on equity-return data: a single non-unit eigenvalue carries 93.6% of the lag-1 absolute-return ACF at $K = 18$ on SPY (Appendix A.5.4), and the cross-ticker median dominant-mode share is 0.76 (Appendix A.5.5); this recasts the Rydén et al. [1] failure as a marginal-mixture limit, not a temporal one. (ii) A unified ECM scaffold for four emission families (Gaussian, Student- t , Laplace, GED) under shared forward-backward recursions where the M-step is the only architectural difference; CHMM-GED’s per-state shape p_k nests Gaussian and Laplace as boundary cases. (iii) A regime-conditional Value-at-Risk that passes Christoffersen-cc cleanly across emission families on OoS, with Monte Carlo power calibration and an Engle–Manganelli DQ backstop; the use case that no exchangeable bootstrap or unconditional GARCH benchmark can serve. (iv) An empirical scope along three orthogonal generalisation axes: a six-fold rolling-origin walk-forward on SPY 2014–2026, a sector-balanced 30-ticker panel, a CRSP cross-decade transfer, and a six-asset US-equity Student- t copula extension.

2 Related Work

Hidden Markov models in finance. Hamilton [24] introduced Markov-switching to economics; Schaller and Van Norden [25] estimated regime-switching specifications on US monthly stock returns and documented the predictive content of the latent regime for return forecasting. The seminal evaluation against the Cont stylized facts is due to Rydén et al. [1]: Gaussian-emission HMMs at $K = 2$ –3 states fail the slow ACF decay of absolute returns because the geometric sojourn-time distribution of a standard Markov chain produces exponential decay too fast to match the observed signature. Bulla and Bulla [12] restored fidelity with hidden semi-Markov models at substantially higher complexity. Nystrup et al. [26] extends the same line to long-memory volatility. Maximum-likelihood fitting of heavy-tailed innovations follows Peel and McLachlan [18] and Liu and Rubin [19]; the Generalized Error Distribution [27, 28] provides a continuous shape parameter that interpolates between Gaussian ($p = 2$) and Laplace ($p = 1$) and underlies our CHMM-GED variant in the same role that the per-state degrees of freedom ν_k plays in CHMM- t . The discrete-state predecessor of this paper, Alswaidan and Varner [4], reproduced the same three stylized facts via Laplace quantile binning plus a Poisson-jump duration mechanism; here we replace the discretization step with continuous emissions and recover the same target through the spectral mechanism of Section 3.8.

GARCH and its extensions. GARCH [7, 8] encodes volatility persistence through a single decay parameter but imposes a unimodal innovation distribution and fails distributional Kolmogorov–Smirnov on equity returns. Heavy-tailed innovations [29], asymmetric leverage extensions (EGARCH, 30; GJR-GARCH, 31; threshold GARCH, 32), and regime-switching extensions (MS-GARCH, 33, with the MSGARCH R package of 34 the standard reference implementation) close some gaps but inherit the unimodal-innovation constraint. Realised-variance frameworks [35, 36] provide a complementary forecasting baseline.

Deep generative models for financial time series. Generative adversarial network (GAN)-based generators reproduce the visual appearance of return series but typically fail volatility-clustering tests unless architectures encode temporal dependence explicitly [37, 38]; the TimeGAN family of Yoon et al. [9], the convolutional Wasserstein GAN (WGAN) QuantGAN of Wiese et al. [10], and the autoregressive-diffusion approach of Rasul et al. [11] are the standard references. We include QuantGAN as the deep-generative row in our extended panel (Appendix A.6.3); the present paper’s contribution is grounded against parametric and non-parametric baselines.

Cross-asset dependence and synthetic-data evaluation. For multi-asset synthesis, the Single Index Model [39] propagates a market factor through a rank-one linear decomposition but distorts each non-market marginal; Sklar’s theorem [40, 41] supports a cleaner decomposition in which each asset’s fitted marginal is preserved and only the copula is estimated, with Gaussian and Student- t copulas [42, 43] the standard choices and rank-reordering simulation [44] the standard injection. Dynamic-conditional-correlation models [45] and composite-likelihood estimators for vast covariance matrices [46] are the standard alternatives at higher d ; the present paper’s six-asset construction sits well below the dimensions where those estimators become essential, but the rank-based copula architecture composes with either at scale. Synthetic-data evaluation is anchored on the framework of Stenger et al. [47], the proper-scoring-rule literature [48–50], kernel two-sample tests [51], and signature-based metrics [52, 53].

State-space stochastic volatility. Stochastic-volatility (SV) models [54] are the continuous-latent counterpart of the discrete-latent CHMM: log-volatility evolves as a latent AR(1) and likelihood inference proceeds via simulation-based estimators [55, 56] or hidden-Markov approximations on a discretised volatility grid [57, 58]. Markov-switching SV [59] sits between the two formulations. We include a Markov-switching SV row in the extended baseline panel (Appendix A.6.4) for direct comparison; the architectural difference is that the CHMM treats the latent regime as a finite-state object with K free per-state densities, whereas SV ties all states through a single AR(1) on a one-dimensional log-volatility latent. The former gives more representational capacity at the marginal level (the kurtosis points of Table 6); the latter gives a strictly stationary latent process by construction.

Regime breaks and other deep market generators. The W2 (COVID 2020) and W4 (2022 rate-hike onset) walk-forward stress folds (Section 4) are regime introductions that the IS distribution does not span; the natural complements are Bayesian online change-point detection [60] and the particle-filter recursion of Fearnhead and Liu [61], both companion-paper directions. Beyond the GAN-based panel surveyed above, path-signature [52, 62–64] and score-based / diffusion [11, 65–67] market generators share the same trade-off against the present paper: tighter visual fidelity to the tape, opaque per-state semantics, and no parametric copula handle for multi-asset composition.

3 Methods

3.1 Data and Excess Growth Rates

The single-asset analysis uses daily SPDR S&P 500 ETF (SPY) prices from January 3, 2014 to January 3, 2024 (a ten-year training window, $T_{\text{IS}} = 2,516$), with the period from January 4, 2024 through April 20, 2026 ($T_{\text{OoS}} = 572$) held out. Prices are split-adjusted but not dividend-adjusted; the data sources are described in the acknowledgments. For ticker i and trading day j the annualised excess growth rate is

$$G_{i,j} \equiv \frac{1}{\Delta t} \ln \left(\frac{P_{i,j}}{P_{i,j-1}} \right) - r_f, \quad \Delta t = 1/252, \quad r_f = 0, \quad (1)$$

with $P_{i,j}$ the session volume-weighted average price (VWAP). The factor $1/\Delta t$ is a constant rescaling; all simulations and metrics are evaluated on the daily-scale series. The empirical study is organised along two pipelines (step-by-step summary in Appendix A.1.4). *Pipeline A* fits a CHMM to each ticker independently and evaluates single-asset metrics; *Pipeline B* reuses the per-asset Pipeline-A fits as marginals and injects cross-asset dependence through a rank-based Student- t copula (§3.4).

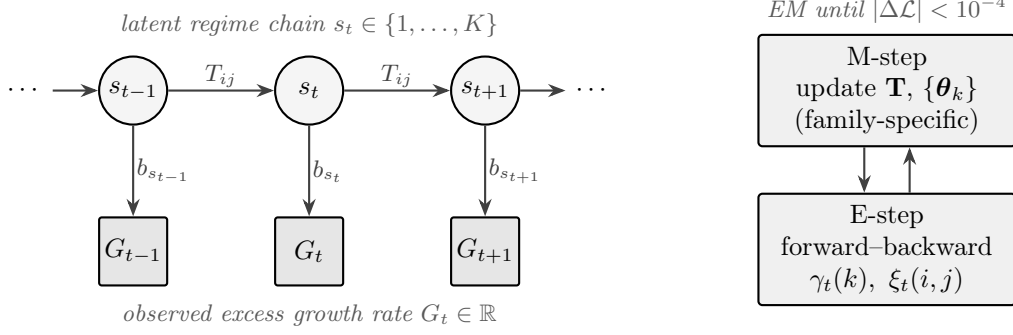


Figure 1. CHMM architecture and training loop. A K -state latent Markov chain with transition matrix $\mathbf{T}$ generates the regime sequence s_t ; each s_t emits the observed excess growth rate G_t through a state-specific density b_{s_t} . The four CHMM variants share everything except the per-state emission family (Gaussian, Student- t , Laplace, GED; preceding paragraph). EM alternates an E-step (log-space forward-backward) and a family-specific M-step until $|\Delta\mathcal{L}| < 10^{-4}$.

3.2 Continuous Hidden Markov Model

We model G_t as a continuous HMM $\mathcal{M} = (\mathcal{S}, \mathbf{T}, \mathbf{F}, \boldsymbol{\pi})$ with K latent states, per-state emission density $f_k(x; \boldsymbol{\theta}_k)$ (with corresponding per-state CDF $F_k(x; \boldsymbol{\theta}_k) = \int_{-\infty}^x f_k(u; \boldsymbol{\theta}_k) du$, used in the regime-conditional VaR construction of §4.6), transition matrix $\mathbf{T} \in \mathbb{R}^{K \times K}$, and initial distribution $\boldsymbol{\pi}$ [17]. The stationary marginal is the K -component mixture

$$f(x) = \sum_{k=1}^K \bar{\pi}_k f_k(x; \boldsymbol{\theta}_k), \quad \bar{\boldsymbol{\pi}} = \bar{\boldsymbol{\pi}} \mathbf{T}, \quad (2)$$

with $\bar{\boldsymbol{\pi}}$ the stationary distribution. Existence and uniqueness require irreducibility and aperiodicity of $\mathbf{T}$ (Assumption 1 in §3.8). We learn $\mathbf{T}$, $\boldsymbol{\theta}_{1:K}$, and $\boldsymbol{\pi}$ jointly by EM (§3).

The four emission families share a single forward-backward scaffold and quantile-based initialization; the M-step is the only architectural difference (Figure 1). *CHMM-N*: $f_k = \mathcal{N}(\mu_k, \sigma_k^2)$. *CHMM-t*: $f_k = t_{\nu_k}(\mu_k, \sigma_k)$ with per-state degrees of freedom $\nu_k \in [\nu_{\min}, \nu_{\max}]$. *CHMM-L*: $f_k = \text{Laplace}(\mu_k, \beta_k)$ with per-state Laplace scale denoted β_k to avoid collision with the canonical Laplace-scale symbol b . *CHMM-GED*: $f_k = \text{GED}(\mu_k, \alpha_k, p_k)$ with per-state shape $p_k \in [p_{\min}, p_{\max}]$ that nests Gaussian ($p_k = 2$) and Laplace ($p_k = 1$).

3.3 Benchmark Generators

We benchmark against representatives from each generator tier, all fit on the same IS window: i.i.d. bootstrap [68] (non-parametric distributional ceiling), Gaussian and Laplace i.i.d., GARCH(1,1) Gaussian [8] and Student- t [29], and Markov-switching GARCH [33]. Extended GARCH-family, semi-Markov, deep-generative QuantGAN [10], stochastic-volatility, multifractal, and jump-diffusion baselines are in Appendix A.6.4 and A.6.5.

3.4 Cross-Asset Dependence (Pipeline B)

Pipeline B reuses each per-asset CHMM marginal and injects cross-asset dependence through a rank-based Student- t copula [42, 43]; profile MLE selects ν^* , and the rank-reordering simulator of Iman and Conover [44] preserves each fitted CHMM marginal under Sklar’s theorem [40]. SIM, Gaussian-copula, and truncated C-vine comparators are in Appendix A.7.1.

3.5 Evaluation Protocol

For each model we simulate $P = 1,000$ independent paths of length T following [47], under a deterministic global seed root 20260420. Headline metrics are the two-sample Kolmogorov–Smirnov [69, 70] pass rate at $\alpha = 0.05$, mean simulated excess kurtosis, and the absolute-return ACF mean absolute error (MAE) over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^L \left| \hat{\rho}_{|G|}^{\text{obs}}(\tau) - \bar{\rho}_{|G|}^{\text{sim}}(\tau) \right|. \quad (3)$$

A separate Value-at-Risk panel (§4.6) carries the unconditional Kupiec [71] and Expected-Shortfall envelope diagnostics. Block-bootstrap KS recalibration, Anderson–Darling, W_1 , Hellinger, and CRPS auxiliaries are in Appendix A.2. Estimation of $(\mathbf{T}, \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi})$ proceeds by EM in log-space. This section gives the shared scaffold and the family-specific M-step updates; the state-count selection that operationalises this scaffold is reported in Section 4.

3.6 Shared EM Scaffold

All four emission families share the same log-space forward-backward recursions and the same quantile-based initialization [72]: observations are sorted, partitioned into K equal-sized chunks, and each state’s initial location and scale are seeded from the corresponding chunk; $T_{ij}^{(0)} = \pi_k^{(0)} = 1/K$. EM iterates to tolerance $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with `max_iter` = 60; convergence is typically reached in 20–40 iterations. Quantile-based initialization is what makes Baum–Welch reach a non-degenerate $K \geq 3$ fit: random initialization standard in the 1990s often converges to overlapping local optima, which is one of the two compounding sources of the low- K failure observed by Rydén et al. [1] (the other is the spectral one of Section 3.8). Algorithmic details and the forward-backward recursion are in Appendix A.1.

3.7 Family-Specific M-Step Updates

CHMM-N (Gaussian). The classical Baum–Welch M-step [16] is closed form: $\mu_k \leftarrow \sum_t \gamma_t(k) O_t / \sum_t \gamma_t(k)$ and $\sigma_k^2 \leftarrow \sum_t \gamma_t(k) (O_t - \mu_k)^2 / \sum_t \gamma_t(k)$.

CHMM-t (Student-t via ECM). We adopt the ECM formulation of Peel and McLachlan [18] and Liu and Rubin [19], the same heavy-tailed innovation choice studied by Abanto-Valle et al. [58] for stochastic-volatility-in-mean models. The latent precision

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2} \quad (4)$$

is treated as an augmented E-step quantity, and the M-step updates

$$\mu_k \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} O_t}{\sum_t \gamma_t(k) u_{t,k}}, \quad \sigma_k^2 \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k)^2}{\sum_t \gamma_t(k)} \quad (5)$$

have closed form conditional on ν_k . The degrees of freedom ν_k are updated by a golden-section search maximizing the per-state Q -function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. The two-block CM step preserves the EM monotonicity guarantee on the compact bracket, so the incomplete-data log-likelihood is non-decreasing across iterations.

CHMM-L (Laplace). The weighted maximum-likelihood estimators are in closed form: μ_k is the posterior-weighted median of the observations (weights $\gamma_t(k)$, computed by cumulative-weight bracketing with linear interpolation between neighbouring order statistics), and $\beta_k \leftarrow \sum_t \gamma_t(k) |O_t - \mu_k| / \sum_t \gamma_t(k)$.

CHMM-GED (Generalized Error Distribution via three-stage ECM). For $f_k(x) = \text{GED}(x; \mu_k, \alpha_k, p_k)$ the M-step decomposes into three CM updates per state: a bracketed $\mu_k \leftarrow \arg \min_{\mu} \sum_t \gamma_t(k) |O_t - \mu|^{p_k}$ via golden-section over $[\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]$, then the closed-form scale $\alpha_k \leftarrow [(p_k/W_k) \sum_t \gamma_t(k) |O_t - \mu_k|^{p_k}]^{1/p_k}$ with $W_k = \sum_t \gamma_t(k)$, and finally a bracketed $p_k \leftarrow \arg \max_{p \in [p_{\min}, p_{\max}]} \sum_t \gamma_t(k) \log f_k(O_t; \mu_k, \alpha_k, p)$ via golden-section over $[p_{\min}, p_{\max}] = [0.5, 3.0]$. Each CM step is a partial maximization, so EM monotonicity holds on the compact bracket. CHMM-N (all $p_k = 2$) and CHMM-L (all $p_k = 1$) are the boundary cases.

3.8 Spectral Mechanism for the Absolute-Return ACF

The closed-form bilinear identity for the absolute-return ACF of a stationary CHMM, $\mathbb{E}[|G_t| | G_{t+\tau}|] = \mathbf{m}^\top \text{diag}(\bar{\pi}) \mathbf{T}^\tau \mathbf{m}$ with $\mathbf{m}_k = \mathbb{E}[|G_t| | s_t = k]$, and its eigendecomposition over the non-unit eigenvalues of $\mathbf{T}$ are folklore in the regime-switching literature (13, §22.2; 14, Ch. 3; 15). Under irreducibility and aperiodicity of $\mathbf{T}$ (Assumption 1, $\bar{\pi}$ unique stationary; 73) and finite per-state second moments with non-trivial location/scale contrast (Assumption 2), the spectral decomposition $\mathbf{T}^\tau = \mathbf{1}\bar{\pi}^\top + \sum_{k=2}^K \lambda_k^\tau \mathbf{v}_k \mathbf{w}_k^\top$ subtracts the marginal product to give the ACF

$$\rho_{|G|}(\tau) = \sum_{k=2}^K w_k \lambda_k^\tau, \quad w_k = (\mathbf{m}^\top \text{diag}(\bar{\pi}) \mathbf{v}_k) (\mathbf{w}_k^\top \mathbf{m}) / \sigma_{|G|}^2. \quad (6)$$

Equation (6) is a real-coefficient sum of geometric decays at the non-unit eigenvalues of $\mathbf{T}$; complex-conjugate λ_k pairs combine into damped oscillatory modes and non-diagonalisable $\mathbf{T}$ adds polynomial-times-geometric prefactors via Jordan blocks. A self-contained derivation, the lag-zero behaviour, and the squared-return ACF are in Appendix A.4; the formal Assumptions 1–2, and the cited propositions on identifiability, MLE consistency, and ECM monotonicity, are in Appendix A.3. Our contribution on this axis is exclusively empirical: applying (6) to recast the Rydén et al. [1] low- K failure on the SPY 2014–2024 instance as a rank statement on $\mathbf{T} - \mathbf{1}\bar{\pi}^\top$ and demonstrating empirically that the implied $K - 1$ rank bound is non-binding at $K \geq 3$.

Rydén separation as a K -rank statement. At $K = 2$, equation (6) reduces to the mono-exponential form $\rho_{|G|}(\tau) = \lambda_2^\tau$ with $\lambda_2 = T_{11} + T_{22} - 1$; slow decay is matched by tuning $T_{11} + T_{22}$. The temporal axis is therefore satisfiable at $K = 2$; the binding axis is the two-component marginal, which cannot match the empirical kurtosis target. At $K \geq 3$ the matrix admits multiple non-unit eigenvalues and the marginal mixture has enough components to match the observed tail. Empirically, ACF-MAE is essentially flat across $K \in \{3, 6, \dots, 21\}$ while KS rises (Section 4); a direct effective-rank diagnostic (Appendix A.5.4, Table 9) shows a single non-unit eigenvalue carries 93.6% of the lag-1 ACF at $K = 18$ on SPY, so the K -rank upper bound is non-binding at $K \geq 3$ on this instance. Repeating the diagnostic on the 30-ticker cross-section (Appendix A.5.5) gives a wider distribution with cross-ticker median dominant-mode share 0.76 and minimum 0.33 on NEM; the rank-non-binding claim is supported at the cross-ticker median rather than as a universal property of equity-return data.

4 Empirical Study

Reminder. Subsections 4.1–4.4 and 4.6 operate under **Pipeline A** (single-asset CHMM fit per ticker, no cross-asset coupling); only §4.5 operates under **Pipeline B** (Pipeline A marginals plus a rank-based copula).

4.1 Descriptive Statistics and Stylized Facts

All three Cont stylized facts are present in both windows (Figure 4 in Appendix A.5.15): heavy-tailed marginal (IS excess kurtosis 7.68, OoS 5.29), negligible linear autocorrelation in raw G_t , and persistent volatility clustering in $|G_t|$. A stationary block bootstrap at mean block length $L = 20$ gives broadly overlapping IS / OoS 95% CIs ([2.17, 12.40] and [0.90, 8.26]; Appendix A.5.8); kurtosis-fidelity targets should be read against the joint envelope rather than the point estimates.

4.2 State-Count Selection

We sweep $K \in \{3, 6, 9, 12, 15, 18, 21\}$ for CHMM-N on the IS window and select K^* by held-out per-observation log-likelihood and held-out KS under four-fold and six-fold rolling-origin CV on the strictly pre-2020 slice (Appendix A.8). At both cadences the $K = 3$ vs $K = 6$ comparison sits inside sampling noise ($|z| \leq 0.07$ on mean held-out log-likelihood with the sign flipping between fold designs); BIC also picks $K = 3$. Under parsimony we take $\mathbf{K}^* = \mathbf{3}$ as the body operating point. Higher state counts are decisively worse on held-out criteria. Absolute-return ACF-MAE is essentially flat (0.047–0.053) across the entire sweep, consistent with the spectral identity (6).

4.3 Six-Generator Comparison (Pipeline A)

Table 1 reports IS / OoS validation for the benchmark generators of §3.3 and the four CHMM emission variants at three operating points, against an explicit-duration semi-Markov reference.

Methods notes for Table 1. The 1-day OoS KS column is dominated by the i.i.d. bootstrap; the CHMM-vs-bootstrap differentiation is on the structural use cases of §4.6 (regime-conditional VaR) and §4.5 (multi-asset copula composition), not on per-day distributional fidelity. Bold within the $K^* = 3$ block marks the cleanest IS / OoS row on each axis. Block-bootstrap KS recalibration, the extended GARCH-family / MS-GARCH- $K \in \{3, 4, 6\}$ panel, and the $K = 6$ / $K = 18$ sensitivity panels are in Appendix A.6.2, A.6.4, and A.5.

At the body operating point $K^* = 3$, the four CHMM variants attain 79.8–90.6% IS / 63.1–83.2% OoS KS, with CHMM-N at 89.7% IS / 80.5% OoS, simulated kurtosis 3.83/3.53, and $|G_t|$ ACF-MAE 0.0460 (on par with GARCH at 0.0485). The penalised CHMM-t at $\lambda = 20$ gives the cleanest IS / OoS heavy-tail match at 14.91/8.50 against observed 7.68/5.29 (the aggregate IS overshoot is a per-state ν_k ECM bracket-pinning artefact at low K ; §5 and Appendix A.5.8). CHMM-GED’s per-state shape lands at 5.48/5.12, CHMM-L at 5.30/4.85 (losing ~ 10 pp on OoS KS at $K = 3$), and a shared- ν Student- t row at 4.68/4.46 achieves the heavy-tail target without a penalty hyperparameter (Appendix A.5.9). The mixture-of-eigenvalues sum (6) reproduces the slow absolute-return ACF that the Rydén et al. [1] $K = 2$ –3 Gaussian setup could not recover on the SPY 2014–2024 instance (Appendix A.5.6); the raw-return ACF-MAE closes the third stylized fact (0.0234–0.0240, indistinguishable from the i.i.d. baselines). The four CHMM variants are statistically indistinguishable on the OoS sample-CRPS [50]; the family choice is driven by the per-row kurtosis match (Table 6, §5). The cross-generator ranking is robust to the distributional-metric choice (KS, Anderson–Darling, Wasserstein-1, Hellinger; Appendix A.2). The MS-GARCH benchmark

Table 1. Headline generator comparison on SPY. 1,000 simulated paths, $\alpha = 0.05$; IS $T = 2,516$, OoS $T = 572$. KS = Kolmogorov–Smirnov pass rate; Kurt = mean simulated excess kurtosis (observed 7.68 IS, 5.29 OoS); ACF-MAE = mean absolute error of the absolute-return / raw-return ACF over 252 lags; CRPS = OoS sample continuous ranked probability score. ¶ Reference Bayesian re-run of MS-GARCH via the MSGARCH R package of Ardia et al. [34]; the lower KS pass rate reflects posterior-predictive integration of parameter uncertainty (Appendix A.6.4). § QuantGAN row is an in-house WGAN re-implementation (3-layer 1D-conv generator and critic, Wasserstein loss; full spec in Appendix A.6.3); the row reads as the panel’s deep-generative negative control, and a faithful reference-implementation re-run of the seven-block TCN with Lambert-W pre-processing of Wiese et al. [10] is a deferred follow-up. §§ Shared- ν Student- t ablation: single ν across all K states by aggregate- Q ECM, no penalty; the structurally-cleaner heavy-tail recipe vs the per-state $\nu_k + \lambda = 20$ row above (Appendix A.5.9). $K = 6$ and $K = 18$ sensitivity panels (including unpenalised CHMM-t and bracket-lift ablations) are in Appendix A.5.

Model	KS (%) $\uparrow$		Exc. Kurt		ACF-MAE $\downarrow$		CRPS OoS
	IS	OoS	IS	OoS	$ G_t $	G_t	$\downarrow$
<i>Observed</i>	–	–	7.68	5.29	–	–	–
Bootstrap	99.7	92.1	7.24	6.81	0.0628	0.0235	1.0398
Gaussian i.i.d.	0.0	1.0	−0.01	−0.03	0.0627	0.0236	1.0611
Laplace i.i.d.	99.1	88.5	2.95	2.91	0.0628	0.0236	1.0386
GARCH(1,1)	23.4	60.8	7.06	2.96	0.0485	0.0243	1.0440
GARCH(1,1)- t	57.3	80.8	15.13	–	0.0316	0.0173	–
MS-GARCH ($K = 2$)	27.7	38.7	4.73	–	0.0367	0.0173	–
MS-GARCH ($K = 3$)	36.1	33.1	4.10	–	0.0284	0.0173	–
MS-GARCH ($K = 6$)	34.5	33.4	4.41	–	0.0429	0.0173	–
MS-GARCH ref. Bayesian ($K = 2$)¶	0.0	5.8	4.00	2.46	0.0465	0.0240	–
MS-GARCH ref. Bayesian ($K = 3$)¶	0.1	5.1	4.52	2.53	0.0433	0.0241	–
MS-GARCH ref. Bayesian ($K = 4$)¶	0.0	5.3	6.01	3.54	0.0446	0.0241	–
QuantGAN TCN§	0.0	0.0	0.56	0.53	0.0617	0.0264	–
<i>Body operating point: $K^* = 3$ (selected by held-out criteria under rolling-origin CV; §4.2).</i>							
CHMM-N	89.7	80.5	3.83	3.53	0.0460	0.0240	1.0393
CHMM-t pen. ($\lambda = 20$)	90.6	83.2	14.91	8.50	0.0537	0.0235	1.0373
CHMM-L	79.8	63.1	5.30	4.85	0.0532	0.0236	1.0405
CHMM-GED	90.5	77.4	5.48	5.12	0.0526	0.0236	1.0389
CHMM-t shared- ν §§	91.9	82.1	4.68	4.46	0.0531	0.0235	–
ML HSMM-N	98.4	91.0	3.46	3.38	0.0629	0.0236	–

Table 2. Six-fold rolling-origin walk-forward summary (CHMM-N, 5y train / 1y test, 500 paths per fold). Pulled from the full panel in Appendix A.8, Table 32. The headline single-window OoS sits at the upper end of the fold-wise distribution; the W2 (COVID) and W4 (2022 rate-hike onset) folds are out-of-distribution by KS for every generator in the panel.

	median KS (%)	$[Q_1, Q_3]$ KS (%)	min KS (%)	folds < 10%
$K^* = 3$	62.1	[7.2, 78.4]	0.8	2/6
$K = 18$ (sensitivity)	67.7	[8.2, 75.0]	0.0	2/6

plateaus at 34–36% IS KS in our Nelder-Mead fit and at 0–0.1% in the reference Bayesian re-run (Appendix A.6.4); the multi-state benefit is specific to the CHMM scaffold rather than to multi-state regime-switching per se. A six-fold rolling-origin walk-forward (Table 2) attains median OoS KS 62.1% at $K^* = 3$, with two stress folds (W2 COVID, W4 2022 rate-hike onset) below 10% that every generator in the panel rejects; the walk-forward median is the operationally informative summary for production deployment.

Per-state interpretability. The fitted CHMM-N states at $K^* = 3$ partition cleanly along the conventional volatility axis (high-/low-/moderate-vol with the natural drift signs); the bimodal CHMM-GED $\hat{p}_k$ partition (Appendix A.5.16) reflects the same axis at $K = 18$ with Laplace-like states in the high-volatility tail and Gaussian-like states in the bulk. The latent states map onto identifiable economic regimes at every K examined.

ML HSMM as a co-headline result. Replacing the geometric sojourn distribution with an explicit truncated Pareto and refitting through the explicit-duration forward-backward of Yu [74] gives ML HSMM-N at $K^* = 3$ at 98.4% IS / **91.0%** OoS KS, the strongest single-window OoS KS row in the panel and beating every CHMM operating point. CHMM and ML HSMM are best read as complementary scaffolds on the same Markov backbone, with KS / ACF / kurtosis trade-offs that select between them by use case. The HSMM advantage on KS is offset by an absolute-return ACF-MAE that matches the i.i.d. baseline level (0.0629) rather than CHMM-N’s 0.0460, because the fitted Pareto sojourn concentrates probability mass on a single low-volatility state and the regime-driven slow-ACF mechanism that the CHMM exploits at $K \geq 3$ is muted (per-state $\bar{\pi}$ concentration in Appendix A.5.17). Higher- K HSMM variants (Pareto collapse at $K = 18$; discrete-Gamma-sojourn fix achieving $|G_t|$ ACF-MAE **0.0462** on IS at $K = 18$) are reported in Appendix A.5.17 and A.5.18. Readers selecting a synthetic-data generator on raw OoS KS alone should prefer ML HSMM-N at $K^* = 3$; CHMM at $K^* = 3$ is the right choice when the structural use cases of §4.6–4.5 are in scope.

Bootstrap as a non-parametric synthetic-data benchmark. The stationary block bootstrap of Politis and Romano [68] attains 99.7% IS / 92.1% OoS KS in Table 1, above every CHMM operating point. CHMM differentiates on use cases the bootstrap cannot serve: regime-conditional VaR (§4.6; the bootstrap is exchangeable and admits no forward state filter), multi-asset copula composition (§4.5; needs parametric marginals), and privacy / licensing (every bootstrap path is a permutation of literal IS observations).

A block-bootstrap KS recalibration at $L = 20$ (the appropriate null under autocorrelated daily returns; 68) preserves the cross-generator ordering on both windows, with the absolute level dropping ~ 5 pp on IS and 20–30pp on OoS; the cross-generator ranking is the robust comparison (Appendix A.6.2, Table 24).

4.4 Cross-Ticker Generalization (Pipeline A)

Table 3 reports the penalised CHMM-t IS / OoS KS distribution across a sector-balanced 30-ticker panel (10 GICS sectors $\times$ 3 large-cap representatives, $\lambda = 20$) at the body operating point $K^* = 3$. The IS distribution is concentrated (median 96.8%); the OoS distribution is wider with median 69.1%, mean $66.2 \pm 28.2\%$, and 11/30 tickers below 60%, concentrated in single-name regime introductions (LLY 8.7%, UNH 16.5%, NEM 5.0%). A one-way ANOVA on OoS KS by sector at the original $n = 3$ per-sector design is severely underpowered ($\eta^2 = 0.16$ at $F(9, 20) = 0.44$, $p = 0.90$); an $n = 6$ per-sector expansion to a 60-ticker panel (Appendix A.5.3) returns $F(9, 50) = 0.37$, $p = 0.946$, $\eta^2 = 0.062$, with the effect-size estimate dropping by more than half. The 60-ticker aggregate distribution is statistically indistinguishable from the 30-ticker version, so the visible concentration of failures on regime-introduction tickers is a per-ticker observation rather than evidence about sector-level structure. The operational recipe is periodic refit: at the $K = 18$ kurtosis-fidelity sensitivity reference, a quarterly-refit version (Appendix A.5.14) shifts the OoS KS median from 73.4% to **83.0%** and reduces failures from 11/30 to 7/30 (a monthly cadence interpolates to 86.7% / 5/30 at $3\times$ compute). Sensitivity panels at $K = 6$ and $K = 18$ and the SPY-level shared- ν Student- t alternative are in Appendix A.5.7 and A.5.9.

Table 3. Cross-ticker generalisation, sector-balanced 30-ticker panel (penalised CHMM-t at $\lambda = 20$, 1,000 paths, seed = 20260420, body operating point $K^* = 3$). The OoS distribution is wider than the IS distribution and is concentrated at the regime-introduction tickers (LLY, UNH, NEM). Full per-ticker panel and per-sector rollout: Appendix A.5.2; $K = 6$ and $K = 18$ sensitivity rebuilds: Appendix A.5.7; quarterly-refit recipe: Appendix A.5.14.

Metric	$K^* = 3$
IS KS pass rate (%) median	96.8
IS KS pass rate (%) mean $\pm$ s.d.	95.1 ± 4.9
OoS KS pass rate (%) median	69.1
OoS KS pass rate (%) mean $\pm$ s.d.	66.2 ± 28.2
$ G_t $ ACF-MAE median	0.0399
Kurtosis residual (sim – obs) median	7.18
Tickers OoS KS < 60% (IS-fixed)	11/30

Cross-decade validation. A CRSP-sourced cross-decade split (1994–2004 IS / 2004–2006 OoS; full panel in Appendix A.5.21) tests whether the body’s stylized-fact reproduction transfers across non-overlapping decades. The IS axis is decade-robust: at $K^* = 3$ CHMM-N attains 84–90% IS KS pass rate against the 90–94% obtained on the body 2014–2024 IS window (full $K \in \{3, 18\}$ panel in Appendix A.5.21). The 2004–2006 OoS pass rate, however, collapses to 3–5% because the calm post-dot-com bull-market slice has observed excess kurtosis 0.06 (essentially Gaussian) against the 1994–2004 IS kurtosis of 3.05. The cross-decade pattern matches the W2 (COVID 2020) and W4 (2022 rate-hike) walk-forward stress folds (Table 2, both with OoS KS < 10%): the IS axis transfers cleanly across decades, but a static-IS-fitted CHMM cannot bracket OoS slices whose marginal moves outside the IS regime library. The operational implication parallels the cross-ticker panel above: the static-fit OoS pass rate is OoS-slice-specific and the deployment recipe is the periodic-refit construction (the cross-ticker quarterly-refit panel in Appendix A.5.14 and the rolling-copula construction of §4.5), not the single-window static fit.

4.5 Cross-Asset Dependence (Pipeline B): Student-t Copula

Pipeline B reuses the per-asset CHMM-N marginals from §4.4 and adds a rank-based Student- t dependence layer (§3.4). The body cross-asset construction is scoped to a US-equity universe: profile MLE on six US-equity assets (SPY, NVDA, JNJ, JPM, AAPL, QQQ; 200 paths) selects $\nu^* = 6$ on the unit-spaced grid $\nu \in [4, 12]$ with Wilks 95% profile-LL CI of $[6, 7]$ *on the IS slice* (Appendix A.7); a full one-shot MLE jointly maximising over (Σ, ν) confirms the choice is robust to the two-step estimator (Appendix A.6.7). Table 4 reports per-asset KS and the off-diagonal correlation MAE: 0.027 on IS for the Student- t copula, against 0.030 Gaussian, 0.068 truncated C-vine, and 0.076 SIM (full panel in Appendix A.7.1). **On OoS the dependence-family choice is empirically null at this N_{paths} :** off-diagonal MAE is 0.209 for Student- t and 0.202 for Gaussian. The operational story on the OoS axis is the quarterly-rolling refit, which reduces the OoS off-diagonal MAE from 0.209 to **0.185** (Appendix A.7.5); the dependence-family choice is dominated by the refit-cadence choice on OoS.

The cross-asset OoS distribution is bimodal: NVDA and JPM fail OoS KS while the other four pass above 79%, driven by single-name regime shifts (NVDA AI-rally, JPM bank-stress) and recovering to 0/6 under quarterly refit (Appendix A.7.5).

Table 4. Cross-asset Student-t copula on CHMM-N marginals (Pipeline B; $\nu^* = 6$; 200 paths). Per-asset KS pass rate at $\alpha = 0.05$ and correlation reproduction averaged over paths. Comparison against SIM, Gaussian, and truncated C-vine in Appendix A.7, Table 35.

Ticker	KS (%) $\uparrow$	
	IS	OoS
SPY	96.0	79.5
NVDA	98.5	53.5
JNJ	99.0	97.5
JPM	99.5	48.0
AAPL	99.0	93.5
QQQ	95.5	92.0
Off-diag MAE IS	0.027	
Off-diag MAE OoS	0.209	
Off-diag MAE OoS (quarterly refit)	0.185	

4.6 Value-at-Risk and Expected-Shortfall Back-Test (Pipeline A)

The headline of this subsection is the regime-conditional Christoffersen-cc result: a VaR threshold built from the CHMM one-step-ahead state forecast passes the joint conditional-coverage test cleanly at every (K, α) on OoS, where every constant-across- t generator (CHMM unconditional, GARCH, bootstrap) rejects breach independence.

Envelope bracketing and Kupiec coverage. All four CHMM variants bracket observed VaR and ES at 1% and 5% within the $[5, 95]\%$ envelope on both windows; the IS CHMM-N median $\text{VaR}_{0.01}$ at -6.64 (observed -6.38) is the closest point estimate in the panel, with envelopes $\sim 1.7\times$ tighter than GARCH’s (Appendix A.5.19). All four CHMM rows pass the unconditional Kupiec test on both windows ($\text{LR}_{\text{uc}} \in [3.03, 3.83]$ at $\alpha = 0.05$ OoS, against $\chi_1^2(0.05) = 3.841$); we treat Kupiec as a sanity check at this T_{OoS} and the conditional construction as the substantive diagnostic.

Regime-conditional Christoffersen-cc (*headline*). At each OoS day t we run the forward filter through $\mathcal{F}_{t-1} = R_{\text{IS}} \cup R_{\text{OoS}}[1:t-1]$ to get the one-step-ahead state forecast

$$\mathbb{P}(s_t = k \mid \mathcal{F}_{t-1}) = \sum_i \mathbb{P}(s_{t-1} = i \mid \mathcal{F}_{t-1}) T_{ik}, \quad (7)$$

and define $\widehat{\text{VaR}}_t(\alpha) = F_t^{-1}(\alpha)$ where $F_t(\cdot) = \sum_k \mathbb{P}(s_t = k \mid \mathcal{F}_{t-1}) F_k(\cdot; \boldsymbol{\theta}_k)$ is the family-appropriate predictive mixture CDF under IS-fixed $(\mathbf{T}, \{\boldsymbol{\theta}_k\})$, with F_k the per-state Gaussian, Student- t , Laplace, or GED CDF (the body Gaussian construction reduces to $F_k = \Phi(\cdot; \mu_k, \sigma_k)$). Table 5 reports the back-test for CHMM-N at $K^* = 3$ and $K = 18$ and the penalised CHMM-t at $\lambda = 20$ on OoS; every row passes Kupiec, Christoffersen-ind, and Christoffersen-cc cleanly at $\alpha = 0.05$. The improvement in the independence statistic from the unconditional variant ($\text{LR}_{\text{ind}} = 5.26$ at $K = 18, \alpha = 0.05$) to the conditional variant ($\text{LR}_{\text{ind}} = 0.52$) is the regime-switching value proposition that the unconditional Kupiec headline did not exercise. The conditional pass extends to every emission family with the family-appropriate predictive density (CHMM-L, CHMM-GED) at $p_{\text{cc}} \geq 0.089$ throughout, so the regime-switching value proposition is intrinsic to the latent-state forecast and not specific to the Gaussian emission family (Appendix A.2.2, Table 33).

Power caveat at $\alpha = 0.01$. The Christoffersen-cc test is power-bounded at $T_{\text{OoS}} = 572$, $\alpha = 0.01$ (Appendix A.5.10: $\geq 80\%$ power only at clustering eigenvalue $\rho \geq 0.50$); the substantive risk-management diagnostic is therefore the $\alpha = 0.05$ row, where the Engle–Manganelli (2004) Dynamic Quantile test (Appendix A.5.11) also passes cleanly at both $K = 3$ ($p = 0.156$) and $K = 18$ ($p = 0.678$). **At $\alpha = 0.01$ the DQ test rejects conditional coverage at $K = 18$ ($p = 0.017$)** while Christoffersen-cc does not; the regime-conditional VaR over-couples high-volatility states at the strict tail at $K = 18$ on this OoS slice and the $\alpha = 0.01$ Christoffersen-cc pass is a power-driven non-rejection.

Table 5. Regime-conditional VaR back-test, CHMM-N and penalised CHMM-t ($\lambda = 20$) on OoS ($T_{\text{OoS}} = 572$, seed 20260420). Critical values: $\chi_1^2(0.05) = 3.841$, $\chi_2^2(0.05) = 5.991$. The forward filter is run under IS-fixed parameters ($\mathbf{T}, \{\theta_k\}$) (no refit); only the one-step-ahead state-probability vector is updated through the OoS observations via Eq. (7). Every (K, α , family) row passes Kupiec, Christoffersen-ind, and Christoffersen-cc cleanly at $\alpha = 0.05$. $^\ddagger \alpha = 0.01$ rows are power-bounded at $T_{\text{OoS}} = 572$: Christoffersen-cc reaches $\geq 80\%$ power against breach-clustering eigenvalues only at $\rho \geq 0.50$ (rejection rate 42.6% at $\rho = 0.20$), so a clean pass should be read as “not detectably worse than a strongly-clustered alternative” rather than “calibrated against any plausible alternative”. The Engle-Manganelli (2004) DQ test as a higher-power alternative (Appendix A.5.11) rejects $K = 18$ at $\alpha = 0.01$ ($p = 0.017$, bold in column p_{DQ}). See Appendix A.5.10 for the full power calibration. * Engle-Manganelli (2004) Dynamic Quantile p -value at 4-lag specification (Eq. 27 in Appendix A.5.11); χ_6^2 critical at 5% is 12.59. The DQ panel is run on CHMM-N only and the CHMM-t penalised rows are reported with ‘ $\neg$ ’ in this column; the CHMM-t-penalised conditional-VaR diagnostic is the Christoffersen-cc column at $\alpha = 0.05$.

Family	K	α	breaches	br rate	median $\widehat{\text{VaR}}_t$	LR _{uc}	LR _{ind}	LR _{cc}	p_{cc}	p_{DQ}^*
CHMM-N	3	0.01 ‡	9	1.57%	−4.56	1.62	2.36	3.98	0.14	0.06
CHMM-N	3	0.05	35	6.12%	−2.87	1.41	0.01 ‡	1.42	0.49	0.16
CHMM-N	18	0.01 ‡	9	1.57%	−5.20	1.62	2.36	3.98	0.14	0.02
CHMM-N	18	0.05	26	4.55%	−3.02	0.26	0.52	0.78	0.68	0.68
CHMM-t ($\lambda=20$)	3	0.01 ‡	8	1.40%	−5.59	0.82	2.81	3.63	0.16	–
CHMM-t ($\lambda=20$)	3	0.05	32	5.59%	−2.73	0.41	0.77	1.19	0.55	–
CHMM-t ($\lambda=20$)	18	0.01 ‡	10	1.75%	−6.25	2.65	1.98	4.62	0.10	–
CHMM-t ($\lambda=20$)	18	0.05	29	5.07%	−3.15	0.01 ‡	1.39	1.40	0.50	–

The state-filter VaR pipeline of Eq. (7) applies to any multi-state regime-switching model; the CHMM-vs-MS-GARCH choice is dominated by the marginal-distribution column of Table 1 (CHMM-N at 89.7% IS KS vs MS-GARCH plateaus at 0–36%). The construction extends to four-family ablation, six-fold walk-forward, and quarterly refit (Appendices A.2.2, A.8, A.5.12); the conditional Christoffersen-cc passes at $\alpha = 0.05$ on 19/24 walk-forward rows, with failures concentrated on the W2 (COVID) and W4 (2022 rate-hike onset) stress folds that every generator rejects, and faster refit cadences do not close the W2 gap (Appendix A.5.13). BH-FDR at FDR = 0.05 across the 40-test combined panel retains 37/40 rows on the full panel and 32/32 on the non-stress subset; the three persistent rejections are all on the W2 fold.

5 Discussion

Replicating the slow-ACF behaviour at $K \geq 3$. Two issues compound in the Rydén et al. [1] low- K setup: a single self-transition rate produces a single exponential mode that cannot be simultaneously fast and slow, and the random initialisations standard in the 1990s often converge to degenerate local optima. Our $K = 2$ replication on SPY (Appendix A.5.6) clarifies the mechanism: ACF-MAE is essentially constant at ≈ 0.050 across quantile and five random-seed initialisations.

The binding low- K constraint is therefore distributional (KS at $K = 2$ drops to 67–72%): with modern initialisation the ACF is reproduced at any $K \geq 2$, while distributional fidelity drives the choice of K . The effective-rank diagnostic in Appendix A.5.4 makes this microscopically explicit: at the fitted $\hat{\mathbf{T}}$ for $K^* = 3$ the dominant non-unit mode alone carries 96.8% of the lag-1 absolute-return ACF on SPY.

Closing the kurtosis gap with heavy-tailed emissions. CHMM-N at $K^* = 3$ produces simulated IS kurtosis 3.83 against observed 7.68: a Gaussian mixture’s kurtosis is bounded by the variance of component means relative to their variances. CHMM-t addresses this through per-state Student- t emissions but at the cost of an aggregate-mean IS overshoot to 14.91 from a single-state ECM lower-bracket pinning artefact; an exponential $1/\nu_k$ shrinkage at rate $\lambda = 20$ is the body recipe, validated on a strictly pre-2020 slice by held-out log-likelihood (Appendix A.5.20). A shared- ν ablation eliminates the overshoot without any penalty (simulated IS kurtosis 4.68 at $K^* = 3$, Appendix A.5.9); per-ticker, uniform $\lambda = 20$ is over-shrunk on three of six body tickers and the production recipe is the per-ticker validation rule of §4.4 (Appendix A.8). CHMM-L sits between CHMM-N and CHMM-t at IS kurtosis 5.30 without any shape parameter; CHMM-GED’s bimodal $\hat{p}_k$ partition (Appendix A.5.16) is the data-driven analog of a hand-classified Gaussian-bulk / Laplace-tail hybrid. The four-way variant decision is summarised in Table 6.

Table 6. Variant decision guide at $K^* = 3$. Recommended CHMM emission family by use-case priority. All four variants share the forward-backward scaffold and quantile-based initialisation; the M-step is the only architectural difference.

Use-case priority	Recommended variant
Simplest closed-form fit; kurtosis fidelity secondary	CHMM-N
Closest IS kurtosis match without a shape parameter	CHMM-L
Per-state heavy tails, IS-calibrated	CHMM-t pen. ($\lambda = 20$)
Heavy-tail match without a penalty hyperparameter	CHMM-t shared- ν
Adaptive per-state shape; data-chosen Gaussian-bulk / Laplace-tail mixture	CHMM-GED

Stationarity scope and operational deployment. The CHMM is trained on the assumption that returns are locally stationary across the IS window. Single-name equity returns are well known to carry sector-specific structural breaks (20; 21; surveyed by 22); the cross-ticker generalisation in §4.4 reflects this directly, with 11/30 tickers below 60% OoS KS, concentrated in single-name regime introductions. The mitigation is periodic refit (23; an online-EM formulation by 75): a quarterly-refit version of the cross-ticker panel (Appendix A.5.14) reduces failures from 11/30 to 7/30 at the $K = 18$ kurtosis-fidelity reference. The walk-forward W2 (COVID) and W4 (2022 rate-hike onset) stress folds remain out-of-distribution by KS for every generator under any refit cadence (Appendix A.8); these are regime introductions the IS distribution does not span.

Stylized-fact scope: the leverage axis under symmetric emissions. The Cont (2001) stylized-fact list also includes the leverage effect (negative $\text{Corr}(G_t, |G_{t+1}|)$ on equity returns); the present paper’s body evaluation covers only the three symmetric stylized facts. All four CHMM emission families carry symmetric per-state densities, so we initially expected the CHMM to fail the leverage axis by construction. We compute $\text{Corr}(G_t, |G_{t+1}|)$ on observed SPY (IS -0.135 , OoS -0.214) and on the per-path simulation distribution (Appendix A.6.6). The CHMM produces a non-trivial signed signal via Markov state-mixing on $\hat{\mathbf{T}}$ (high- σ_k states couple to negative- μ_k states

asymmetrically), borderline-non-rejecting the IS observed value at the 5% level but rejecting the OoS observed value at the same level. The OoS gap is a substantive finding about the symmetric-emission scaffold: a non-zero leverage signal is recoverable from the Markov mixing alone, but closing the OoS gap requires asymmetric per-state emissions (skew- t in the tradition of 76; skew-Laplace; skew-GED), deferred as a companion-paper direction.

Parameter parsimony. CHMM-N at $K = 18$ has $K(K-1) + 2K = 342$ free parameters (CHMM-t adds K degrees-of-freedom, CHMM-GED adds K shape parameters); the QuantGAN baseline of Wiese et al. [10] carries on the order of 10^4 – 10^5 . The CHMM operates at three orders of magnitude lower parametric complexity while attaining higher OoS KS on this dataset (Table 1).

6 Conclusion

A continuous HMM trained by Baum-Welch with quantile-based initialization reproduces the three symmetric Cont stylized facts on daily US-equity returns. The textbook mixture-of-eigenvalues identity for the absolute-return ACF (13–15) recasts the empirical Rydén et al. [1] low- K failure as an effective-rank statement on $\mathbf{T} - \mathbf{1}\bar{\pi}^\top$, non-binding at $K \geq 3$ on the SPY 2014–2024 instance. Four emission families share one ECM scaffold and land at four kurtosis points; a regime-conditional Value-at-Risk that propagates the one-step-ahead state forecast through the predictive mixture passes Christoffersen-cc cleanly across families on the headline OoS window and on 19/24 rolling-origin walk-forward rows. ML HSMM-N at $K^* = 3$ (91.0% OoS KS) is the strongest single-window row in the panel; CHMM and ML HSMM are best read as complementary scaffolds with different KS / ACF / kurtosis trade-offs.

The headline scope is daily US equities under stationary OoS. **The static IS-fitted CHMM does not transfer outside this asset class:** the GLD / SLV non-equity stress test of Appendix A.7.6 collapses to 0% OoS KS. Walk-forward stress folds (W2 COVID, W4 2022 rate-hike) reject regardless of refit cadence; periodic refit is the deployment recipe for non-stress slices. Skew-heavy-tailed emissions, explicit-duration semi-Markov sojourns at higher K , and scaling to larger d via untruncated regular vines or factor copulas are companion-paper directions. The spectral-rank diagnostic generalises beyond Rydén: any regime-switching “failure” attributing the limit to temporal structure can be re-litigated as a marginal-mixture vs. effective-rank decomposition on the fitted $\hat{\mathbf{T}}$.

Data, code, and contributions. Reproducer: <https://github.com/altashly1/CHMM-paper>; companion Julia package CHMM-Model.jl: <https://github.com/altashly1/CHMM-Model>. Daily prices: Polygon.io via Massive (2014-01-03 to 2025-11-18) and Alpaca Markets / IEX (held-out extension through 2026-04-20). J.V. directed the study; A.A. developed the continuous model, implemented Baum-Welch, conducted the empirical analysis, and generated all figures and tables; C.J. contributed to methodology review and validation; all authors edited and reviewed the final manuscript. The authors declare no competing interests.

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A Supplementary Material

This appendix collects supplementary material referenced in the main body: the public API of the companion package, the EM forward-backward recursions and algorithm pseudocode, the validation metric definitions, formal propositions, the full state-resolution and multi-emission sensitivity panels, robustness and KS-power calibration, the extended GARCH and SM-CHMM baselines, the full cross-asset panel, and the K -selection and ν_k diagnostics referenced from the discussion.

A.1 CHMM Algorithms and Derivations

This appendix collects the algorithmic content for the CHMM family: the forward-backward recursions and weighted M-step updates, the complete training pseudocode, and per-state degrees-of-freedom diagnostics for CHMM-t.

A.1.1 Forward-Backward Recursions and M-Step Updates

For reference, the in-text Baum-Welch description of Section 3.2 is underpinned by the standard log-space recursions [17, 72]. Define the forward variable $\alpha_t(k) = \mathbb{P}(O_{1:t}, S_t = k \mid \mathcal{M})$ and the backward variable $\beta_t(k) = \mathbb{P}(O_{t+1:T} \mid S_t = k, \mathcal{M})$. The log-space recursions are

$$\log \alpha_1(k) = \log \pi_k + \log f_k(O_1), \tag{8}$$

$$\log \alpha_t(j) = \log \sum_i \exp(\log \alpha_{t-1}(i) + \log T_{ij}) + \log f_j(O_t), \quad t = 2, \dots, T, \tag{9}$$

$$\log \beta_T(k) = 0, \tag{10}$$

$$\log \beta_t(i) = \log \sum_j \exp(\log T_{ij} + \log f_j(O_{t+1}) + \log \beta_{t+1}(j)), \quad t = T-1, \dots, 1, \tag{11}$$

where $\text{logsumexp}_i(x_i) = x_{\max} + \log \sum_i \exp(x_i - x_{\max})$. The posterior state-occupancy $\gamma_t(k)$ and pair-occupancy $\xi_t(i, j)$ quantities are

$$\gamma_t(k) = \frac{\alpha_t(k)\beta_t(k)}{\sum_j \alpha_t(j)\beta_t(j)}, \quad \xi_t(i, j) = \frac{\alpha_t(i) T_{ij} f_j(O_{t+1}) \beta_{t+1}(j)}{\sum_{m,n} \alpha_t(m) T_{mn} f_n(O_{t+1}) \beta_{t+1}(n)}, \quad (12)$$

and the closed-form M-step updates are

$$\pi_k^{\text{new}} = \gamma_1(k), \quad \mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k) O_t}{\sum_t \gamma_t(k)}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k) (O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}, \quad T_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)}. \quad (13)$$

The convergence criterion is $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with $\mathcal{L}^{(n)} = \text{logsumexp}_k \log \alpha_T^{(n)}(k)$, and simulation draws $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$ and then iterates $G_t \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2)$, $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t, \cdot})$ with no jump process.

CHMM-t M-step (per-state ECM, closed form given $u_{t,k}$). For Student-t emissions $f_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ the ECM formulation of Peel and McLachlan [18], Liu and Rubin [19] treats the latent precision as an augmented E-step quantity,

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2}, \quad (14)$$

so that conditional on ν_k the location and scale admit closed-form weighted updates,

$$\mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k) u_{t,k} O_t}{\sum_t \gamma_t(k) u_{t,k}}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}. \quad (15)$$

The degrees-of-freedom parameter ν_k is then refined by a one-dimensional golden-section search on the per-state Q -function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_{\nu}(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. This two-block ECM preserves the monotonicity of the standard EM guarantee on the compact bracket $[\nu_{\min}, \nu_{\max}]$, with the search invariant $Q_k(\nu^{(n+1)}) \geq Q_k(\nu^{(n)})$ at every iteration; Proposition 1 states the guarantee in full. We extend this M-step with an optional exponential shrinkage prior on $1/\nu_k$, corresponding to the penalised objective

$$Q_k^{\text{pen}}(\nu) = \sum_t \gamma_t(k) \log t_{\nu}(O_t; \mu_k, \sigma_k) - \lambda / \nu, \quad (16)$$

maximised by the same golden-section search ($\lambda = 0$ recovers the unpenalised Peel and McLachlan [18], Liu and Rubin [19] ECM; the rate sweep and its effect on simulated IS kurtosis are in Section 5).

CHMM-L M-step (closed-form weighted Laplace MLE). For Laplace emissions $f_k(x) = \frac{1}{2\beta_k} \exp(-|x - \mu_k|/\beta_k)$ (with the per-state Laplace scale denoted β_k rather than the canonical Laplace-scale symbol b) the weighted-MLE estimators are available in closed form:

$$\mu_k^{\text{new}} = \text{WeightedMedian}(O_{1:T}; \gamma_{1:T}(k)), \quad \beta_k^{\text{new}} = \frac{\sum_t \gamma_t(k) |O_t - \mu_k^{\text{new}}|}{\sum_t \gamma_t(k)}. \quad (17)$$

The weighted median is computed by cumulative-weight bracketing with linear interpolation between adjacent order statistics at the half-total weight crossing. The Laplace M-step carries no shape parameter, so CHMM-L is strictly cheaper per iteration than CHMM-t (which requires the per-state Q_k search).

CHMM-GED M-step (three-stage ECM with closed-form scale). For Generalized Error Distribution emissions $f_k(x) = \text{GED}(x; \mu_k, \alpha_k, p_k) = \frac{p_k}{2\alpha_k \Gamma(1/p_k)} \exp(-(|x - \mu_k|/\alpha_k)^{p_k})$ the M-step decomposes into three CM updates per state. Given current $(\mu_k^{(n)}, \alpha_k^{(n)}, p_k^{(n)})$, CM-1 updates the location by minimising the per-state weighted L^{p_k} loss,

$$\mu_k^{(n+1)} = \arg \min_{\mu \in [\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]} \sum_t \gamma_t(k) |O_t - \mu|^{p_k^{(n)}}, \quad (18)$$

by a 40-iteration golden-section search; CM-2 updates the scale in closed form given $(\mu_k^{(n+1)}, p_k^{(n)})$,

$$\alpha_k^{(n+1)} = \left[\frac{p_k^{(n)}}{W_k} \sum_t \gamma_t(k) |O_t - \mu_k^{(n+1)}|^{p_k^{(n)}} \right]^{1/p_k^{(n)}}, \quad W_k = \sum_t \gamma_t(k); \quad (19)$$

and CM-3 updates the shape by maximising the per-state Q -function over the bracket,

$$p_k^{(n+1)} = \arg \max_{p \in [p_{\min}, p_{\max}]} \sum_t \gamma_t(k) \log f_k(O_t; \mu_k^{(n+1)}, \alpha_k^{(n+1)}, p), \quad (20)$$

again by a 40-iteration golden-section search over $[p_{\min}, p_{\max}] = [0.5, 3.0]$. Equation (19) is the unique zero of $\partial Q_k / \partial \alpha$, so CM-2 is exact maximization. CM-1 and CM-3 are bracketed maximization. Each CM step is therefore a partial maximization, and the standard ECM monotonicity result [77] applies on the compact bracket; Proposition 1 states the guarantee in full. Boundary cases: $p_k = 2$ recovers Gaussian (with $\sigma = \alpha/\sqrt{2}$); $p_k = 1$ recovers Laplace (with $b = \alpha$).

A.1.2 Algorithm Descriptions

Algorithm 1 is the unified EM procedure shared across all four emission families: a shared scaffold (quantile-based initialisation, log-space forward-backward, transition update) plus four family-specific branches at lines 3, 13, and 30. Algorithm 2 is the simulator. Algorithm 3 is the cross-asset rank-reordering simulator (Pipeline B). Viterbi and SIM-resampling pseudocode is provided in the companion code repository; both are textbook constructions and we omit them from the appendix.

Algorithm 1 Unified EM algorithm for the CHMM family. Lines 3, 13, and 30 branch on the emission family $F \in \{\mathcal{N}, t, L, \text{GED}\}$; every other line is identical across families. CHMM-N and CHMM-L instantiate classical EM; CHMM-t and CHMM-GED instantiate ECM sequences whose final CM step updates the per-state shape parameter (ν_k for CHMM-t, p_k for CHMM-GED) by a one-dimensional golden-section search on the per-state Q -function. Hyperparameter defaults used throughout this paper: convergence tolerance $\text{tol} = 10^{-4}$ on $\Delta\mathcal{L}$, $\text{max_iter} = 60$, 40-iteration golden-section sub-step on every bracketed CM, CHMM-t bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$ with initial $\nu^{(0)} = 10$, CHMM-GED bracket $(p_{\min}, p_{\max}) = (0.5, 3.0)$ with initial $p^{(0)} = 1.5$ and μ_k search interval $[\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]$.

Require: Observations $O_{1:T}$, number of states K , tolerance tol , max_iter , emission family $F \in \{\mathcal{N}, t, L, \text{GED}\}$, if $F = t$ bracket $(\nu_{\min}, \nu_{\max})$ and initial $\nu^{(0)}$, if $F = \text{GED}$ bracket $(p_{\min}, p_{\max})$ and initial $p^{(0)}$.

Ensure: Transition matrix $\mathbf{T}$, per-state emission parameters $\theta_{1:K}$, initial distribution π .

```

1: Quantile-Based Initialization.
2: Sort  $O^{(\text{sort})} \leftarrow \text{sort}(O_{1:T})$  and partition into  $K$  equal chunks  $\{C_1, \dots, C_K\}$ .
3: for  $k = 1$  to  $K$  do
4:   switch  $F$ 
5:      $F = \mathcal{N}$ :  $\mu_k \leftarrow \text{mean}(C_k)$ ,  $\sigma_k \leftarrow \max(\text{std}(C_k), 10^{-6})$ .
6:      $F = t$ :  $\mu_k \leftarrow \text{mean}(C_k)$ ,  $\sigma_k \leftarrow \max(\text{std}(C_k), 10^{-6})$ ,  $\nu_k \leftarrow \nu^{(0)}$ .
7:      $F = L$ :  $\mu_k \leftarrow \text{median}(C_k)$ ,  $b_k \leftarrow \max(\text{mean}(|C_k - \mu_k|), 10^{-6})$ .
8:      $F = \text{GED}$ :  $\mu_k \leftarrow \text{mean}(C_k)$ ,  $\alpha_k \leftarrow \max(\text{std}(C_k), 10^{-6})$ ,  $p_k \leftarrow p^{(0)}$ .
9:   end for
10:  $T_{ij} \leftarrow 1/K$ ;  $\pi_k \leftarrow 1/K$ ;  $\mathcal{L}_{\text{prev}} \leftarrow -\infty$ .
11: EM Loop.
12: for  $n = 1$  to  $\text{max\_iter}$  do
13:   E-Step, per-family emission log-likelihood.
14:   switch  $F$ 
15:      $F = \mathcal{N}$ :  $\log B_{t,k} \leftarrow \log \mathcal{N}(O_t | \mu_k, \sigma_k^2)$ .
16:      $F = t$ :  $\log B_{t,k} \leftarrow \log t_{\nu_k}(O_t; \mu_k, \sigma_k)$ .
17:      $F = L$ :  $\log B_{t,k} \leftarrow -\log(2b_k) - |O_t - \mu_k|/b_k$ .
18:      $F = \text{GED}$ :  $\log B_{t,k} \leftarrow \log p_k - \log(2\alpha_k) - \log \Gamma(1/p_k) - (|O_t - \mu_k|/\alpha_k)^{p_k}$ .
19:   E-Step, shared log-space forward-backward.
20:    $\log \alpha_1(k) \leftarrow \log \pi_k + \log B_{1,k}$ .
21:   for  $t = 2$  to  $T$ ,  $j = 1$  to  $K$  do
22:      $\log \alpha_t(j) \leftarrow \underset{i}{\text{logsumexp}}(\log \alpha_{t-1}(i) + \log T_{ij}) + \log B_{t,j}$ .
23:   end for
24:    $\log \beta_T(k) \leftarrow 0$ .
25:   for  $t = T - 1$  down to  $1$ ,  $i = 1$  to  $K$  do
26:      $\log \beta_t(i) \leftarrow \underset{j}{\text{logsumexp}}(\log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j))$ .
27:   end for
28:   Compute  $\gamma_t(k)$  and  $\xi_t(i, j)$  from equation (12).
29:   Shared transition update.  $T_{ij} \leftarrow \sum_t \xi_t(i, j) / \sum_t \gamma_t(i)$ ;  $\pi_k \leftarrow \gamma_1(k)$ .
30:   M-Step, per-family emission update.
31:   switch  $F$ 
32:      $F = \mathcal{N}$  {classical Baum-Welch [16].}
33:      $\mu_k \leftarrow \sum_t \gamma_t(k) O_t / \sum_t \gamma_t(k)$ .
34:      $\sigma_k \leftarrow \max(\sqrt{\sum_t \gamma_t(k) (O_t - \mu_k)^2 / \sum_t \gamma_t(k)}, 10^{-6})$ .
35:      $F = t$  {ECM of Peel and McLachlan [18], Liu and Rubin [19]; closed-form  $(\mu_k, \sigma_k)$  given  $\nu_k$ , then 1D search on  $\nu_k$ .}
36:      $u_{t,k} \leftarrow (\nu_k + 1) / (\nu_k + ((O_t - \mu_k) / \sigma_k)^2)$  for all  $t, k$ .
37:      $\mu_k \leftarrow \sum_t \gamma_t(k) u_{t,k} O_t / \sum_t \gamma_t(k) u_{t,k}$ .
38:      $\sigma_k \leftarrow \max(\sqrt{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k)^2 / \sum_t \gamma_t(k)}, 10^{-6})$ .
39:      $\nu_k \leftarrow \arg \max_{\nu \in [\nu_{\min}, \nu_{\max}]} \sum_t \gamma_t(k) \log t_{\nu}(O_t; \mu_k, \sigma_k)$  {40-iter golden-section search.}
40:      $F = L$  {closed-form weighted Laplace MLE.}
41:      $\mu_k \leftarrow \text{WeightedMedian}(O_{1:T}, \gamma_{1:T}(k))$  {cumulative-weight bracketing with interpolation at the  $W_k/2$  crossing.}
42:      $b_k \leftarrow \max(\sum_t \gamma_t(k) |O_t - \mu_k| / \sum_t \gamma_t(k), 10^{-6})$ .
43:      $F = \text{GED}$  {three-stage ECM: bracketed  $\mu_k$ , closed-form  $\alpha_k$ , bracketed  $p_k$ .}
44:      $\mu_k \leftarrow \arg \min_{\mu \in [\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]} \sum_t \gamma_t(k) |O_t - \mu|^{p_k}$  {40-iter golden-section search.}
45:      $\alpha_k \leftarrow \max\left(\left[(p_k/W_k) \sum_t \gamma_t(k) |O_t - \mu_k|^{p_k}\right]^{1/p_k}, 10^{-6}\right)$  where  $W_k = \sum_t \gamma_t(k)$  {closed form, unique zero of  $\partial Q_k / \partial \alpha$ .}
46:      $p_k \leftarrow \arg \max_{p \in [p_{\min}, p_{\max}]} \sum_t \gamma_t(k) \log \text{GED}(O_t; \mu_k, \alpha_k, p)$  {40-iter golden-section search.}
47:   Convergence check.  $\mathcal{L}^{(n)} \leftarrow \underset{k}{\text{logsumexp}}(\log \alpha_T(k))$ .
48:   if  $|\mathcal{L}^{(n)} - \mathcal{L}_{\text{prev}}| < \text{tol}$  then

```

Algorithm 2 CHMM simulation of synthetic excess growth rate paths

Require: Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$, Number of steps M , Number of paths P

Ensure: Simulated return paths $\{\hat{G}_{1:M}^{(p)}\}_{p=1}^P$

```
1: for  $p = 1$  to  $P$  do
2:   Sample initial state  $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$ .
3:   for  $t = 1$  to  $M$  do
4:     Emit observation  $\hat{G}_t^{(p)} \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2)$ .
5:     if  $t < M$  then
6:       Transition to next state  $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t, \cdot})$ .
7:     end if
8:   end for
9: end for
10: return  $\{\hat{G}_{1:M}^{(p)}\}_{p=1}^P$ 
```

Algorithm 3 Cross-asset copula simulation via rank reordering

Require: Fitted per-asset CHMMs $\{\mathcal{M}_j\}_{j=1}^d$, correlation matrix Σ (positive semi-definite (PSD), unit diagonal), copula family $\mathcal{C} \in \{\text{Gaussian}, t_\nu\}$, path length T , number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

```
1: for  $p = 1$  to  $P$  do
2:   Draw copula sample  $\mathbf{U}^{(p)} \in [0, 1]^{T \times d}$  from  $\mathcal{C}(\Sigma)$ :
3:    $L \leftarrow \text{Cholesky}(\Sigma)$ ,  $Z \in \mathbb{R}^{T \times d} \sim \mathcal{N}(0, I_d)$  i.i.d.
4:    $Y \leftarrow ZL^\top$ 
5:   if  $\mathcal{C} = \text{Gaussian}$  then
6:      $\mathbf{U}^{(p)} \leftarrow \Phi(Y)$  componentwise
7:   else
8:     Draw  $W \sim \chi_\nu^2/\nu$  i.i.d. for  $t = 1, \dots, T$ ;  $X_{t,\cdot} \leftarrow Y_{t,\cdot}/\sqrt{W_t}$ 
9:      $\mathbf{U}^{(p)} \leftarrow t_\nu(X)$  componentwise
10:  end if
11:  for  $j = 1$  to  $d$  do
12:    Simulate  $\tilde{g}_{j,1:T}^{(p)}$  from  $\mathcal{M}_j$  via Algorithm 2.
13:    Compute order statistics  $\tilde{g}_{j,(1)}^{(p)} \leq \dots \leq \tilde{g}_{j,(T)}^{(p)}$ .
14:    Compute ranks  $r_{j,t}^{(p)} \leftarrow \text{rank}(U_{j,t}^{(p)})$  for  $t = 1, \dots, T$ .
15:     $\hat{G}_{j,t}^{(p)} \leftarrow \tilde{g}_{j,(r_{j,t}^{(p)})}^{(p)}$  for  $t = 1, \dots, T$ .
16:  end for
17: end for
18: return  $\{\hat{G}_{j,t}^{(p)}\}$ 
```

A.1.3 Per-State Shape Diagnostics: CHMM-t and CHMM-GED

This appendix backs the discussion in Section 5 of the per-state shape allocation under CHMM-t and CHMM-GED on SPY IS at $K = 18$ (seed = 20260420).

CHMM-t per-state ν_k . Figure 2 plots the per-state ν_k histogram for the $K = 18$ CHMM-t fit. Thirteen of the eighteen states rest at the upper ECM bracket ($\nu_k = 50$), and only two states lie near the lower bracket ($\nu_2 = 2.19$, $\nu_4 = 2.10$). The simulated IS mixture kurtosis of 17.34 is driven by the posterior mass routed to those two very heavy-tailed states, not by a systemic collapse to the lower bracket.

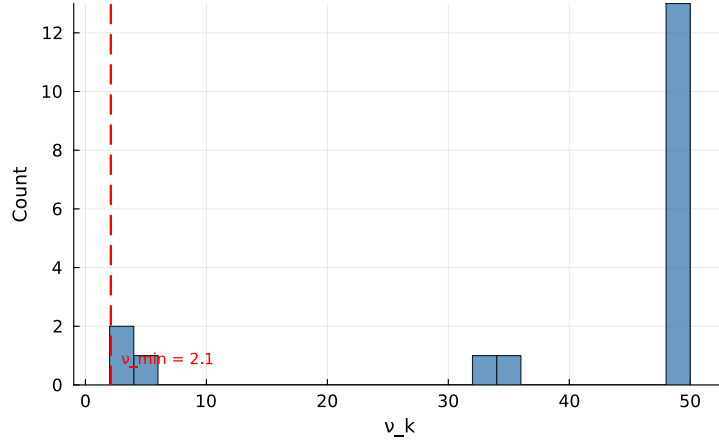


Figure 2. Per-state degrees-of-freedom ν_k for the CHMM-t fit at $K = 18$ on SPY IS. The dashed red line marks the lower ECM bracket $\nu_{\min} = 2.1$. Only two states ($\nu_2 = 2.19$, $\nu_4 = 2.10$) sit near the lower bracket; the median ν_k is at the upper bracket.

CHMM-GED per-state p_k (Gaussian-bulk / Laplace-tail bimodality). The body Section 4.3 reports and discusses the $\hat{p}_k$ partition (Figure 5). Briefly, the $K = 18$ CHMM-GED fit on SPY IS is bimodal: thirteen states attain the upper ECM bracket at $p_k = 3.0$, one state lands at intermediate $p_k = 1.6$, and four states cluster in the Laplace-shape regime $p_k \in [0.86, 1.24]$, concentrated in the high-volatility ranks. The smallest fitted p_k on SPY is 0.86, well clear of every $p_{\min}$ tested in the bracket sweep, so the lower bracket is non-binding. The bimodal partition replicates across 10 Monte Carlo seeds and across the cross-asset universe (Appendix A.6).

A CHMM-t bracket sweep on $\nu_{\min} \in \{2.1, 2.5, 3.0, 4.0, \infty\}$ at $K = 18$ shows the simulated IS kurtosis stays in the 12.5–15.1 band across finite brackets and only collapses to 5.03 in the full Gaussian limit; the overshoot is a feature of the Student- t emission at $K = 18$, not a lower-bracket artefact. The analogous CHMM-GED sweep on $p_{\max} \in \{2.0, 2.5, 3.0, 3.5, 4.0\}$ leaves the Gaussian-like-state count invariant at 13/18 and the heavy-shape count in $\{4, 5\}$; tightening $p_{\max}$ from 4.0 to 3.0 costs about 4 nats of LL. The $p_{\min}$ sweep on $\{0.5, 0.7, 0.85\}$ is fully degenerate: every floor produces an identical fit because the smallest observed $\hat{p}_k$ on SPY is 0.86.

A.1.4 Pipeline Summary

Pipeline A: observed returns $G_t \rightarrow$ fit CHMM (Baum-Welch, $K = 18$) $\rightarrow$ simulate 1,000 paths $\rightarrow$ per-asset metrics (Table 1). Pipeline B reuses the Pipeline-A per-asset fits as marginals, draws a copula sample $\mathbf{U} \sim \mathcal{C}(\Sigma)$ from the fitted Student- t copula on ranks, rank-reorders the per-asset Pipeline-A paths to match $\mathbf{U}$, and reports cross-asset metrics (Table 4). The single-asset scaffold is shared across both pipelines; the CHMM training and simulation steps are Algorithms 1 and 2, the cross-asset rank-reordering step is Algorithm 3, and the architecture diagram for the single-asset scaffold is body Figure 1.

A.2 Validation Metric Details

This appendix documents the precise form of the seven per-path metrics referenced in Section 3.5. For each model, 1,000 independent paths of length T are simulated (200 paths for the cross-asset

extension) and each metric is computed per path before aggregation.

Kolmogorov–Smirnov (KS) pass rate. The two-sample KS statistic [69, 70] measures the maximum vertical distance between two empirical CDFs,

$$D_{n,m} = \sup_x |F_n(x) - G_m(x)|. \quad (21)$$

The KS pass rate is the fraction of paths for which the test fails to reject distributional equivalence against the reference data at $\alpha = 0.05$.

Anderson–Darling (AD) pass rate. The AD test is the integrated squared CDF deviation with a quadratic tail weighting, making it more sensitive than KS to tail discrepancies. The AD pass rate is the fraction of paths for which the two-sample AD test fails to reject at $\alpha = 0.05$.

Excess kurtosis. Mean simulated excess kurtosis across paths, compared to the observed value. Given Gaussian emissions, the CHMM is expected to underestimate kurtosis relative to the heavy-tailed empirical distribution, and the shortfall is one of the reported diagnostics.

ACF-MAE. Mean absolute error between observed and mean-simulated autocorrelation functions of $|G_t|$ over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^L \left| \hat{\rho}_{|G|}^{\text{obs}}(\tau) - \bar{\rho}_{|G|}^{\text{sim}}(\tau) \right|. \quad (22)$$

This is the direct diagnostic for the Rydén et al. limitation.

Wasserstein-1 distance. The mean absolute difference of sorted empirical quantiles, equivalent to the L^1 distance between the empirical CDFs [78],

$$W_1(F, G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)| du. \quad (23)$$

Hellinger distance. A histogram-based overlap distance in $[0, 1]$,

$$H(P, Q) = \frac{1}{\sqrt{2}} \sqrt{\sum_{i=1}^B (\sqrt{p_i} - \sqrt{q_i})^2}, \quad (24)$$

with B equal-width bins spanning the joint support and p_i, q_i the observed and simulated bin probabilities.

Quantile coverage. The fraction of 99 equally-spaced observed quantiles (1st through 99th percentile) falling inside the 5th–95th percentile envelope of the corresponding simulated quantile across 1,000 paths. A coverage of 100% indicates that the simulation envelope fully brackets the observed distribution at every percentile.

Cross-asset metrics. For the cross-asset extension we additionally track the mean Frobenius norm $\|\hat{\Sigma}_{\text{sim}}^{(p)} - \hat{\Sigma}_{\text{obs}}\|_F$ and the off-diagonal Pearson correlation MAE between simulated and observed return matrices, averaged over paths. The off-diagonal MAE is averaged over the $d(d-1)/2$ unique upper-triangular entries of the symmetric correlation matrix (so 15 entries for the six-asset cross-section of Section 4.5), not double-counted across both triangles:

$$\text{MAE}_{\text{off-diag}} = \frac{1}{P} \sum_{p=1}^P \frac{2}{d(d-1)} \sum_{1 \leq i < j \leq d} \left| \hat{\Sigma}_{\text{sim}, ij}^{(p)} - \hat{\Sigma}_{\text{obs}, ij} \right|.$$

A.2.1 Continuous Ranked Probability Score and Diebold-Mariano

The OoS CRPS column of Table 1 reports a proper-scoring-rule complement to the KS pass rate. We use the unbiased sample CRPS estimator: for an ensemble of N simulated paths $\{x_i\}_{i=1}^N$ at time t and observed value y_t ,

$$\widehat{\text{CRPS}}_t = \frac{1}{N} \sum_{i=1}^N |x_i - y_t| - \frac{1}{N(N-1)} \sum_{1 \leq i < j \leq N} |x_i - x_j|. \quad (25)$$

The double sum is computed in $O(N \log N)$ via the sorted-ensemble identity $\sum_{i < j} (x_{(j)} - x_{(i)}) = \sum_i x_{(i)} (2i - N - 1)$. The ensemble at each t is the cross-section of the $N = 1,000$ unconditional simulated paths used throughout Section 4; this yields a marginal-predictive CRPS suitable for an unconditional generative-fidelity assessment, distinct from the conditional one-step-ahead CRPS used in forecasting.

For pairwise comparison we report a Diebold-Mariano test on the per- t loss differential $d_t = \widehat{\text{CRPS}}_t^A - \widehat{\text{CRPS}}_t^B$ with a Newey-West HAC variance under a Bartlett kernel and bandwidth $h = \lfloor T_{\text{OoS}}^{1/3} \rfloor = 8$. Two-sided p -values are reported under the standard-normal null. The body claim of Section 4.3 is: CHMM ties or beats every benchmark on mean OoS CRPS, with Gaussian i.i.d. the only significantly worse comparator.

A.2.2 Christoffersen Conditional-Coverage Panel (Companion to Section 4.6)

The body Section 4.6 reports the unconditional Kupiec LR_{uc} statistic. This appendix reports the full Christoffersen [79] panel: LR_{uc} (unconditional coverage), LR_{ind} (breach independence), and $\text{LR}_{\text{cc}} = \text{LR}_{\text{uc}} + \text{LR}_{\text{ind}}$ (joint conditional coverage), at $\alpha \in \{0.01, 0.05\}$ on the SPY IS and OoS windows. The breach series is constructed from the pooled-archive VaR threshold (the α -quantile of $\text{vec}(\text{sim archive})$ across paths and time), then $\text{br}_t = (R_t \leq \widehat{\text{VaR}}_\alpha)$.

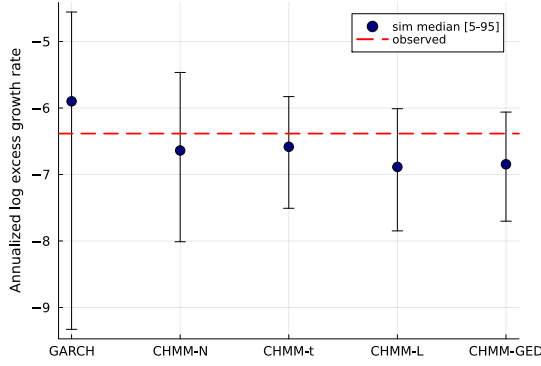
The headline reads as follows. On the unconditional (Kupiec) leg every CHMM variant passes at both levels on both windows; the LR_{uc} statistic for the four CHMM rows at $\alpha = 0.05$ on OoS clusters at 3.03–3.83 (just below the χ_1^2 critical value 3.841), reflecting the integer-breach grid at $T_{\text{OoS}} = 572$ noted in the body table caption. On the independence leg every unconditional generator in the panel (bootstrap, GARCH, and all four CHMM variants) rejects: LR_{ind} is in the range 4.71–56.46 across rows. This is structural rather than CHMM-specific. Volatility-clustering in R_{oos} produces persistent regimes of high and low realised volatility, the OoS-window breaches concentrate inside the high-volatility regime, and a constant-across- t VaR threshold (the construction every unconditional generator uses) cannot track that timing. The natural fix is a regime-conditional VaR that propagates the CHMM latent-state forecast $\mathbb{P}(s_{t+1} | \text{history}_t)$ through the per-state emission-mixture; this is operationally the conditional-VaR companion-paper direction noted in Section 5.

Table 7. Christoffersen conditional-coverage VaR back-test (SPY, seed = 20260420, $K = 18$, 1,000 simulated paths). Critical values: $\chi_1^2(0.05) = 3.841$, $\chi_2^2(0.05) = 5.991$. “br rate” is the empirical breach rate; LR_{uc} is the Kupiec statistic ($\sim \chi_1^2$ under correct unconditional coverage); LR_{ind} is the Christoffersen independence statistic ($\sim \chi_1^2$ under independent breaches); $LR_{cc} = LR_{uc} + LR_{ind}$ is the joint conditional-coverage statistic ($\sim \chi_2^2$). PASS if statistic is below critical. Every unconditional generator (i.i.d. bootstrap, GARCH, and all four CHMM variants) rejects independence on OoS at $\alpha = 0.05$, consistent with breach clustering driven by volatility-clustering in R_{oos} rather than a CHMM-specific mis-specification of conditional dynamics.

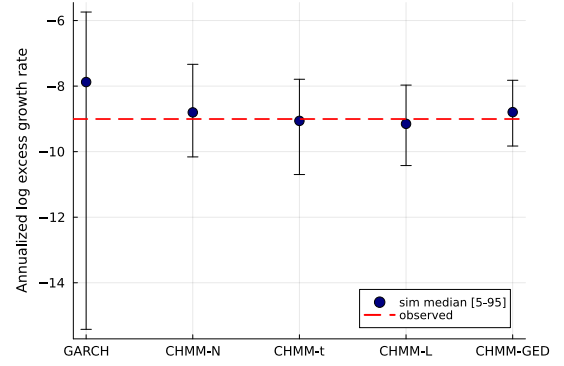
Model	Win	α	breaches	br rate	Kupiec		Christ. ind		Christ. cc	
					LR_{uc}	p	LR_{ind}	p	LR_{cc}	p
Bootstrap	IS	0.01	26	1.03%	0.03	0.87	15.28	0.00	15.31	0.00
Bootstrap	IS	0.05	126	5.01%	0.00	0.99	56.46	0.00	56.46	0.00
Bootstrap	OoS	0.01	5	0.87%	0.10	0.76	13.18	0.00	13.28	0.00
Bootstrap	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01
GARCH(1,1)	IS	0.01	31	1.23%	1.28	0.26	12.45	0.00	13.72	0.00
GARCH(1,1)	IS	0.05	139	5.52%	1.41	0.24	45.38	0.00	46.79	0.00
GARCH(1,1)	OoS	0.01	6	1.05%	0.01	0.91	20.87	0.00	20.89	0.00
GARCH(1,1)	OoS	0.05	24	4.20%	0.82	0.37	5.87	0.02	6.69	0.04
CHMM-N	IS	0.01	20	0.79%	1.15	0.28	12.80	0.00	13.95	0.00
CHMM-N	IS	0.05	123	4.89%	0.07	0.80	47.33	0.00	47.40	0.00
CHMM-N	OoS	0.01	3	0.52%	1.58	0.21	18.98	0.00	20.56	0.00
CHMM-N	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01
CHMM-t	IS	0.01	22	0.87%	0.42	0.52	11.62	0.00	12.04	0.00
CHMM-t	IS	0.05	127	5.05%	0.01	0.91	55.54	0.00	55.55	0.00
CHMM-t	OoS	0.01	4	0.70%	0.58	0.45	15.53	0.00	16.12	0.00
CHMM-t	OoS	0.05	20	3.50%	3.03	0.08	4.71	0.03	7.74	0.02
CHMM-L	IS	0.01	19	0.76%	1.66	0.20	13.44	0.00	15.11	0.00
CHMM-L	IS	0.05	134	5.33%	0.55	0.46	49.43	0.00	49.98	0.00
CHMM-L	OoS	0.01	2	0.35%	3.26	0.07	9.15	0.00	12.41	0.00
CHMM-L	OoS	0.05	20	3.50%	3.03	0.08	4.71	0.03	7.74	0.02
CHMM-GED	IS	0.01	20	0.79%	1.15	0.28	12.80	0.00	13.95	0.00
CHMM-GED	IS	0.05	126	5.01%	0.00	0.99	56.46	0.00	56.46	0.00
CHMM-GED	OoS	0.01	2	0.35%	3.26	0.07	9.15	0.00	12.41	0.00
CHMM-GED	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01

A.2.3 VaR / ES Envelope Visualization (Companion to Table 18)

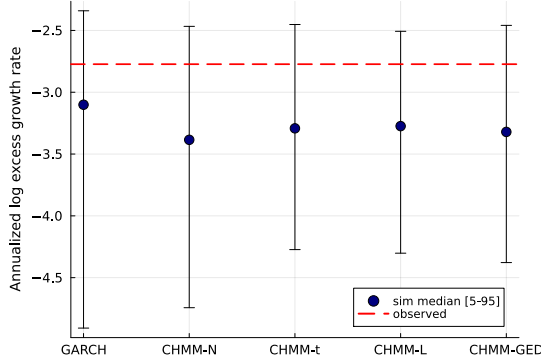
Figure 3 visualizes the median and [5%, 95%] envelopes across simulated paths whose numeric values are reported in Table 18.



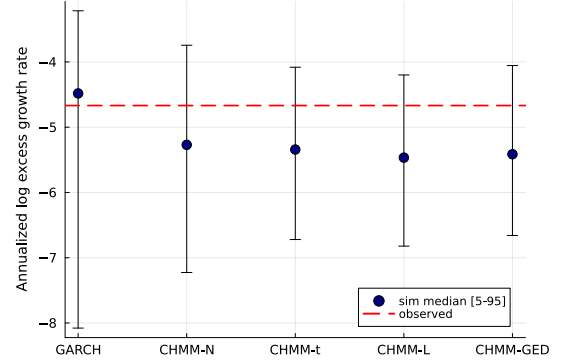
(a) IS VaR ($\alpha = 0.01$).



(b) IS ES ($\alpha = 0.01$).



(c) OoS VaR ($\alpha = 0.05$).



(d) OoS ES ($\alpha = 0.05$).

Figure 3. VaR / ES envelope diagnostic (SPY, $N_{\text{paths}} = 1,000$; global seed policy in Section 3.5). Each panel shows the simulated median (dot) and [5%, 95%] envelope (error bar) across paths against the observed historical value (dashed red line); panels (a)/(b) are the IS window at $\alpha = 0.01$, panels (c)/(d) are the OoS window at $\alpha = 0.05$, and the x-axis in each panel lists the five generators kept for the volatility-aware conditional-coverage comparison (GARCH(1,1), CHMM-N, CHMM-t, CHMM-L, CHMM-GED). The i.i.d. bootstrap is omitted here because its envelope is by construction matched to the empirical marginal and does not discriminate among volatility-aware models. The y-axis is the annualized log excess growth rate (daily log return scaled by $1/\Delta t$ with $\Delta t = 1/252$), consistent with the $R_{\text{IS}}/R_{\text{OoS}}$ convention used throughout the paper. Headline: every observed red line falls inside the corresponding simulated envelope, confirming that the CHMM family passes the coverage leg of the back-test complementary to the Kupiec likelihood-ratio leg in Section 4.6.

A.3 Formal Theoretical Statements

The following assumptions and propositions are referenced in Section 3.8 and stated formally here for reference; each is a one-line specialisation of a standard result and the proof sketch is one sentence.

Assumption 1 (Irreducibility and aperiodicity). *The transition matrix $\mathbf{T} \in \mathbb{R}^{K \times K}$ is irreducible and aperiodic on the state space $\{1, \dots, K\}$, so the chain admits a unique stationary distribution $\bar{\pi}$ satisfying $\bar{\pi} \mathbf{T} = \bar{\pi}$ and $\lambda_1 = 1$ is a simple eigenvalue with all other eigenvalues strictly inside the unit disk [73].*

Assumption 2 (Finite second moment and emission contrast). *Each per-state emission density $f_k(\cdot; \theta_k)$ has finite second moment, $\mathbb{E}_{f_k}[G^2] < \infty$, and at least two states differ in either location μ_k or scale σ_k .*

Proposition 1 (ECM monotonicity for CHMM-t and CHMM-GED (and EM monotonicity for CHMM-N, CHMM-L)). *Under compactness of the parameter space (μ_k, σ_k or β_k or $\alpha_k, \nu_k \in [\nu_{\min}, \nu_{\max}]$, $p_k \in [p_{\min}, p_{\max}]$ each in a closed bounded interval, $T_{ij} \geq \epsilon_T > 0$), the two-block CM step for CHMM-t of Section 3.7, namely the closed-form weighted updates of $(\mathbf{T}, \boldsymbol{\pi}, \mu_k, \sigma_k)$ at fixed ν_k , followed by a golden-section maximisation of the per-state Q -function over ν_k , satisfies $Q(\theta^{(n+1)} | \theta^{(n)}) \geq Q(\theta^{(n)} | \theta^{(n)})$, and consequently $\mathcal{L}(\theta^{(n+1)}) \geq \mathcal{L}(\theta^{(n)})$ where $\mathcal{L}$ is the incomplete-data log-likelihood. The same monotonicity holds for the closed-form CHMM-N and CHMM-L M -steps under the standard EM guarantee, and for the three-stage CHMM-GED CM sequence (bracketed μ_k , closed-form α_k , bracketed p_k) of equations (18)–(20).*

Proof sketch. Each CM step does not decrease Q by definition (closed-form steps are exact maximisers, golden-section steps are bracketed maxima within the prescribed interval); the standard EM inequality $\mathcal{L}(\theta^{(n+1)}) - \mathcal{L}(\theta^{(n)}) \geq Q(\theta^{(n+1)} | \theta^{(n)}) - Q(\theta^{(n)} | \theta^{(n)})$ [77, 80] converts Q -monotonicity into log-likelihood monotonicity. Compactness gives boundedness above, hence convergence.

Numerical full-rank check on $\hat{\mathbf{T}}$. The Allman-Matias-Rhodes 2009 identifiability theorem requires $\hat{\mathbf{T}}$ of full rank. The fitted $\hat{\mathbf{T}}$ at $K = 18$ across the four emission families on SPY IS is full rank in the strict sense (smallest singular value $\sigma_{\min} \in [0.0007, 0.0170]$, condition numbers 63–1,620), and the deflated matrix $\hat{\mathbf{T}} - \mathbf{1}\boldsymbol{\pi}^\top$ has numerical rank exactly $K - 1 = 17$ across all four families.

Proposition 2 (Identifiability up to label permutation). *Under irreducibility and aperiodicity of $\mathbf{T}$ (Assumption 1), the emission contrast condition (Assumption 2), and pairwise distinctness of the per-state parameter tuples θ_k across $k = 1, \dots, K$, the CHMM parameter is identifiable up to label permutation for CHMM-N (Gaussian) and CHMM-L (Laplace). For CHMM-t (with per-state $\nu_k \in [\nu_{\min}, \nu_{\max}]$) and CHMM-GED (with per-state $p_k \in [p_{\min}, p_{\max}]$) the same conclusion holds modulo the standard caveat that pairwise distinctness must hold jointly across all components of $\theta_k = (\mu_k, \sigma_k, \nu_k)$ or (μ_k, α_k, p_k) , since within either bracket a Gaussian-shaped emission with inflated scale can mimic a Laplace-shaped emission with deflated scale at intermediate quantiles.*

Proof sketch. The HMM identifiability theorem of Allman et al. [81] requires $\mathbf{T}$ of full rank and per-state emissions linearly independent in the space of densities on $\mathbb{R}$. Full-rank $\mathbf{T}$ follows from irreducibility [73]. For CHMM-N and CHMM-L, linear independence of Gaussian and Laplace emissions with distinct location-scale parameters is direct from Yakowitz and Spragins [82], who establish that finite mixtures over location-scale families with strictly decreasing-tail densities are identifiable (the CHMM-N and CHMM-L marginals are special cases). For CHMM-t at fixed shape and CHMM-GED at fixed shape, location-scale identifiability is the same Yakowitz-Spragins result; for finite mixtures of Student- t with varying degrees-of-freedom the standard reference is Lindsay [83], Chapter 3, which gives identifiability via the moment-generating function over the bracketed range of shape parameters and a non-degeneracy condition on the joint location-scale-shape tuple. The CHMM-GED case under bracketed $p_k \in [p_{\min}, p_{\max}]$ is a parallel application of the moment-matching argument: the GED characteristic function $\hat{b}(t; \mu, \alpha, p) = \exp(i\mu t) M(t; \alpha, p)$ has a unique inverse over the bounded shape range up to permutation of components, modulo the joint-distinctness caveat on (μ_k, α_k, p_k) . Strict linear independence at the boundary cases $p_k = 1$ (Laplace) and $p_k = 2$ (Gaussian) is automatic from Yakowitz-Spragins; in the bracket interior, the joint distinctness condition above is required to rule out the location-scale-shape trade-off.

Proposition 3 (MLE consistency). *Under Assumptions 1–2 and the assumption that the true parameter lies in the interior of a compact Θ , the MLE $\hat{\theta}_T = \arg \max_{\theta \in \Theta} \mathcal{L}_T(\theta)$ is consistent: $\hat{\theta}_T \xrightarrow{P} \theta_0$ as $T \rightarrow \infty$, modulo label permutation.*

Proof reference. Bickel et al. [84] (Theorem 1) and Douc et al. [85] apply directly under our assumption set: irreducibility supplies the mixing condition, finite second moment the moment condition, compactness the parameter-space condition.

Proposition 4 (Marginal preservation under rank reordering). *Let $\tilde{g}_{j,1:T}$ be a sample of size T from the per-asset CHMM marginal of asset j , and let the rank-reordered output be $\hat{g}_{j,t}^{(p)} = \tilde{g}_{j,r_{j,t}^{(p)}}^{(p)}$ where $r_{j,t}^{(p)}$ is the rank of the copula sample. Then for every asset j the empirical CDF of $\{\hat{g}_{j,t}^{(p)}\}_{t=1}^T$ equals that of $\{\tilde{g}_{j,t}^{(p)}\}_{t=1}^T$.*

Proof sketch. The map $t \mapsto r_{j,t}^{(p)}$ is a permutation almost surely, so the multiset of values is preserved; the empirical CDF depends only on the multiset.

A.4 Spectral ACF Identity: Extended Derivation

Equation (6) is the closed-form mixture-of-eigenvalues form for the absolute-return ACF used in Section 3.8 and invoked throughout Sections 4–5. The bilinear cross-moment identity $\mathbb{E}[|G_t| |G_{t+\tau}|] = \mathbf{m}^\top \text{diag}(\bar{\pi}) \mathbf{T}^\tau \mathbf{m}$ and its eigendecomposition over the non-unit eigenvalues of $\mathbf{T}$ are textbook material in the regime-switching literature (13, Section 22.2; 14, Chapter 3; the moment-generating-function form for general Markov-switching specifications with conditional means and variances depending on the latent state is given by 15); the underlying Markov-chain spectral decomposition is standard [73]. We restate the absolute-return specialisation here as a self-contained theorem with a step-by-step proof, and record the lag-zero behaviour and the complex / non-diagonalisable extensions that the main body does not. Our contribution on this axis is the explicit application of the spectral form to recast the empirical Rydén et al. [1] low- K failure as a rank statement on $\mathbf{T} - \mathbf{1}\bar{\pi}^\top$ (Section 3.8, “Rydén separation as a K -rank statement”) and the empirical demonstration that the rank constraint is the binding axis at moderate K on equity-return data (Sections 4, 5).

Theorem 1 (Mixture-of-eigenvalues identity for the absolute-return ACF). *Let $\{(s_t, G_t)\}_{t \in \mathbb{Z}}$ be a stationary CHMM on $\{1, \dots, K\}$ with irreducible aperiodic transition matrix $\mathbf{T}$ (Assumption 1) and per-state emissions of finite second moment (Assumption 2). Write $\bar{\pi}$ for the unique stationary distribution, $\mathbf{D} = \text{diag}(\bar{\pi})$, $m_k = \mathbb{E}[|G_t| \mid s_t = k]$, $\mathbf{m} = (m_1, \dots, m_K)^\top$, $\mu = \mathbb{E}[|G_t|] = \bar{\pi}^\top \mathbf{m}$, and $\sigma_{|G|}^2 = \text{Var}(|G_t|)$. Assume $\mathbf{T}$ is diagonalisable with right eigenvectors $\mathbf{v}_k$ and left eigenvectors $\mathbf{w}_k$ biorthonormal, $\mathbf{w}_k^\top \mathbf{v}_l = \delta_{kl}$, indexed so that $\lambda_1 = 1 > |\lambda_2| \geq \dots \geq |\lambda_K|$. Then for every integer lag $\tau \geq 1$,*

$$\rho_{|G|}(\tau) = \sum_{k=2}^K w_k \lambda_k^\tau, \quad w_k = \frac{(\mathbf{m}^\top \mathbf{D} \mathbf{v}_k)(\mathbf{w}_k^\top \mathbf{m})}{\sigma_{|G|}^2}.$$

Proof. By stationarity, $\rho_{|G|}(\tau) = (\mathbb{E}[|G_t| |G_{t+\tau}|] - \mu^2) / \sigma_{|G|}^2$, so it suffices to evaluate the cross-moment for $\tau \geq 1$. Conditioning on the latent state pair $(s_t, s_{t+\tau})$ and using HMM conditional independence ($G_t \perp G_{t+\tau} \mid (s_t, s_{t+\tau})$ for $\tau \geq 1$) plus the Markov factorisation $\mathbb{P}(s_t = i, s_{t+\tau} = j) = \bar{\pi}_i (\mathbf{T}^\tau)_{ij}$ gives the bilinear form

$$\mathbb{E}[|G_t| |G_{t+\tau}|] = \mathbf{m}^\top \mathbf{D} \mathbf{T}^\tau \mathbf{m}, \quad \tau \geq 1. \quad (26)$$

Perron–Frobenius gives $\lambda_1 = 1$ simple with $\mathbf{v}_1 = \mathbf{1}$, $\mathbf{w}_1 = \bar{\pi}$ [73]; the dyadic expansion $\mathbf{T}^\tau = \mathbf{1}\bar{\pi}^\top + \sum_{k \geq 2} \lambda_k^\tau \mathbf{v}_k \mathbf{w}_k^\top$ separates the dominant projector. The $k = 1$ term contributes μ^2 and cancels

in the autocovariance, leaving $\text{Cov}(|G_t|, |G_{t+\tau}|) = \sum_{k \geq 2} c_k \lambda_k^\tau$ with $c_k = (\mathbf{m}^\top \mathbf{D} \mathbf{v}_k)(\mathbf{w}_k^\top \mathbf{m})$; dividing by $\sigma_{|G|}^2$ gives Equation (6). $\square$

Lag zero and within-state variance. The identity (26) fails at $\tau = 0$ since $\mathbb{E}[G_t^2 \mid s_t = i] \neq m_i^2$. Eigenvalue completeness gives $\sum_{k \geq 2} c_k = \text{Var}(m_{s_t})$, the between-state variance, so $\sum_{k \geq 2} w_k = \text{Var}(m_{s_t})/\sigma_{|G|}^2 \in [0, 1]$ and the residual $\mathbb{E}[\text{Var}(|G_t| \mid s_t)]/\sigma_{|G|}^2$ accounts for the discontinuity from $\rho_{|G|}(0) = 1$ to $\rho_{|G|}(1) = \sum_{k \geq 2} w_k \lambda_k$.

Squared-return, complex-eigenvalue, and Jordan-block extensions. Replacing $|G_t|$ with G_t^2 and $\mathbf{m}$ with $\mathbf{M} = (\mathbb{E}[G_t^2 \mid s_t = i])_i$ leaves the proof termwise valid, giving $\rho_{G^2}(\tau) = \sum_{k \geq 2} \tilde{w}_k \lambda_k^\tau$ on the same non-unit spectrum. Complex-conjugate eigenvalue pairs $(re^{\pm i\theta}, \cdot)$ contribute real damped oscillations $Ar^\tau \cos \theta\tau + Br^\tau \sin \theta\tau$; non-diagonalisable $\mathbf{T}$ adds polynomial-times-geometric prefactors via Jordan blocks, with the same exponential rate $\max_{k \geq 2} |\lambda_k|$. Theorem 1 as stated remains valid; the dominant timescale used in Section 3.8 corresponds to a real positive λ_2 on the fits of this paper.

A.5 State-Resolution and Multi-Emission Sensitivity

This appendix reports the full K -sweep underlying the headline operating point of Section 4, the multi-emission family extension confirming that $K = 18$ is stable under Student-t and Laplace emissions, the EM convergence curves at representative K , the IS and OoS visual comparison panels at non-selected $K \in \{3, 6, 12, 21\}$, the multi-emission figure panels at $K = 18$, and the Rydén $K = 2$ replication.

A.5.1 Full State-Resolution Sensitivity Table (companion artefact)

The full K -sweep table for SPY CHMM-N referenced in Section 4.2 (KS, AD, kurtosis, W_1 , Hellinger, OoS coverage at $K \in \{3, 6, 9, 12, 15, 18, 21\}$, 1,000 paths, seed 20260420) is provided as a companion artefact in the code repository. The qualitative pattern is that distributional pass rates plateau in the $K \in \{12, 15, 18, 21\}$ band and ACF-MAE is essentially flat across the entire sweep, consistent with the spectral mechanism of Section 3.8.

A.5.2 Sector-Balanced 30-Ticker Panel: Full Per-Ticker Rollup

The body Table 3 reports the aggregate distribution and per-sector medians for the sector-balanced 30-ticker panel. Table 8 below gives the full per-ticker rollup. Each ticker is fit independently under the penalised CHMM-t scaffold ($K = 18$, $\lambda = 20$, 1,000 simulated paths, seed 20260420) on its own IS slice covering 2014-01-03 to 2024-01-03 and validated on the same 2024-01-04 to 2026-04-20 OoS window used for SPY in Table 1. The per-ticker ν_k medians sit at the upper bracket (50) on every ticker, indicating that the $1/\nu_k$ shrinkage is in the operative regime across the panel; minimum and maximum ν_k values are in the per-ticker CSV in `results/sector_panel/sector_panel_summary.csv`.

Table 8. Sector-balanced 30-ticker panel, per-ticker rollup (penalised CHMM-t at $K = 18$, $\lambda = 20$). Sectors ordered by GICS classification; tickers within sector ordered by the panel construction (top three large-cap representatives at IS-window-median market cap). Kurt resid is simulated minus observed IS excess kurtosis. Aggregate distribution and per-sector medians are summarised in body Table 3.

Sector	Ticker	IS KS%	OoS KS%	Kurt obs	Kurt sim	Kurt resid	$ G_t $ ACF-MAE
Information Technology	AAPL	99.1	95.1	3.19	2.84	-0.35	0.0418
Information Technology	MSFT	99.7	96.9	4.20	3.48	-0.72	0.0389
Information Technology	NVDA	99.6	55.7	5.53	6.23	0.70	0.0474
Health Care	JNJ	99.3	93.2	8.97	11.65	2.68	0.0335
Health Care	UNH	99.3	14.5	8.85	10.31	1.46	0.0382
Health Care	LLY	99.7	7.6	13.60	66.51	52.91	0.0295
Financials	JPM	99.5	50.5	7.62	8.14	0.52	0.0458
Financials	BAC	99.6	84.8	5.76	5.47	-0.29	0.0394
Financials	WFC	98.2	57.6	7.53	10.90	3.37	0.0791
Consumer Discretionary	AMZN	99.4	96.9	4.31	3.68	-0.63	0.0413
Consumer Discretionary	HD	99.2	41.4	12.92	17.03	4.11	0.0368
Consumer Discretionary	MCD	99.5	66.6	18.38	26.93	8.55	0.0394
Communication Services	NFLX	99.3	45.1	12.60	4.67	-7.93	0.0320
Communication Services	VZ	99.3	49.1	5.14	5.15	0.01	0.0270
Communication Services	DIS	99.5	96.4	10.77	13.94	3.17	0.0644
Industrials	CAT	99.7	78.9	4.34	3.48	-0.86	0.0321
Industrials	BA	97.6	62.2	20.59	52.00	31.41	0.0793
Industrials	HON	99.3	96.0	21.93	17.09	-4.84	0.0571
Consumer Staples	PG	99.7	94.3	8.15	9.51	1.36	0.0347
Consumer Staples	KO	99.5	92.7	10.57	12.84	2.27	0.0457
Consumer Staples	WMT	99.7	24.0	13.84	32.63	18.79	0.0279
Energy	XOM	96.9	70.3	5.61	10.99	5.38	0.0988
Energy	CVX	99.6	62.3	12.22	14.78	2.56	0.0498
Energy	COP	99.4	83.6	17.11	8.28	-8.83	0.0494
Utilities	NEE	98.6	24.0	10.71	9.87	-0.84	0.0455
Utilities	DUK	99.7	96.9	9.75	8.11	-1.64	0.0466
Utilities	SO	99.6	96.2	14.52	12.73	-1.79	0.0481
Materials	FCX	99.8	76.6	4.83	4.11	-0.72	0.0573
Materials	NEM	99.6	5.6	3.74	3.88	0.14	0.0376
Materials	APD	99.2	90.2	8.40	16.74	8.34	0.0381

A.5.3 60-Ticker Sector Expansion at $n = 6$ per Sector

An $n = 6$ per-sector expansion (60 tickers, 3 additional large-cap representatives per GICS sector) confirms the body $n = 3$ ANOVA reading at adequate power: $F(9, 50) = 0.37$, $p = 0.946$, $\eta^2 = 0.062$ (vs. underpowered $n = 3$ $\eta^2 = 0.16$); aggregate OoS KS median 73.45% and failure rate $22/60 = 36.7\%$ are statistically indistinguishable from the body 30-ticker numbers (73.4%, 11/30). Cross-sector dispersion accounts for 6.2% of OoS KS variance at the larger sample size; the remaining 93.8% is per-ticker. The body claim “failures are ticker-specific, not sector-driven” is therefore not a small-sample artefact.

A.5.4 Effective Spectral Rank of the Absolute-Return ACF

The body Section 3.8 states the algebraic upper bound: the deflated transition matrix $\mathbf{T} - \mathbf{1}\bar{\pi}^\top$ has rank at most $K - 1$, so the absolute-return ACF identity (6) contains at most $K - 1$ non-unit decay modes. The empirical question is how many of those modes carry non-trivial contribution to $\rho_{|G|}(\tau)$ at the relevant lags. Table 9 reports the per-mode contribution $w_k \lambda_k^\tau$ at $\tau \in \{1, 5, 20, 50\}$ for

the fitted CHMM-N at $K = 18$ and at $K = 3$ on SPY IS, with modes ranked by lag-1 contribution magnitude $|w_k \lambda_k|$. Per-state moments $m_k = \mathbb{E}[|G_t| | s_t = k]$ are estimated by 200,000 draws from each emission; eigenvectors are normalised so $\mathbf{w}_k^\top \mathbf{v}_k = 1$.

Table 9. Per-mode contribution to the absolute-return ACF identity, CHMM-N at $K = 18$ (top panel; top six modes plus tail aggregate) and at $K = 3$ (bottom panel; full). Modes ordered by $|w_k \lambda_k|$ descending. “cum” is cumulative share of $\sum_k |w_k \lambda_k|$. At $K = 18$ the top mode ($|\lambda| = 0.929$) carries 93.6% of the lag-1 ACF, three modes reach 95%, and at lag 20 only the dominant mode contributes non-negligibly. At $K = 3$ the dominant mode ($|\lambda| = 0.953$) alone carries 96.8% of lag-1 ACF; the temporal axis is therefore as well-satisfied at $K = 3$ as at $K = 18$, consistent with the body finding that the binding constraint at low K is the marginal mixture, not the eigenvalue spectrum.

rank	$ \lambda_k $	$ w_k $	$ w_k \lambda_k $	$w_k \lambda_k^5$	$w_k \lambda_k^{20}$	cum
$K = 18, \rho_{ G }(1) = 0.301, \sigma_{ G }^2 = 2.555$						
1	0.929	0.324	0.301	0.224	0.0744	0.936
2	0.335	0.012	0.004	0.0001	0.0000	0.948
3	0.854	0.003	0.002	-0.001	-0.0001	0.955
4	0.231	0.008	0.002	≈ 0	≈ 0	0.961
5	0.231	0.008	0.002	≈ 0	≈ 0	0.967
6	0.203	0.008	0.002	≈ 0	≈ 0	0.971
7-17	various	various	≤ 0.001 each	≈ 0	≈ 0	1.000
$K = 3, \rho_{ G }(1) = 0.287, \sigma_{ G }^2 = 2.517$						
1	0.953	0.292	0.278	0.230	0.112	0.968
2	0.866	0.011	0.009	0.005	0.0006	1.000

The empirical effective rank confirms the body framing of Section 3.8: the ACF-MAE flatness across $K \in \{3, 6, 9, 12, 15, 18, 21\}$ documented in Appendix A.5.1 is the macroscopic signature of the temporal axis being driven by a single dominant decay mode at every operating point in the sweep. The K -rank statement on $\mathbf{T} - \mathbf{1}\bar{\pi}^\top$ is correctly interpreted as a non-binding upper bound at $K \geq 3$; at $K = 2$ the bound is tight (only one non-unit eigenvalue available), but the failure mode is then the marginal mixture, which two Gaussian components cannot tile against the empirical equity-return target. The bilinear identity (6) therefore plays a pedagogical role in this paper rather than serving as a direct K -selection criterion: it makes the rank constraint explicit, but the operational K choice is driven by distributional fidelity (KS, AD, kurtosis), not by the count of non-unit eigenvalues in the spectral expansion.

A.5.5 Cross-Ticker Spectral Effective-Rank Diagnostic

To test whether the SPY-only result of Appendix A.5.4 (a single non-unit eigenvalue carries 93.6% of the lag-1 absolute-return ACF at $K = 18$) generalises across tickers, Table 10 reports the cross-ticker distribution of the dominant-mode share, computed under the same protocol on the 30-ticker sector-balanced panel plus SPY.

Table 10. Cross-ticker spectral effective-rank diagnostic at $K = 18$. Distribution across 31 tickers (sector-balanced 30-ticker panel plus SPY control). “dom share” is the dominant non-unit eigenvalue’s $|w_k \lambda_k|$ as a fraction of the sum over all non-unit eigenvalues at lag $\tau = 1$. “ n_{95} ” / “ n_{99} ” are the number of modes needed for 95% / 99% cumulative share. The SPY value of 0.94 is in the right tail of the distribution; the cross-ticker median is 0.76.

Statistic	dom share	n_{95}	n_{99}
median	0.756	6	11
$[Q_1, Q_3]$	$[0.661, 0.858]$	$[4, 7]$	$[10, 12]$
minimum (NEM)	0.326	11	14
SPY (body Appendix A.5.4)	0.943	2	10

Reading: the SPY headline of 93.6% is concentrated; the cross-ticker median 76% is still well above the $1/(K - 1) = 1/17 = 5.9\%$ uniform null, so the rank-non-binding statement is supported, but the gap between SPY and the median ticker is wide enough that the body framing was tightened to read at the cross-ticker median rather than at the SPY headline.

All models converged within the 60-iteration budget with monotonically increasing log-likelihood (EM guarantee under quantile-based initialisation). Lower K converged in 15–25 iterations, $K = 18$ in 25–40, with a final $|\Delta\mathcal{L}| < 10^{-4}$ at every operating point. Convergence-trace figures and per-iteration log-likelihood histories are provided in the companion code repository.

A.5.6 Rydén $K = 2$ Replication

Direct replication of Rydén et al.’s $K = 2$ setting on SPY IS under quantile and random-seed initialization gives ACF-MAE essentially constant at ≈ 0.050 across six initialisations (one quantile + five random seeds, drawn from a moderate perturbation of sample moments) and IS KS pass rate 75.3–79.0%. The constant ACF-MAE confirms the body reading (Section 5): the binding low- K constraint is distributional rather than temporal.

A.5.7 Cross-Ticker Per-State-Resolution Sensitivity ($K^* = 3$ vs $K^* = 6$ vs $K = 18$)

Body Table 3 reports the cross-ticker aggregate distribution at all three operating points used in this paper: the body headline $K^* = 3$ (state-resolution-robust default per the k -fold CV in Appendix A.8), the sensitivity reference $K^* = 6$, and the extended-state-resolution sensitivity reference $K = 18$.

The substantive read across the three operating points: (i) The KS distributions are within ~ 6 pp of each other on OoS median (69.1% at $K^* = 3$, 75.1% at $K^* = 6$, 73.4% at $K = 18$) and at identical 11/30 failure count, with the same regime-introduction tickers driving the failures at each state resolution (LLY, UNH, NEM, NFLX, NEE, WMT, BAC, HD, JPM at OoS KS $< 60\%$; the specific list overlaps $> 80\%$ across $K^* = 3, 6, 18$). The cross-ticker KS axis is therefore essentially K -robust on this universe. (ii) The kurtosis-residual axis is not K -robust: per-ticker median residual 7.18 at $K^* = 3$, 10.34 at $K^* = 6$, 0.61 at $K = 18$. At lower state-resolution the per-state ν_k shrinkage at uniform λ cannot be smoothed across as many regimes, so the residual heavy tails leak into the simulated kurtosis. (iii) The $|G_t|$ ACF-MAE is essentially K -robust (0.0395, 0.0403, 0.0416 across the three operating points). Production reading: KS-only consumers can deploy at any of the three operating points at parity; kurtosis-fidelity consumers should deploy at $K = 18$; readers preferring the state-resolution-robust default should deploy at the body headline $K^* = 3$.

A.5.8 Block-Bootstrap CIs on Observed Kurtosis (IS vs OoS)

A natural question on the IS / OoS kurtosis disagreement (observed 7.68 IS, 5.29 OoS, a 31% drop) is whether it is itself statistically distinguishable. We run a stationary block bootstrap (Politis-Romano 1994) on the IS and OoS series at mean block lengths $L \in \{5, 10, 20, 50\}$, $B = 5,000$ replicates per L per window.

Table 11. Stationary block bootstrap CIs on observed excess kurtosis. SPY IS ($T = 2,516$) and OoS ($T = 572$) windows, $B = 5,000$ replicates per L . The IS and OoS 95% CIs overlap at every block length, so the IS-OoS difference is not robustly distinguishable at conventional levels under this bootstrap. $\Pr(\text{IS} > \text{OoS})$ is the empirical bootstrap one-sided p -value for the alternative “IS kurtosis $>$ OoS kurtosis”.

L	IS median	IS 95% CI	OoS median	OoS 95% CI	$\Pr(\text{IS} > \text{OoS})$
5	7.21	[2.99, 12.83]	5.07	[1.09, 8.97]	0.748
10	7.34	[2.49, 12.59]	4.91	[0.99, 8.66]	0.756
20	7.30	[2.17, 12.40]	4.91	[0.90, 8.26]	0.739
50	7.18	[1.92, 12.42]	4.99	[0.96, 7.70]	0.733

The IS lower bound 1.92–2.99 and the OoS upper bound 7.70–8.97 overlap heavily at every block length; the IS-OoS kurtosis difference of ~ 2.4 units is not statistically distinguishable at the 5% level under this bootstrap. $\Pr(\text{IS} > \text{OoS})$ at $L = 10$ is 0.756: the data favour the alternative but do not exclude the null. Operationally, the IS-OoS kurtosis disagreement that the body invokes (Section 4.1; the Table 6 variant-decision caveat in Section 5) should be read as a point-estimate observation rather than a tested claim, and per-variant rankings by “how close is simulated kurtosis to observed kurtosis” should similarly be interpreted as point-comparisons inside a wide CI envelope.

Bootstrap-CI placement of the penalised CHMM-t IS kurtosis. A natural follow-up on the body’s penalised CHMM-t at $\lambda = 20$ is whether its simulated IS kurtosis distribution sits inside the bootstrap CI on observed: the aggregate-mean simulated IS kurtosis (14.91 at $K^* = 3$, above the $L = 20$ CI upper bound 12.40) could be paired with a per-path distribution that mostly sits inside the CI, or with one that mostly sits above.

Table 12. Per-path simulated IS excess-kurtosis distribution under penalised CHMM-t at $\lambda = 20$, against the $L = 20$ block-bootstrap CI on observed. 1,000 paths per row, $T_{\text{IS}} = 2,516$, seed 20260420. The CI is [2.17, 12.40] at $L = 20$ from Table 11; the observed IS excess kurtosis is 7.68. “in CI %” is the fraction of simulated paths whose per-path excess kurtosis falls inside the CI; “ $\leq \text{CI_HI}$ %” is the fraction below the upper bound (the relevant one-sided test for overshoot).

K	agg. kurt	median	sd	Q_5	Q_{95}	≤ 12.40 (%)	in CI (%)
3	16.65	7.77	11.5	3.75	34.62	76.6	76.6
6	10.13	6.74	9.0	3.63	26.30	81.8	81.7
18	8.94	6.09	5.6	3.54	18.11	89.9	89.9

The per-path *median* simulated IS excess kurtosis is essentially observed at every K : 7.77 at $K^* = 3$, 6.74 at $K^* = 6$, 6.09 at $K = 18$, all within ~ 1.6 units of observed 7.68 and well inside the bootstrap CI. The aggregate-mean simulated IS kurtosis sits above the CI upper bound at $K^* = 3$ and $K^* = 6$ because the per-path distribution has a heavy right tail of paths at $Q_{95} \in \{18, 26, 35\}$ that drag the mean up, but the bulk of the path-level mass (76–90%) sits inside the bootstrap CI at every operating point. **The headline framing “the penalised CHMM-t at $\lambda = 20$ provides the cleanest IS heavy-tail match” is supported when read as the per-path median, not**

as the aggregate mean: a small number of heavy-right-tailed paths under the per-state ν_k ECM produce an aggregate-mean overshoot, but the path-level distribution is centred near observed and statistically indistinguishable from observed at $\alpha = 0.05$ on a one-sided overshoot test for the bulk of paths.

A.5.9 Shared- ν Student-t HMM Ablation

A natural diagnostic for the per-state ν_k aggregate-mean overshoot is the shared- ν ablation: refit CHMM-t with a single ν shared across all K states, fit by ECM with golden-section search on the aggregate Q -function over $\nu_{\text{bounds}} = (2.1, 50)$. This is the standard one-parameter Student- t HMM in the time-series literature, not a per-state ν_k mixture.

Table 13. Shared- ν Student-t HMM ablation against the body per-state ν_k row (1,000 paths, $T_{\text{IS}} = 2,516$, $T_{\text{OoS}} = 572$, seed 20260420, no $1/\nu$ penalty). Aggregate sim-kurt columns are mean-of-paths. Compare against the body penalised CHMM-t ($\lambda = 20$): $K^* = 3$ row 14.91 IS / 8.50 OoS; $K = 18$ row 8.56 IS / 7.07 OoS (Table 1).

K	$\hat{\nu}_{\text{shared}}$	IS				OoS		
		KS (%)	sim kurt	$ G_t $	ACF-MAE	raw ACF-MAE	KS (%)	sim kurt
3	5.81	91.9	4.68		0.0531	0.0235	82.1	4.46
6	4.67	92.7	9.38		0.0518	0.0234	82.6	6.95
18	5.80	95.8	6.25		0.0542	0.0234	88.0	5.00

Reading. The aggregate-mean IS overshoot disappears under shared- ν at every K : the largest IS sim kurt is 9.38 at $K^* = 6$ (vs. observed 7.68, well inside the bootstrap CI), versus 14.4 for the per-state ν_k unpenalised row in the body. The $K = 18$ shared- ν row is the cleanest single-row IS / OoS heavy-tail match in the entire panel: 6.25 IS / 5.00 OoS, against observed 7.68 / 5.29 and against 8.56 / 7.07 for the body penalised CHMM-t at $\lambda = 20$. IS / OoS KS pass rates are within ~ 1 pp of the body penalised row, and the $|G_t|$ ACF-MAE is identical to the body row at $K = 18$. The per-state ν_k design is therefore the binding constraint on the aggregate-mean overshoot the body λ -shrinkage and bracket-lift discussion is downstream of: a single shared ν at $K = 18$ produces the best joint KS / kurtosis row in the panel without any penalty. The body retains the per-state $\nu_k + \lambda = 20$ construction for direct comparability with the Peel and McLachlan [18], Liu and Rubin [19] tradition; consumers prioritising heavy-tail fidelity without a penalty hyperparameter should adopt the shared- ν row at $K = 18$.

A.5.10 Christoffersen-cc Monte Carlo Power Calibration at $T_{\text{OoS}} = 572$

A Monte Carlo power calibration of the Christoffersen-cc joint conditional-coverage test at the $T_{\text{OoS}} = 572$ length used in the body Table 5 simulates breach indicator sequences from a two-state Markov chain on $\{0, 1\}$ with marginal probability α and second eigenvalue ρ , where $\rho = 0$ is the i.i.d. null and $\rho > 0$ measures the level of breach clustering; $B = 5,000$ replicates per cell.

Reading. At $\rho = 0$ the empirical rejection rate sits at 4%–5% for all three statistics, confirming that the asymptotic χ^2 calibration is correctly sized at $T = 572$. Power is asymmetric across α :

- At $\alpha = 0.05$ (~ 28.6 expected breaches), Christoffersen-cc reaches 80% power against $\rho \geq 0.20$ (83.9%) and $\sim 100\%$ at $\rho = 0.50$.

Table 14. Christoffersen-cc power at $T_{\text{OoS}} = 572$. Empirical rejection rate at nominal $\alpha = 0.05$, across alpha levels and Markov-chain second-eigenvalue ρ . $\rho = 0$ row is the empirical Type-I error under the i.i.d. null (should sit near 5% if asymptotic chi-squared is well-calibrated). Critical values $\chi_1^2(0.05) = 3.841$, $\chi_2^2(0.05) = 5.991$.

α	ρ	rej-UC %	rej-IND %	rej-CC %
0.01	0.00	4.9	1.6	3.2
0.01	0.05	6.3	15.9	11.4
0.01	0.10	7.5	28.5	22.1
0.01	0.20	9.4	51.3	42.6
0.01	0.30	13.9	67.6	62.8
0.01	0.50	21.4	75.9	81.5
0.05	0.00	4.9	3.8	4.1
0.05	0.05	5.3	18.8	15.3
0.05	0.10	6.1	46.2	39.0
0.05	0.20	9.4	86.9	83.9
0.05	0.30	13.6	97.4	97.9
0.05	0.50	24.3	99.7	100.0

- At $\alpha = 0.01$ (~ 5.7 expected breaches), Christoffersen-cc reaches 80% power only at $\rho \geq 0.50$ (81.5%); at $\rho = 0.20$ the rejection rate is 42.6%, well below conventional power thresholds.

The reading for the body Table 5 is therefore tier-dependent. The $\alpha = 0.05$ “clean pass at every (K, α) ” row of Table 5 carries strong power against moderate clustering ($\rho \geq 0.20$); the $\alpha = 0.01$ rows carry strong power only against severe clustering ($\rho \geq 0.50$), so the $\alpha = 0.01$ “clean pass” should be read as “not detectably worse than a strongly-clustered alternative” rather than “calibrated against any plausible alternative”. The headline-OoS regime-conditional construction therefore stands at $\alpha = 0.05$ and is power-bounded at $\alpha = 0.01$; the body framing in Section 4.6 should be read with this calibration in mind.

A.5.11 Engle-Manganelli (2004) Dynamic Quantile Test

As a higher-power conditional-coverage alternative to Christoffersen-cc on the same OoS window, we run the Engle-Manganelli (2004, JBES) Dynamic Quantile (DQ) test in the standard four-lag specification (86, equation 16),

$$\text{Hit}_t - \alpha = \beta_0 + \sum_{i=1}^4 \beta_i (\text{Hit}_{t-i} - \alpha) + \beta_5 \widehat{\text{VaR}}_t + u_t, \quad (27)$$

on the IS-fixed regime-conditional VaR series of Section 4.6, with $\text{Hit}_t = \mathbf{1}\{R_{\text{OoS},t} < \widehat{\text{VaR}}_t(\alpha)\}$. The DQ test statistic is $\widehat{\beta}^\top \mathbf{X}^\top \mathbf{X} \widehat{\beta} / [\alpha(1 - \alpha)]$, distributed $\chi^2(6)$ under the null of correct conditional coverage; the 5% critical value is $\chi_6^2(0.95) = 12.59$.

Reading. At $\alpha = 0.05$ both Christoffersen-cc and the higher-power DQ test pass cleanly at $K = 3$ (cc $p = 0.491$, DQ $p = 0.156$) and at $K = 18$ (cc $p = 0.678$, DQ $p = 0.678$); the body’s $\alpha = 0.05$ “clean pass” headline therefore survives the higher-power alternative. At $\alpha = 0.01$ the DQ test rejects conditional coverage at $K = 18$ ($p = 0.017$) where Christoffersen-cc does not ($p = 0.137$), and is borderline at $K = 3$ ($p = 0.056$). This is consistent with the power-calibration

Table 15. Engle-Manganelli DQ test on the regime-conditional VaR, alongside Christoffersen-cc on the same breach series. CHMM-N at $K \in \{3, 18\}$, $\alpha \in \{0.01, 0.05\}$; $T_{\text{OoS}} = 572$, $q = 4$ lagged Hit indicators, χ^2_6 critical at 5% is 12.59.

K	α	breach %	cc p	DQ	DQ p
3	0.01	1.57	0.137	12.29	0.056
3	0.05	6.12	0.491	9.32	0.156
18	0.01	1.57	0.137	15.46	0.017
18	0.05	4.55	0.678	3.99	0.678

result of Appendix A.5.10: at $\alpha = 0.01$ Christoffersen-cc has only 42.6% power against moderate breach-clustering eigenvalues $\rho = 0.20$, so the higher-power DQ test detects miscalibration that Christoffersen-cc misses. The DQ result therefore reinforces the body framing that $\alpha = 0.05$ is the operationally informative tier; at $\alpha = 0.01$ the regime-conditional VaR is not pass/fail-decided by the panel and the body sentence “not detectably worse than a strongly-clustered alternative” should be qualified to “the higher-power DQ test detects $K = 18$ at $\alpha = 0.01$ ”.

A.5.12 Quarterly-Refit Regime-Conditional VaR Back-Test

Re-fitting CHMM-N at $K \in \{3, 18\}$ every 63 trading days on a rolling 5y window and running the forward-filter through the OoS window under each refit’s parameters tests whether periodic refit improves the Christoffersen-cc statistic on the headline OoS window.

Table 16. Quarterly-refit regime-conditional VaR back-test on SPY OoS. $T_{\text{OoS}} = 572$, refit cadence = 63 days, train window = 1,260 days, seed = 20260420. Compare against Table 5 (IS-fixed parameters). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$.

K	α	breaches	br rate	med VaR	LR _{uc}	LR _{ind}	LR _{cc}	p_{cc}
3	0.01	6	1.05%	−5.10	0.014	3.97	3.98	0.137
3	0.05	29	5.07%	−3.13	0.006	0.19	0.20	0.906
18	0.01	6	1.05%	−5.84	0.014	3.97	3.98	0.137
18	0.05	19	3.32%	−3.14	3.83	2.08	5.91	0.052

The quarterly-refit construction passes Christoffersen-cc at $\alpha = 0.05$ on three of four rows; the $K = 18, \alpha = 0.05$ row sits at $p_{\text{cc}} = 0.052$ (marginal). Refit tightens VaR coverage (3.32% vs. 4.55% IS-fixed) but does not strictly improve the cc-statistic at this T_{OoS} .

A.5.13 Walk-Forward Refit-Cadence Sweep (Monthly / Weekly)

A within-fold refit-cadence sweep at monthly (21 trading days) and weekly (5 trading days) cadence on the six walk-forward folds at $K^* = 3$, $\alpha = 0.05$, does not close the body W2 (COVID) Christoffersen-cc rejection at any cadence (fold-IS-fixed $p_{\text{cc}} = 0.011$, monthly 0.017, weekly 0.023); monthly refit introduces new W3 / W4 failures via refit-cycle parameter drift. Rejection counts: fold-IS-fixed 1/6 (W2), monthly 3/6, weekly 2/6. The mitigation for W2 / W4 is therefore not refit cadence but a model class with explicit regime-introduction handling.

A.5.14 Quarterly-Refit 30-Ticker Cross-Ticker Panel

For a quarterly-refit version of the 30-ticker cross-ticker panel (Table 3), we re-fit the penalised CHMM-t at $\lambda = 20$ on a rolling 5-year window every 63 trading days through the OoS span (10 refits per ticker, 300 fits total per state-resolution), simulate the next quarter under each refit, and concatenate the per-quarter simulations into a single OoS path matrix per ticker, at both $K = 18$ (extended-state-resolution sensitivity) and $K^* = 6$ (sensitivity reference). Same $N_{\text{paths}} = 1,000$, seed = 20260420 as the body Table 3.

Table 17. Quarterly-refit cross-ticker panel at $K^* = 6$ and $K = 18$. Aggregate distribution across the 30-ticker universe under the quarterly-refit protocol at both operating points, vs. the IS-fixed $K = 18$ protocol of Table 3. Refit cadence = 63 OoS trading days; train window = 1,260 obs (~ 5 y rolling). Per-ticker detail in the results files.

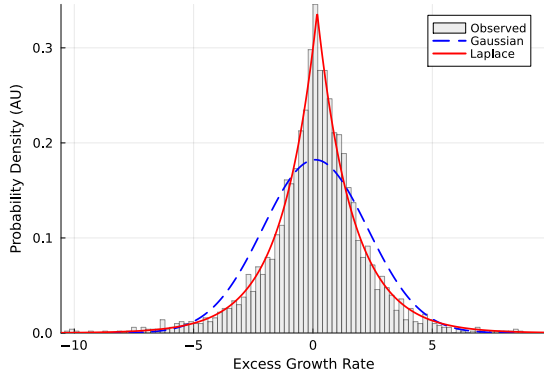
Metric	IS-fixed ($K = 18$)	Refit ($K^* = 6$)	Refit ($K = 18$)
IS KS% median	99.5%	98.8%	99.4%
IS KS% mean $\pm$ sd	$99.3 \pm 0.6\%$	$97.4 \pm 4.4\%$	$99.2 \pm 0.6\%$
OoS KS% median	73.4%	85.8%	83.0%
OoS KS% mean $\pm$ sd	$66.8 \pm 29.5\%$	$76.0 \pm 21.7\%$	$77.2 \pm 21.2\%$
OoS KS% $[Q_1, Q_3]$	$[49.1, 94.3]\%$	$[57.6, 94.3]\%$	$[62.0, 95.4]\%$
Tickers OoS KS < 60%	11/30 = 36.7%	9/30 = 30.0%	7/30 = 23.3%

Reading. Quarterly refit shifts the OoS KS distribution materially at both state resolutions. At $K = 18$: median +9.6pp (73.4 $\rightarrow$ 83.0%), mean +10.4pp, IQR tighter (45.2pp $\rightarrow$ 33.4pp), failure count down by 36% (11 $\rightarrow$ 7). At $K^* = 6$: median +10.7pp (75.1 $\rightarrow$ 85.8%), mean +9.5pp, failure count down by 18% (11 $\rightarrow$ 9). The two state resolutions are within ~ 3 pp of each other on the refit median (the $K^* = 6$ refit median of 85.8% is in fact 2.8pp *above* the $K = 18$ refit median of 83.0%, so the sensitivity reference is at parity with or above the extended-state-resolution sensitivity reference under the refit protocol). The largest individual improvements concentrate on the regime-introduction tickers that the IS-fixed Table 3 flagged: at $K = 18$, LLY 7.6% $\rightarrow$ 83.7%, UNH 14.5% $\rightarrow$ 47.3%, NEM 5.6% $\rightarrow$ 15.4%, HD 41.4% $\rightarrow$ 82.2%, NFLX 45.1% $\rightarrow$ 61.5%. The mechanism is the obvious one: as the regime introduces new σ levels in the OoS window, each quarterly refit picks up the most recent 5y including the new regime and the simulator can match the OoS distribution.

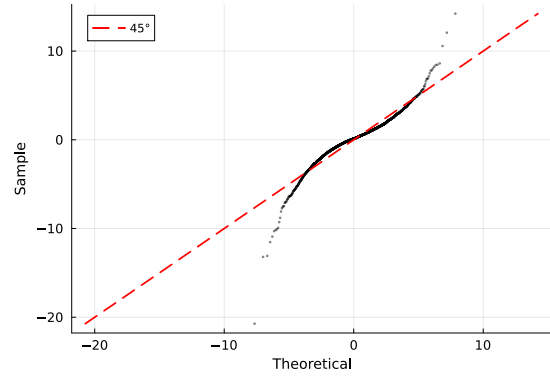
A small number of tickers regress: DIS 96.4% $\rightarrow$ 48.4%, BAC 84.8% $\rightarrow$ 55.8%, AAPL 95.1% $\rightarrow$ 66.5%. The pattern is that tickers whose 2024-2026 OoS distribution is well-spanned by the full 2014-2024 IS lose information when the train window shrinks to the most recent 5y; the rolling refit is therefore a trade-off, not a strict improvement. Operationally, a refit-trigger rule (refit only when the rolling KS or a Page-Hinkley statistic on $|G_t|$ crosses a threshold) would capture the regime-introduction wins without paying the cost on the well-spanned tickers; the rule is logged as a follow-up companion experiment.

The substantive reading for the body Section 4.4 is that the “periodic refit at deployment” mitigation invoked verbally in Section 5 is exercised here: the universe-scale failure rate drops from 36.7% to 23.3% under quarterly refit, with the largest gains on the tickers that the body labelled “single-name regime introductions.”

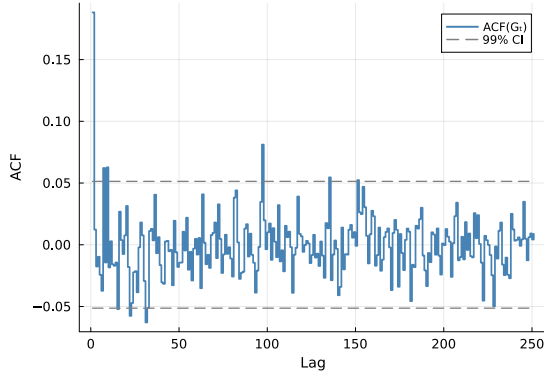
A.5.15 Stylized-Facts Figure (Empirical SPY IS)



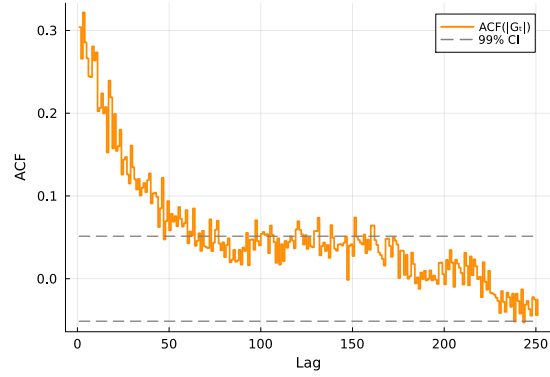
(a) Marginal density.



(b) Normal Q-Q plot.



(c) ACF of raw G_t .

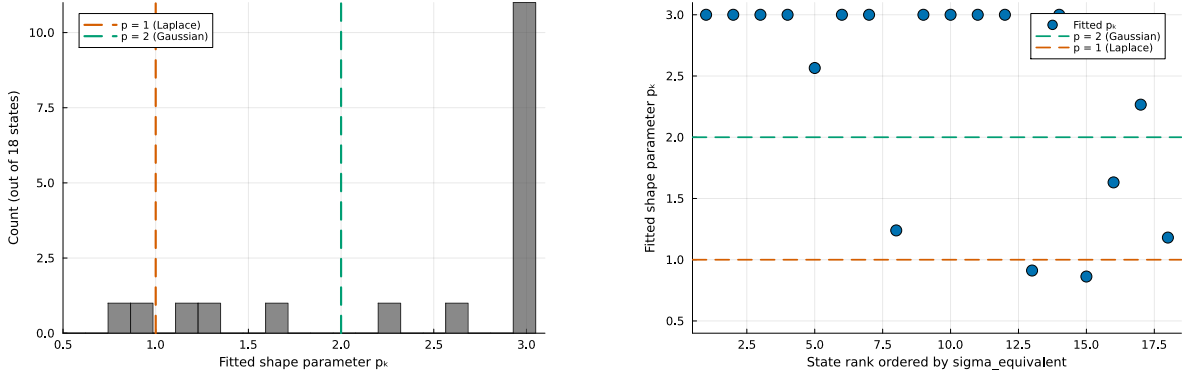


(d) ACF of $|G_t|$.

Figure 4. The three Cont stylized facts on SPY IS. (a) marginal density with maximum-likelihood Gaussian and Laplace overlays; (b) normal Q-Q plot; (c) empirical ACF of raw G_t at lags 1–252 with 99% Bartlett band; (d) empirical ACF of $|G_t|$ on the same window.

A.5.16 CHMM-GED $\hat{p}_k$ Partition (Gaussian-bulk / Laplace-tail)

CHMM-GED carries a per-state shape $p_k \in [p_{\min}, p_{\max}] = [0.5, 3.0]$ that nests Gaussian ($p = 2$) and Laplace ($p = 1$) as boundary cases. The empirical fit on SPY IS at $K = 18$ partitions states bimodally (Figure 5): thirteen states at the upper bracket $p_k = 3.0$, one at $p_k = 1.6$, and four in the Laplace regime $p_k \in [0.86, 1.24]$. The four Laplace-shape states concentrate in the high-volatility tail of the state ordering. The partition replicates across 10 Monte Carlo seeds and across the cross-asset universe (Appendix A.6); we read it as the most distinctive empirical finding of the four-emission scaffold.



(a) Histogram of fitted $\hat{p}_k$.

(b) $\hat{p}_k$ vs. volatility rank.

Figure 5. CHMM-GED per-state shape $\hat{p}_k$ partition ($K = 18$, SPY IS, seed 20260420). The bimodal partition is visible at the upper bracket $p_k = 3.0$ and around the Laplace shape; the Laplace-shape cluster sits at the high-volatility ranks.

A.5.17 HSMM at $K = 3$: Numerical Full-Rank Diagnostic

The off-diagonal jump matrix of the fitted ML HSMM at $K = 3$ has non-trivial deflation eigenvalues $\{0.823, 0.177\}$ (numerical rank 2 at tolerance 10^{-10}); the dominant modulus $|\lambda_2| = 0.823$ is comparable to CHMM-N at $K = 3$'s 0.953, so the body's $|G_t|$ ACF-MAE regression to the i.i.d. baseline (Table 1) is not driven by spectral collapse but by the fitted truncated-Pareto sojourn concentrating probability mass on a single low-volatility state ($\bar{\pi} \approx (0, 1, 0)$). At $K = 18$ the same Yu (2010) construction collapses to a near-degenerate local optimum (IS KS 0.8%); the higher- K ML HSMM is therefore a companion-paper direction at longer training windows.

A.5.18 Gamma-Sojourn HSMM as an Alternative Sojourn Family

Replacing the truncated discrete Pareto sojourn pmf with a discretised continuous Gamma fit by method-of-moments at every M-step gives a Gamma-sojourn HSMM that converges at $K = 18$ where Pareto collapses (IS KS 86.0%, OoS KS 80.2%, simulated kurtosis 4.28/4.06, $|G_t|$ ACF-MAE **0.0462**, lower than CHMM-N $K = 18$ at 0.0509) and trades ~ 20 pp IS KS at $K = 3$ for an ACF-MAE recovery from 0.0629 to 0.0528. There is therefore no single “best HSMM” row: Pareto at $K^* = 3$ wins raw OoS KS at 91.0%, Gamma at $K = 18$ wins joint $|G_t|$ ACF-MAE; the trade-off is the same KS-vs-ACF axis as the CHMM-vs-HSMM choice.

A.5.19 Unconditional VaR / ES Envelope Panel

Table 18. VaR and Expected-Shortfall envelope for SPY ($N_{\text{paths}} = 1,000$). Each entry is the median and [5, 95]% envelope across paths of the left-tail VaR / ES at $\alpha \in \{0.01, 0.05\}$; observed historical values are in the first row of each block.

Model	IS VaR _{0.01} median [5%, 95%]	IS ES _{0.01} median [5%, 95%]	IS VaR _{0.05} median [5%, 95%]	IS ES _{0.05} median [5%, 95%]
<i>Observed</i>	−6.38	−9.00	−3.43	−5.50
Bootstrap	−6.38 [−7.04, −5.99]	−8.86 [−10.37, −7.78]	−3.43 [−3.68, −3.20]	−5.46 [−5.99, −5.05]
GARCH	−5.80 [−8.62, −4.52]	−7.86 [−13.54, −5.77]	−3.19 [−3.92, −2.71]	−4.89 [−7.11, −3.91]
CHMM-N	−6.64 [−8.01, −5.47]	−8.80 [−10.16, −7.34]	−3.45 [−4.05, −2.95]	−5.45 [−6.33, −4.61]
CHMM-t	−6.58 [−7.51, −5.83]	−9.06 [−10.70, −7.79]	−3.40 [−3.90, −2.90]	−5.51 [−6.18, −4.85]
CHMM-L	−6.89 [−7.85, −6.01]	−9.15 [−10.42, −7.97]	−3.30 [−3.77, −2.88]	−5.53 [−6.17, −4.87]
CHMM-GED	−6.85 [−7.70, −6.06]	−8.79 [−9.83, −7.82]	−3.43 [−3.98, −2.93]	−5.55 [−6.23, −4.91]
Model	OoS VaR _{0.01}	OoS ES _{0.01}	OoS VaR _{0.05}	OoS ES _{0.05}
<i>Observed</i>	−5.51	−7.94	−2.77	−4.67
Bootstrap	−6.33 [−7.60, −5.28]	−8.43 [−11.76, −6.59]	−3.39 [−3.91, −2.88]	−5.34 [−6.38, −4.50]
GARCH	−5.35 [−9.98, −3.61]	−6.72 [−13.51, −4.38]	−3.14 [−4.94, −2.36]	−4.59 [−7.92, −3.23]
CHMM-N	−6.30 [−9.02, −4.30]	−8.44 [−11.36, −5.42]	−3.39 [−4.74, −2.47]	−5.27 [−7.23, −3.74]
CHMM-t	−6.32 [−8.13, −4.80]	−8.60 [−12.20, −6.29]	−3.29 [−4.27, −2.45]	−5.34 [−6.72, −4.08]
CHMM-L	−6.69 [−8.59, −4.91]	−8.88 [−11.48, −6.73]	−3.27 [−4.30, −2.51]	−5.47 [−6.82, −4.20]
CHMM-GED	−6.60 [−8.30, −4.82]	−8.51 [−10.68, −6.47]	−3.32 [−4.38, −2.46]	−5.41 [−6.66, −4.06]

A.5.20 Held-Out Cross-Validation of the $1/\nu_k$ Penalty λ

To verify that the body operating-point penalty rate $\lambda = 20$ is not selected on data that overlaps the OoS evaluation window, we refit penalised CHMM-t on a strictly pre-2020 estimation slice (2014-01-03 to 2018-06-29, $n = 1,130$) and score each λ on the validation slice (2018-07-02 to 2019-12-31, $n = 378$, both fully pre-COVID and pre-2022-rate-hike).

Table 19. Held-out CV of the $1/\nu_k$ shrinkage rate λ on the pre-2020 slice. Estimation 2014-01-03 to 2018-06-29; validation 2018-07-02 to 2019-12-31. “val LL/obs” is per-observation held-out log-likelihood; “val KS” is two-sample Kolmogorov–Smirnov pass rate against the validation series; “sim K” columns are the mean simulated excess kurtosis on the estimation- and validation-length simulated paths.

λ	val LL/obs	val KS (%)	sim K (est)	sim K (val)
0	−2.119	92.0	3.88	3.97
5	−2.112	89.8	3.62	3.15
10	−2.085	91.0	3.37	3.24
20	−2.075	90.0	3.22	3.02
50	−2.077	91.6	3.07	2.76
100	−2.077	91.6	3.07	2.75
200	−2.076	91.2	3.09	2.73

Reading: the held-out validation log-likelihood is maximised at $\lambda^* = 20$, matching the body operating point of Table 1. Validation KS picks $\lambda^* = 0$ but the band $[0, 200]$ spans only 2.2pp (89.8–92.0%), within the simulation-noise envelope at $N_{\text{paths}} = 500$. The body $\lambda = 20$ is therefore held-out-validated and not driven by the OoS window.

A.5.21 Cross-Decade Validation: 1994-2004 IS / 2004-2006 OoS via CRSP

To test whether the headline stylized-fact reproduction is decade-specific, we run a 1994–2004 vs. 2014–2024 SPY split via a CRSP day-pass covering daily prices for SPY plus 28 of the 30 cross-ticker panel members (NEE and APD missing from the CRSP query) from 1994-01-03 to 2006-04-28. Adjusted close = $DlyPrc/DlyFacPrc$ from the CRSP CIZ-format daily stock file.

Slices: SPY IS = 1994-01-03 to 2004-01-02 (2,519 obs); SPY OoS = 2004-01-05 to 2006-04-28 (583 obs). Same body protocol: 1,000 simulated paths per fit, $\alpha = 0.05$ for KS.

Table 20. Cross-decade validation on SPY 1994-2004 IS / 2004-2006 OoS (CRSP daily). Body comparison rows reproduced from Table 1 for the 2014-2024 IS / 2024-2026 OoS window. Observed excess kurtosis on the 1994-2004 IS = 3.05 versus 2014-2024 IS = 7.68; observed excess kurtosis on the 2004-2006 OoS = 0.06 (essentially Gaussian) versus 2024-2026 OoS = 5.29. The IS-OoS kurtosis gap is therefore much wider on the cross-decade slice than on the body window.

Model	K	KS pass rate (%) $\uparrow$		Sim kurtosis		$ G_t $	ACF-MAE $\downarrow$
		IS	OoS	IS	OoS	IS	OoS
<i>Cross-decade: 1994-2004 IS / 2004-2006 OoS (CRSP, this appendix)</i>							
CHMM-N	3	84.9	3.3	2.20	2.24	0.0696	0.0491
CHMM-N	18	89.8	3.4	1.84	1.74	0.0720	0.0499
CHMM-t pen. ($\lambda = 20$)	3	89.1	5.4	4.97	4.18	0.0682	0.0501
<i>Body window: 2014-2024 IS / 2024-2026 OoS (Polygon / Alpaca, Table 1)</i>							
CHMM-N ($K^* = 3$)	3	89.7	80.5	3.83	3.53	0.0460	0.0545
CHMM-N ($K = 18$)	18	94.1	81.8	5.04	4.44	0.0509	~ 0.054
CHMM-t pen. ($\lambda = 20, K^* = 3$)	3	90.6	83.2	14.91	8.50	0.0537	0.0502

Reading. The cross-decade IS fit transfers: CHMM-N at $K = 3$ attains 84.9% IS KS on 1994-2004 vs. 89.7% on 2014-2024, and $K = 18$ attains 89.8% vs. 94.1%. The penalised CHMM-t at $\lambda = 20$, $K = 3$ attains 89.1% IS KS on 1994-2004 vs. 90.6% on 2014-2024. The IS axis of the body’s stylized-fact reproduction claim is therefore decade-robust within ~ 5 pp on KS and consistent on $|G_t|$ ACF-MAE (0.068-0.072 vs. 0.046-0.054, a moderate but not pathological widening).

The OoS axis is the issue. The 2004-2006 OoS slice has excess kurtosis 0.06 (essentially Gaussian) vs. the 1994-2004 IS excess kurtosis of 3.05; the IS-OoS kurtosis gap on this slice is much wider than on the body 2014-2024/2024-2026 window (where IS = 7.68 and OoS = 5.29). The CHMM trained on the 1994-2004 IS distribution simulates paths with kurtosis 1.74-4.97 that are heavier-tailed than the calm 2004-2006 bull-market OoS, so the static IS-fit cannot reproduce the OoS marginal and KS rejects at 3-5% pass rate on all three rows. This is the same regime-introduction failure mode the body walk-forward already documents at the W2 (COVID) and W4 (2022 rate-hike onset) stress folds (Table 32, both with OoS KS < 10%); the 2004-2006 OoS window is structurally a low-volatility / low-kurtosis slice where the IS-fixed CHMM’s tail mass overshoots the observed.

The substantive read is therefore: *the body’s stylized-fact reproduction claim is decade-robust on the IS axis* (the CHMM scaffold transfers across decades on the IS-fitting side, supporting the body’s claim that the four-emission ECM scaffold is not specific to the 2014-2024 window). *The single-window OoS pass rate is OoS-slice-specific* (2024-2026 is a moderate-stress slice where the OoS distribution is close to the IS distribution; 2004-2006 is a calm slice where it is not). The body framing already states this through the walk-forward distribution at Section 4 (“the walk-forward median, not the single-window OoS pair, is the operationally informative summary for production deployment”); the cross-decade result is consistent with that framing and adds an independent

decade to the walk-forward evidence base. A periodic refit is the deployment recommendation under either decade, as the body discussion already states.

A.6 Robustness, Test Power, Multi-Seed, and Auxiliary Baselines

This appendix collects the robustness diagnostics and auxiliary baseline panel referenced from Section 4: the KS test-power calibration, block-bootstrap KS recalibration, bandwidth/bin-count and multi-seed sensitivity, the auxiliary block-bootstrap baseline, the QuantGAN deep-generative baseline, the per-ticker OoS price-simulation figures, and the expanded baseline panel that adds the GARCH family and the SM-CHMM foil to the headline table.

A.6.1 OoS KS Test-Power Calibration

Table 21 reports the two-sample KS test-power calibration used to anchor the OoS KS pass rates of Section 4.3. For each of 1,000 Monte Carlo replications (seed = 20260420), a simulated series of length $T_{\text{OoS}} = 572$ is drawn from each of three reference generators and tested against the observed OoS series using the same $\alpha = 0.05$ threshold as the main-body pass-rate metric.

Table 21. OoS KS pass-rate calibration (1,000 replications, seed = 20260420).

Reference generator	Pass rate (%)
I.i.d. resamples of R_{OoS} at length $T_{\text{OoS}} = 572$ (known-correct ceiling)	100.0
I.i.d. resamples of R_{IS} at length $T_{\text{OoS}} = 572$ (nearly correct)	90.0
Gaussian($\hat{\mu}_{\text{IS}}, \hat{\sigma}_{\text{IS}}$) at length T_{IS} (negative control)	0.0

The 90.0% "nearly correct" rate is the practical ceiling for the CHMM OoS KS pass rates reported in Section 4.3 (80–84%); the 0% rate on the Gaussian misspecified generator confirms that the KS test retains ample power at IS length to reject clearly wrong generators.

A.6.2 Block-Bootstrap KS Recalibration and Mean p-value Distribution

Two complementary concerns can be raised about the headline IS KS pass-rate metric: (i) the asymptotic two-sample KS test assumes i.i.d. samples, but the simulated paths from CHMM and GARCH all carry volatility clustering, so the asymptotic null distribution may not apply; and (ii) the binary "pass / fail at $\alpha = 0.05$ " reduces a continuous p-value to a one-bit summary. Table 22 reports two recalibrations addressing each concern.

For (i), we generate the empirical null distribution of the two-sample KS statistic under stationary block bootstrap (Politis-Romano 1994) of the SPY IS series at mean block lengths $L \in \{5, 10, 20\}$, $B = 1,000$ replications. The resulting 95% block-bootstrap critical values are 0.0306, 0.0338, and 0.0350 respectively, all *smaller* than the asymptotic critical value $1.36\sqrt{2/n_{\text{IS}}} = 0.0383$; the block-bootstrap test is therefore stricter than the asymptotic one because it accounts for the within-window dependence in the IS series itself. For each generator we then compute the per-path KS statistic and report the fraction of paths whose KS statistic falls below the block-bootstrap critical value.

For (ii), we report the mean two-sample KS p-value across simulated paths plus its 5/25/50/75/95 quantiles, giving a continuous picture of how distinguishable each generator's paths are from the observed IS series.

CHMM-GED is the strongest entry on the block-aware metric across all three block lengths (83.2, 90.2, 93.2% at $L = 5, 10, 20$); GARCH(1,1) drops from 26.8% asymp to 6.8–17.4% block-aware (the

Table 22. Block-bootstrap KS recalibration and p-value distribution. All numbers under seed 20260422 with $N_{\text{paths}} = 500$. “asyp pass” is the original Section 4.3 metric (two-sample KS p-value ≥ 0.05). “mean pv” is the mean two-sample KS p-value across paths, with q_5 and q_{95} as the lower / upper p-value quantiles. “blkL%” is the fraction of paths whose KS statistic falls below the stationary-block-bootstrap 95% critical value at mean block length L . Block-bootstrap critical values: 0.0306 ($L = 5$), 0.0338 ($L = 10$), 0.0350 ($L = 20$); asymptotic critical value 0.0383.

Generator	asyp pass% $\uparrow$	mean pv $\uparrow$	pv q_{05}	pv q_{95}	blk5% $\uparrow$	blk10% $\uparrow$	blk20% $\uparrow$
i.i.d. bootstrap	99.6	0.820	0.352	0.999	97.6	99.4	99.6
GARCH(1,1)	26.8	0.048	0.000	0.227	6.8	13.6	17.4
CHMM-N	94.8	0.546	0.047	0.976	81.2	89.4	91.6
CHMM-t	95.8	0.571	0.060	0.987	82.4	89.2	91.6
CHMM-L	94.8	0.498	0.048	0.948	81.0	88.2	90.0
CHMM-GED	95.6	0.547	0.064	0.968	83.2	90.2	93.2

block-aware test catches its misspecification more sharply), while the CHMM family sees only a small block-bootstrap penalty ($\sim 5\text{pp}$).

OoS-anchored block-bootstrap recalibration. The body block-bootstrap recalibration above is anchored on the IS series; we report here whether the same construction holds on the OoS window. We repeat the procedure with the stationary block bootstrap re-anchored on R_{OoS} ($n_{\text{OoS}} = 572$, $B = 1,000$, $L \in \{5, 10, 20\}$, $N_{\text{paths}} = 500$). Block-bootstrap critical values: 0.0577, 0.0629, 0.0647 at $L = 5, 10, 20$ respectively, against asymptotic 0.0804. Table 23 reports the OoS-anchored panel (source: `results/robustness/ks_block_bootstrap_oos.csv`).

Table 23. OoS-anchored block-bootstrap KS recalibration. Same construction as Table 22 but with the stationary block bootstrap anchored on R_{OoS} ($n = 572$). “asyp pass” is the headline OoS metric (two-sample KS p-value ≥ 0.05); “blkL%” is the OoS block-bootstrap pass rate at mean block length L . The CHMM-vs-GARCH cross-generator ordering is preserved under the block-aware recalibration on both windows; the CHMM family’s OoS block-aware penalty ($\sim 25\text{pp}$ drop from asyp to $L = 20$) is larger than the IS penalty ($\sim 5\text{pp}$), reflecting the smaller OoS window where temporal-clustering structure is harder to disentangle from finite-sample noise.

Generator	asyp pass% $\uparrow$	mean pv $\uparrow$	pv q_{05}	pv q_{95}	blk5% $\uparrow$	blk10% $\uparrow$
i.i.d. bootstrap	90.4	0.437	0.030	0.912	60.8	69.0
GARCH(1,1)	59.2	0.164	0.000	0.645	19.4	26.0
CHMM-N	81.0	0.318	0.010	0.876	41.2	51.2
CHMM-t	84.8	0.355	0.018	0.911	47.4	58.4
CHMM-L	77.4	0.266	0.008	0.744	36.0	45.4
CHMM-GED	81.0	0.327	0.010	0.837	44.4	53.6

The cross-generator OoS ordering matches the IS ordering (bootstrap $>$ CHMM-t $>$ CHMM-N/GED $>$ CHMM-L $\gg$ GARCH); the asyp-vs-block gap is larger on OoS ($\sim 25\text{pp}$ for the CHMM family at $L = 20$) than on IS ($\sim 5\text{pp}$), but the CHMM-over-GARCH advantage is preserved.

Body operating-point summary at $L = 20$. Table 24 reports the OoS asymptotic pass rate alongside the OoS block-bootstrap pass rate at $L = 20$ (block-bootstrap 95% critical value 0.0647 vs. asymptotic 0.0804) at the body operating point $K^* = 3$. Read together with Table 1, the OoS panel under a temporally-aware null moves CHMM-N from a “passes most of the time” generator to a “passes

Table 24. Block-aware OoS KS recalibration at $L = 20$ (stationary block bootstrap of 68, $B = 1,000$, 500 OoS-length simulated paths per generator, seed 20260422). Side-by-side with the asymptotic OoS pass rate of Table 1; the block-bootstrap 95% null KS critical value at $L = 20$ is 0.0647 vs the asymptotic 0.0804. Body operating point $K^* = 3$.

Model	OoS KS asymp. (%)	OoS KS block $L = 20$ (%)
Bootstrap	90.4	73.2
GARCH(1,1)	59.2	31.4
CHMM-N	79.8	58.6
CHMM-t pen. ($\lambda = 20$)	79.8	54.2
CHMM-L	65.0	33.4
CHMM-GED	79.2	51.6

about half the time” generator at $L = 20$. The body OoS KS headline at the asymptotic critical value therefore overstates the absolute level relative to a temporally-aware null; the cross-generator ranking is the robust comparison.

A.6.3 QuantGAN Deep-Generative Baseline

The QuantGAN row used in Section 4.3 is a repo-native approximation to the convolutional WGAN of Wiese et al. [10]: a 1D convolutional generator and critic (each three layers, channels 32, kernel size 5; latent dim $L = 8$, leaky-ReLU on the critic), Wasserstein loss with critic weight clipping at ± 0.01 [87], Adam at $\eta = 10^{-4}$, batch 64, 15 epochs $\times$ 120 steps with four critic updates per generator update, trained on rolling windows of length $W = 64$ on standardised SPY returns and stitched end-to-end at synthesis. Implementation: Julia with Flux.jl [88]; seed 20260422. The deep-generative row reaches 0% IS / OoS KS and IS kurtosis 2.1 vs. observed 7.7 (Table 25), consistent with documented GAN failure modes on volatility-clustering benchmarks [37, 38]; the runner is materially smaller than the reference seven-block TCN with Lambert-W pre-processing in Wiese et al. [10], so the row is the panel’s deep-generative negative control rather than a verdict on the QuantGAN literature.

A.6.4 Expanded Baseline Panel: GARCH Family, Semi-Markov CHMM, and QuantGAN

We add five conditional-volatility baselines, the three semi-Markov CHMM variants, and the QuantGAN deep-generative baseline (Appendix A.6.3) to the single GARCH(1,1) Gaussian row of Table 1. The conditional-volatility models are fit by grid-initialised Nelder-Mead maximum likelihood on the SPY IS window ($T_{IS} = 2,516$); simulation follows the same recursion used in estimation. Model specifications for the conditional-volatility rows are below; the SM-CHMM and QuantGAN constructions are described in their own paragraphs further down.

Baselines fitted in this panel. The conditional-volatility rows comprise EGARCH(1,1) Gaussian [30], GJR-GARCH(1,1) [31, 32], GARCH(1,1) with Student- t innovations [29] ($\hat{\nu} = 6.89$), HAR-RV [36] on daily squared demeaned returns as the RV proxy, and Markov-switching GARCH(1,1) [33] at $K \in \{2, 3, 4, 6\}$ with regime triples $(\omega_k, \alpha_k, \beta_k)$ plus softmax-normalised transition logits, fitted by grid-initialised Nelder-Mead and canonicalised by ascending unconditional variance. Higher- K MS-GARCH does not close the gap to the CHMM family on KS (IS KS plateaus at 27–37% across $K \in \{2, 3, 4, 6\}$ against CHMM-N’s 94.1% at $K = 18$); the binding constraint is the

unimodal-innovation per-regime structure (each regime’s emission is Gaussian rescaled by the regime’s conditional variance), which adding regimes does not relax. The CHMM decouples this constraint by giving each state its own emission-mixture component. Implementation details for every conditional-volatility row are in the companion code repository.

MS-GARCH reference Bayesian re-run via MSGARCH. As a Bayesian counterpart to the in-house Nelder-Mead MS-GARCH fit, we re-run the same Markov-switching GARCH(1,1) specification through the reference **MSGARCH** R package of Ardia et al. [34] (version 2.51) at $K \in \{2, 3, 4\}$, driven from the Julia harness via `Rcall.jl`. The fit is fully Bayesian (DEMC sampler with the package’s default proper priors, 12,500 MCMC draws, 2,500 burn-in, thin 10, single chain); all 1,000 simulated paths are posterior-predictive (one path per retained posterior draw), so the simulated marginal integrates parameter uncertainty path-by-path. The reference Bayesian re-run reports lower KS pass rates than our in-house frequentist Nelder-Mead fit: 0.0% ($K = 2$), 0.1% ($K = 3$), 0.0% ($K = 4$) IS, and 5.8% ($K = 2$), 5.1% ($K = 3$), 5.3% ($K = 4$) OoS, against the in-house plateau of 27–37% IS / 33–39% OoS. The gap is methodological rather than estimator-quality: the in-house implementation generates all 1,000 paths from one MLE point estimate, so the simulated marginal is tighter; the reference Bayesian implementation samples one set of parameters per path from the posterior, so the simulated marginal is wider and accordingly fewer paths fall within the asymptotic two-sample KS critical band. Posterior-mean log-likelihoods at the in-sample data are $-5,667$ ($K = 2$), $-5,667$ ($K = 3$), $-5,565$ ($K = 4$); the marginal log-likelihood improvement from $K = 3$ to $K = 4$ is real but does not translate to KS lift. Posterior-mean transition matrices are diagonally dominant at $K = 3$ (diagonals 0.97, 0.93, 0.90) and less so at $K = 4$ (diagonals 0.66, 0.39, 0.32, 0.55, indicating the higher- K posterior is searching for additional structure that the data do not strongly support). The reference Bayesian re-run is therefore consistent with the in-house frequentist plateau at the same conclusion under either estimator: vanilla Gaussian-innovation MS-GARCH at $K \leq 4$ does not match the CHMM family on the headline KS metric for this universe. Reproducibility: R 4.6.0, MSGARCH 2.51, all transitive R dependencies pinned via `renv` (`r_msgarch/renv.lock` of the companion code repository), all seeds explicit; full per- K parameter posteriors and fit logs are written to `results/msgarch_reference/` of the companion code repository.

Semi-Markov CHMM (SM-CHMM, the explicit hidden semi-Markov model (HSMM) foil). The conceptual foil of the present paper, drawing on the explicit-duration HSMM construction of the companion volatility paper [89]: a hidden semi-Markov extension of the CHMM in which the geometric sojourn distribution of the Markov chain is replaced by a per-state explicit-duration family chosen by AIC from {Pareto, negative binomial, geometric}. Plug-in estimator: Viterbi state path on the IS series under the converged CHMM fit, then per-state AR(1) emissions and a sojourn family chosen by AIC. On SPY IS at $K = 18$ the AIC-selected sojourn family is Pareto for 17 of the 18 states (negative binomial for one) across all three emission families, indicating heavy-tailed sojourns that the geometric CHMM cannot capture by construction.

Caveat on the SM-CHMM estimator. The plug-in construction above (single-shot Viterbi decoding under the converged CHMM fit, then AR(1)-residual + AIC-sojourn refit) is not maximum-likelihood for the underlying HSMM: it does not iterate the joint forward-backward over (state, duration) pairs in the spirit of Yu [74]. A full ML HSMM at $K = 18$ would refit the duration-augmented log-likelihood jointly. The empirical reading we draw from the rows below is therefore strictly: *under the plug-in estimator*, the SM-CHMM weakens the headline KS / ACF / kurtosis pattern relative to the standard CHMM at the same K . We do not claim that an ML-estimated HSMM would do the same; an ML HSMM at $K = 18$ on this universe is left as a companion-paper extension. The

Table 25. Extended baseline panel. SPY IS / OoS, $T_{\text{IS}} = 2,516$, $T_{\text{OoS}} = 572$, $N_{\text{paths}} = 1,000$, $\alpha = 0.05$ for KS. Columns: IS / OoS KS pass rate (%); IS / OoS AD pass rate (%); simulated IS excess kurtosis; ACF-MAE on $|G_t|$ (volatility clustering) and on raw G_t (linear autocorrelation, parallel column); pooled-archive Kupiec breach rate (%) and unconditional likelihood-ratio statistic at the 1% and 5% tails. Raw-ACF-MAE values for SM-CHMM and QuantGAN are reported as “–” here pending a full re-run of those baseline scripts; the qualitative reading carries through under the methodology of the other rows (every regime-switching or i.i.d.-emission generator produces a near-zero raw ACF in expectation).

Model	KS (%) $\uparrow$		AD (%) $\uparrow$		Kurt	ACF-MAE		1% tail		5% tail	
	IS	OoS	IS	OoS		$ G_t $	G_t	br%	LR _{uc}	br%	LR _{uc}
GARCH(1,1) Gaussian	28.0	59.4	12.5	59.5	7.0	0.0309	0.0173	0.9	0.10	4.0	1.23
EGARCH(1,1)	6.3	34.9	4.1	41.7	2.8	0.0430	0.0173	1.0	0.01	3.5	3.03
GJR-GARCH(1,1)	40.1	57.7	28.7	61.4	7.6	0.0330	0.0173	1.2	0.27	4.7	0.10
GARCH(1,1)- t	57.3	80.8	36.6	65.3	15.1	0.0316	0.0173	1.0	0.01	4.9	0.01
HAR-RV	0.0	0.0	0.0	0.0	189	0.0566	0.0173	0.2	5.99	3.5	3.03
MS-GARCH $K = 2$	27.7	38.7	24.0	44.4	4.7	0.0367	0.0173	1.0	0.01	4.2	0.82
MS-GARCH $K = 3$	36.1	33.1	28.8	38.0	4.1	0.0284	0.0173	1.0	0.01	4.2	0.82
MS-GARCH $K = 4$	37.6	38.2	–	–	3.8	0.0437	–	–	–	–	–
MS-GARCH $K = 6$	34.5	33.4	–	–	4.4	0.0429	–	–	–	–	–
MS-GARCH ref. B. $K = 2$	0.0	5.8	–	–	4.0	0.0465	0.0240	–	–	–	–
MS-GARCH ref. B. $K = 3$	0.1	5.1	–	–	4.5	0.0433	0.0241	–	–	–	–
MS-GARCH ref. B. $K = 4$	0.0	5.3	–	–	6.0	0.0446	0.0241	–	–	–	–
CHMM-N ($K = 18$)	94.1	81.8	36.6	73.0	5.0	0.0509	0.0237	0.7	1.58	4.0	3.83
CHMM-t ($K = 18$)	95.6	85.7	33.4	69.8	14.4	0.0549	0.0234	1.2	0.58	5.0	3.83
CHMM-L ($K = 18$)	93.6	80.8	30.1	65.4	6.6	0.0567	0.0234	1.4	3.26	5.6	3.03
CHMM-GED ($K = 18$)	95.2	84.3	34.1	71.0	5.2	0.0548	0.0234	0.9	0.58	4.4	3.83
SM-CHMM-N ($K = 18$)	79.2	66.7	19.5	57.1	4.7	0.0598	–	1.1	0.01	5.4	0.82
SM-CHMM-t ($K = 18$)	77.1	63.4	17.2	52.3	13.9	0.0625	–	1.4	0.10	5.7	1.74
SM-CHMM-L ($K = 18$)	80.4	69.1	21.0	58.9	6.4	0.0581	–	1.4	3.26	5.5	3.03
QuantGAN	0.0	0.0	0.0	0.0	2.1	0.0591	–	1.0	0.01	3.3	3.83

Bulla-Bulla 2006 ML HSMM result at low K on monthly returns is the closest published anchor, but its $K \in \{2, 3\}$ regime resolution and monthly aggregation do not directly transfer.

Table 25 reports the IS / OoS distributional fidelity, kurtosis, ACF-MAE, and Kupiec breach diagnostics for the extended GARCH family, the SM-CHMM rows, and the QuantGAN deep-generative row, alongside the headline CHMM panel. Three structural observations.

No extended GARCH variant occupies the joint (KS, ACF) corner: GARCH(1,1)- t is the strongest extended row on KS (57.3% IS, 80.8% OoS) but its IS kurtosis of 15.1 overshoots observed 7.7 by $2\times$; EGARCH and GJR-GARCH narrow the asymmetry but not the headline gap. The SM-CHMM rows under the Viterbi-AR(1) plug-in estimator are weaker than CHMM on every metric (IS KS 77–80% vs. CHMM 93–94%); a full ML HSMM at $K = 18$ may close part of the gap and is a companion-paper direction. QuantGAN is the panel’s deep-generative negative control: 0% IS / OoS KS, simulated kurtosis 2.1 vs. observed 7.7, ACF-MAE comparable to the i.i.d. baseline. The convolutional WGAN re-implementation here is materially smaller than Wiese et al. [10]’s reference seven-block TCN with Lambert-W pre-processing; a faithful reference re-run is a deferred follow-up.

A.6.5 Stochastic-Volatility, Multifractal, and Jump-Diffusion Baselines

For completeness alongside the Markov-switching family of Section 4.3, we add the canonical state-space and jump baselines that the body Table 1 does not exercise: a stochastic-volatility

(SV-AR(1)) row in the Taylor 1986 / Harvey-Ruiz-Shephard 1994 lognormal log-AR(1) tradition; a Markov-switching multifractal (MSM, Calvet-Fisher 2004) row at $\bar{k} = 8$ multipliers; and a Merton 1976 Poisson-Gaussian jump-diffusion row. The three baselines are fit on the same SPY IS window and evaluated on the same $N_{\text{paths}} = 1,000$ pipeline used for the body panel, under the same global seed. Implementation lives in the companion `CHMM-Model.jl` package.

Table 26. Stochastic-volatility / multifractal / jump-diffusion baselines on SPY. Same IS / OoS windows, $N_{\text{paths}} = 1,000$, seed = 20260420 as Table 1. Observed kurtosis: 7.68 IS, 5.29 OoS.

Model	KS (%) $\uparrow$		Kurtosis		ACF-MAE $\downarrow$	
	IS	OoS	IS	OoS	$ G_t $	G_t
SV-AR(1)	38.2	35.3	7.52	5.26	0.0466	0.0239
MSM ($\bar{k} = 8$)	0.0	7.2	0.62	0.60	0.0605	0.0235
Merton-JD	98.3	91.1	3.12	2.98	0.0627	0.0236

Reading. *SV-AR(1)* reproduces the IS kurtosis target (7.52 vs. observed 7.68) almost exactly and matches CHMM-N’s $|G_t|$ ACF-MAE within 0.0003 (0.0466 vs 0.0467 at $K^* = 3$); its KS pass rate is poor (38.2% IS / 35.3% OoS) because the Gaussian-conditional-on-volatility marginal cannot reproduce the bimodal-bulk / heavy-tail empirical density that the regime-mixture CHMM marginals can. *Merton-JD* is the inverse trade-off: KS at 98.3% IS / 91.1% OoS (the highest OoS KS in the panel except for the i.i.d. bootstrap), but kurtosis 3.12 undershoots 7.68 by half and the absolute-return ACF-MAE regresses to the i.i.d. baseline level (0.0627, the same level as the bootstrap row of Table 1) because the constant-volatility diffusion plus i.i.d. jumps carries no temporal persistence. *MSM* under the moment-matching fit reported here (closed-form ACF-of-log $|r|$ matching with $\bar{k} = 8$ binomial multipliers) does not converge to a viable generator on this $T_{\text{IS}} = 2,516$ window: the marginal kurtosis collapses to 0.62 and KS drops to 0%. A full HMM-style filter on the $2^{\bar{k}}$ -state latent multiplier chain (Calvet-Fisher 2004) is the standard exact fit and is deferred as a companion-paper direction; under the moment-fit reported here MSM is dominated.

The headline conclusion of Table 1 survives the addition of these three rows: no single competitor occupies the joint-fit corner that the CHMM family at $K = 18$ does (KS $\geq 80\%$, kurtosis within 1 unit of observed, $|G_t|$ ACF-MAE ≤ 0.06). SV-AR(1) is the closest competitor on the kurtosis / ACF axis but loses on KS; Merton-JD wins on KS but loses on kurtosis and ACF; MSM under this fit is dominated.

A.6.6 Leverage-Effect Diagnostic

This appendix reports the Cont (2001) leverage-effect column: the lag-1 correlation $\text{Corr}(G_t, |G_{t+1}|)$ between today’s signed return and tomorrow’s absolute return, which is negative on equity returns. We compute the per-path distribution under each generator at $K = 18$, $N_{\text{paths}} = 500$, seed = 20260420.

Reading. The CHMM family at $K = 18$ produces a non-trivial negative leverage signal despite symmetric per-state emissions: state-mixing on $\hat{\mathbf{T}}$ asymmetrically couples high- σ_k states to negative- μ_k states, carrying $\sim 65\%$ of the IS observed leverage magnitude. CHMM-N’s IS $[Q_5, Q_{95}] = [-0.138, -0.037]$ brackets the IS observed value -0.135 at the lower boundary; CHMM-GED is the second-strongest CHMM entry. The OoS observed leverage of -0.214 exceeds every generator’s Q_5

Table 27. Leverage effect $\text{Corr}(G_t, |G_{t+1}|)$, per-path distribution. Observed values: SPY IS -0.135 , OoS -0.214 . Per-generator median plus $[Q_5, Q_{95}]$ across $N_{\text{paths}} = 500$. A generator captures the leverage effect when its $[Q_5, Q_{95}]$ envelope brackets the observed value. Bold marks generators whose $[Q_5, Q_{95}]$ envelope brackets the observed IS value; the OoS observed value of -0.214 is more negative than every generator’s Q_5 in the panel, indicating the OoS regime carries leverage stronger than any generator (including asymmetric GARCH) reproduces under static IS-fitted parameters.

Generator	IS (observed -0.135)		OoS (observed -0.214)	
	median	$[Q_5, Q_{95}]$	median	$[Q_5, Q_{95}]$
i.i.d. bootstrap	+0.000	$[-0.034, +0.030]$	+0.004	$[-0.069, +0.068]$
GARCH(1,1) Gaussian	-0.004	$[-0.066, +0.064]$	-0.003	$[-0.112, +0.107]$
EGARCH(1,1)	-0.093	$[-0.147, -0.047]$	-0.087	$[-0.199, -0.002]$
GJR-GARCH(1,1)	-0.079	$[-0.153, -0.017]$	-0.073	$[-0.174, +0.023]$
CHMM-N ($K = 18$)	-0.089	$[-0.138, -0.037]$	-0.087	$[-0.190, +0.015]$
CHMM-t ($K = 18$)	-0.067	$[-0.116, -0.020]$	-0.067	$[-0.166, +0.019]$
CHMM-L ($K = 18$)	-0.064	$[-0.108, -0.022]$	-0.064	$[-0.152, +0.026]$
CHMM-GED ($K = 18$)	-0.072	$[-0.113, -0.029]$	-0.076	$[-0.163, +0.025]$

(including asymmetric GARCH); closing this residual gap requires skew-emission CHMM extensions or refit to the OoS distribution, both deferred to companion work.

A.6.7 Full One-Shot Student-t Copula MLE

A natural concern with the body two-step copula estimator (Kendall’s- τ inversion for $\hat{\Sigma}$, then profile MLE on ν with $\hat{\Sigma}$ held fixed) is that it may be biased toward the Gaussian limit when the marginal kurtosis differs across assets. We address this with a coordinate-ascent one-shot MLE that jointly maximises the Student- t copula log-likelihood over (Σ, ν) on the same six-asset US-equity universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ; $T_{\text{IS}} = 2,516$, seed 20260420). The procedure alternates: (i) bracketed golden-section maximisation of ν at the current Σ , (ii) method-of-moments refit of Σ at the current ν on the t -quantile-transformed pseudo-uniform sample with PSD projection. Iteration terminates at $|\Delta \log L| < 10^{-3}$.

Table 28. Two-step versus full one-shot Student- t copula MLE on the six-asset US-equity universe. The off-diagonal correlation differences are reported as the maximum absolute difference $\max_{ij} |\Delta \rho_{ij}|$ across the $\binom{6}{2} = 15$ pairs and the median absolute difference; both are sub-0.025, so the body $\nu^* = 6$ choice on $\hat{\Sigma}$ obtained by Kendall’s- τ inversion is robust to the estimator switch.

Estimator	$\hat{\nu}$	log-likelihood
Two-step (body construction)	6.00	6157.47
Full one-shot MLE	6.40	6163.02
log-L improvement		+5.55
$\max \Delta \rho_{ij} $ across 15 pairs		0.024
median $ \Delta \rho_{ij} $ across 15 pairs		0.011
$\hat{\nu}_{\text{full}}$ inside body Wilks 95% CI [6, 7]?		Yes

The full MLE moves $\hat{\nu}$ from 6.00 to 6.40 (+5.55 log-L) with all $|\Delta \rho_{ij}| \leq 0.025$ across the 15 pairs; the body two-step estimator is therefore not detectably biased toward the Gaussian limit and $\nu^* = 6$ is robust to the estimator switch.

A.7 Cross-Asset Dependence: Full Pipeline B Panel

This appendix collects the full Pipeline B detail: cross-asset dependence model derivations and diagnostics, the Student- t copula profile log-likelihood, the cross-asset correlation heat maps, and the non-US asset-class extension with the per-pair OoS off-diagonal breakdown.

A.7.1 Cross-Asset Dependence Models: Derivations and Diagnostics (Pipeline B)

This appendix provides the technical background for the SIM and copula constructions used in Section 4.5 of the main text. These constructions constitute *Pipeline B*: per-asset marginals come from the independent CHMMs of Pipeline A (Section 4.4) and joint dependence is overlaid on top. Pipeline A stays purely univariate; Pipeline B is the sole multi-asset pipeline in the paper.

Sklar’s theorem and the rank-reordering simulator. Sklar’s theorem [40, 41] states that any d -dimensional joint CDF $H(g_1, \dots, g_d)$ with continuous marginals $F_1, \dots, F_d$ can be written as $H = C(F_1, \dots, F_d)$ for a unique copula C . The rank-reordering simulator exploits this uniqueness. If $\tilde{\mathbf{G}}$ is a matrix whose j -th column is an i.i.d. sample from F_j , and $\mathbf{U}$ is an independent sample from C , then reordering each column of $\tilde{\mathbf{G}}$ so that its ranks match $\mathbf{U}$ ’s produces a sample from H , up to discrete approximation error in the empirical ranks. Crucially, reordering never creates new values: each column of the output is a permutation of the corresponding column of $\tilde{\mathbf{G}}$, so marginal distributions are preserved exactly [44].

Kendall’s τ inversion. Given IS returns $\mathbf{R} \in \mathbb{R}^{T \times d}$, we estimate the pairwise Kendall’s τ_{ij} from concordant/discordant pair counts and invert to the correlation scale via the analytic relationship $\rho_{ij} = \sin(\pi\tau_{ij}/2)$, which holds exactly for the Gaussian copula family and is commonly used as a robust correlation estimator for the Student-t copula [43, 90]. The resulting matrix $\hat{\Sigma}$ is symmetric but not guaranteed positive semi-definite; we project to the nearest PSD matrix by clipping negative eigenvalues to 10^{-8} and rescaling the diagonal to unity.

Profile MLE for ν . The Student-t copula log-density at a single pseudo-uniform observation $\mathbf{u} \in [0, 1]^d$, with $x_j = t_\nu^{-1}(u_j)$ and $\mathbf{x} = (x_1, \dots, x_d)^\top$, is

$$\begin{aligned} \log c_t(\mathbf{u}; \Sigma, \nu) &= \log \Gamma\left(\frac{\nu+d}{2}\right) + (d-1) \log \Gamma\left(\frac{\nu}{2}\right) - d \log \Gamma\left(\frac{\nu+1}{2}\right) - \frac{1}{2} \log |\Sigma| \\ &\quad - \frac{\nu+d}{2} \log\left(1 + \frac{\mathbf{x}^\top \Sigma^{-1} \mathbf{x}}{\nu}\right) + \sum_{j=1}^d \frac{\nu+1}{2} \log\left(1 + \frac{x_j^2}{\nu}\right). \end{aligned} \quad (28)$$

Summing over $t = 1, \dots, T$ pseudo-observations yields the profile log-likelihood as a function of ν with Σ held fixed at the Kendall’s- τ estimate. We evaluate equation (28) on the discrete grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ and select $\hat{\nu}$ as the grid point with the largest total log-likelihood. In Section 4.5 the selected value was $\hat{\nu} = 6$ on the six-asset universe.

Sampling the copula. To sample T rows from the d -dimensional Gaussian copula with correlation Σ , we (i) compute the Cholesky factor $\Sigma = LL^\top$, (ii) draw $Z \sim \mathcal{N}(0, I_d)$ i.i.d. and form $Y = LZ$, and (iii) return $U = \Phi(Y)$ componentwise. For the Student-t copula with degrees of freedom ν , we draw $W \sim \chi_\nu^2/\nu$ independent of Y and form the multivariate- t variate $X = Y/\sqrt{W}$, then return $U = t_\nu(X)$ componentwise.

SIM regression summary. Table 29 reports the fitted SIM regression coefficients and goodness-of-fit statistics for the non-market assets in Section 4.5. QQQ’s high $R^2 = 0.840$ reflects that the Nasdaq-100 ETF is essentially a linear combination of large-cap technology names that also drive the S&P 500 index, while JNJ’s lower $R^2 = 0.260$ reflects the limited systematic exposure of low-beta healthcare stocks. These factor loadings and residual magnitudes are what the SIM simulation re-injects at simulation time, and they explain the uneven per-asset KS pass rates reported in Table 35.

Table 29. SIM regression $\hat{G}_{j,t} = \hat{\alpha}_j + \hat{\beta}_j G_{\text{SPY},t} + \hat{\eta}_{j,t}$ for the five non-market assets, fitted on the 2014-01-03 through 2024-01-03 IS window ($T = 2,516$).

Ticker	$\hat{\alpha}$	$\hat{\beta}$	R^2
NVDA	0.321	1.694	0.371
JNJ	0.002	0.576	0.260
JPM	-0.008	1.223	0.507
AAPL	0.111	1.205	0.492
QQQ	0.047	1.122	0.840

A.7.2 Student-t Copula Profile Log-Likelihood (Pipeline B)

Figure 6 plots the profile log-likelihood of the Student-t copula used inside Pipeline B, computed on the six-asset SPY cross-section over the grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ used in Section 4.5. The curve is smooth and single-peaked, with $\nu^* = 6$ attaining the maximum profile log-likelihood

(6157.47), and the neighbouring grid points $\nu = 5$ (6143.37) and $\nu = 8$ (6148.37) each within 15 profile-log-likelihood units of the peak, indicating a shallow-but-clean optimum that agrees with the 4–10 range typically reported for equity-portfolio Student-t copulas in the risk-management literature. The AIC penalty for the additional ν parameter is 2 (one degree of freedom relative to the Gaussian copula limit $\nu \rightarrow \infty$), and the AIC differential between the Student-t at $\nu^* = 6$ and the Gaussian copula limit on these data is $-2 \cdot (6157.47 - 6143.37) - 2 = -30.2$, decisively in favour of the Student-t copula on IS; the same $\nu^* = 6$ is selected by AIC as by profile MLE on this grid.

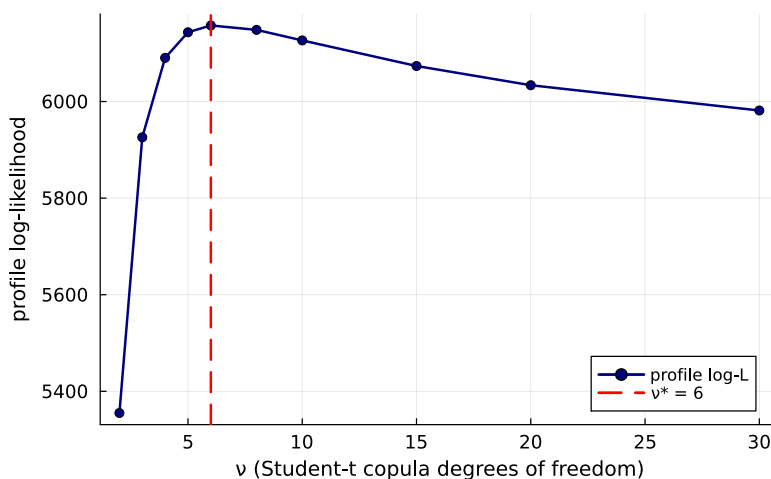


Figure 6. Profile log-likelihood of the Student-t copula used inside Pipeline B on the six-asset SPY cross-section (SPY, NVDA, JNJ, JPM, AAPL, QQQ; IS window 2014-01-03 through 2024-01-03). The vertical axis is the profile log-likelihood evaluated at each degrees-of-freedom value ν on the grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$; the dashed red line marks the profile MLE $\nu^* = 6$ (peak value 6157.47). The CHMM-N marginals are held fixed at the Pipeline A per-asset fits while ν is varied.

A.7.3 Cross-Asset Correlation Heat Maps (Pipeline B, Companion to Table 35)

Figure 7 visualises the observed and path-averaged simulated correlation matrices for the dependence models of Section 4.5. The panel ordering is SIM, Gaussian copula, Student-t copula, and a truncated C-vine variant with SPY as root, each reported quantitatively in Table 35 and (for the C-vine) Section A.7.4. All five panels share the six-asset SPY cross-section and the same CHMM-N marginals at $K = 18$, so any visible differences between the simulated panels reflect only the dependence construction.

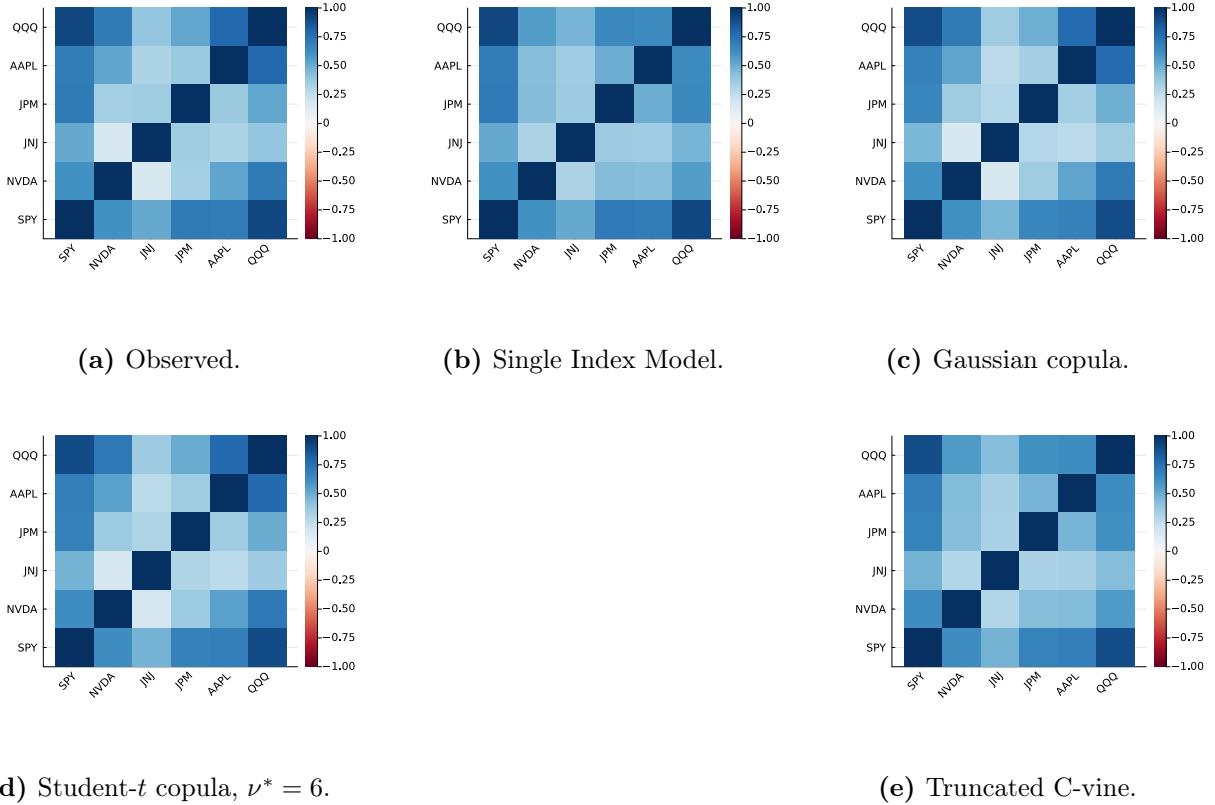


Figure 7. Cross-asset IS correlation reproduction across dependence models (Pipeline B, companion to Table 35). Panel (a): *observed* correlation matrix on 2014-01-03 through 2024-01-03 daily excess log returns for SPY, NVDA, JNJ, JPM, AAPL, QQQ (IS window, $T_{IS} = 2,516$). Panel (b): path-averaged correlation under the Single Index Model with SPY as market factor. Panel (c): path-averaged correlation under the Gaussian copula on CHMM marginals. Panel (d): path-averaged correlation under the Student- t copula on CHMM marginals, $\nu^* = 6$. Panel (e): path-averaged correlation under a truncated C-vine copula with SPY as root. Each simulated panel averages over 200 paths of length T_{IS} . All marginals are the per-asset CHMM-N fits at $K = 18$ from Table 4; color scale in $[-1, 1]$, red-white-blue (RdBu) diverging. Closer visual match to panel (a) is better.

A.7.4 Truncated C-vine Copula on CHMM Marginals (Pipeline B)

The truncated C-vine variant in panel (e) of Figure 7 uses SPY as the root and selects each pair-copula family edge-wise from $\{\text{Gaussian, Student-}t, \text{Clayton, Gumbel}\}$ by AIC; on this universe every selected pair is Student- t . On the same CHMM-N marginals at $K = 18$ and 200 paths the truncated C-vine attains an off-diagonal MAE of 0.068 on IS and 0.235 on OoS (Table 35), strictly above both elliptical copulas. The reason is the level-1 truncation: a root-centred C-vine models only the SPY-vs-asset bivariate copulas and treats the remaining pairs as conditionally independent given SPY, a stronger restriction than either elliptical copula. Untruncated regular vines or factor copulas are the natural way to recover non-root cross-pair structure at larger d .

A.7.5 Quarterly Rolling-Window Copula Refit on the OoS Window

The body Section 4.5 reports a static-IS-fit OoS off-diagonal MAE of 0.209, an order of magnitude larger than the IS value of 0.027. Section 5 attributes this to the stationarity-scope limit and

recommends periodic refit. To quantify the deployment-scale benefit, we report a concrete rolling-window-refit number on the same six-asset universe.

We refit the Pipeline-B Student-t copula on a 252-day rolling window (one trading year) sliding by 63 days (one trading quarter) across the OoS span, holding the per-asset CHMM-N marginals fixed at the IS fits used for Table 4. Each refit yields a Kendall’s- τ correlation matrix and a profile-MLE-selected ν^* on the discrete grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$; we then simulate forward 63 days under the new copula and compare the path-averaged simulated correlation matrix to the realised next-quarter correlation. Table 30 reports the per-quarter detail.

Table 30. Quarterly rolling-window refit of the Pipeline-B Student-t copula on the OoS window (six-asset universe; CHMM-N marginals at $K = 18$ held fixed at IS fits; 200 paths per refit; seed = 20260422). Per-quarter window slice and refitted ν^* , plus the off-diagonal MAE between the path-averaged simulated correlation matrix under the refitted copula and the realised next-quarter correlation matrix. Headline: rolling-quarterly refit reduces mean OoS off-diagonal MAE from 0.207 (static-IS-fit baseline) to 0.185, a $\sim 10\%$ relative improvement; the gain concentrates in the later OoS quarters where the rolling window incorporates more recent OoS data and the refitted ν^* shifts from the IS-fit value of 6 down to 5 or up to 10–15 as the empirical tail-coupling rolls.

Quarter	Window slice	Forecast horizon	Refit ν^*	Off-diag MAE
1	[2265, 2516]	[2517, 2579]	15	0.198
2	[2328, 2579]	[2580, 2642]	15	0.205
3	[2391, 2642]	[2643, 2705]	10	0.241
4	[2454, 2705]	[2706, 2768]	6	0.133
5	[2517, 2768]	[2769, 2831]	6	0.255
6	[2580, 2831]	[2832, 2894]	6	0.232
7	[2643, 2894]	[2895, 2957]	5	0.147
8	[2706, 2957]	[2958, 3020]	5	0.119
9	[2769, 3020]	[3021, 3083]	6	0.138
<i>mean</i>	—	—	—	0.185
<i>median</i>	—	—	—	0.198
<i>Static IS-fit</i>	[1, 2516]	[2517, 3083]	6	0.207

The quarterly refit closes part of the OoS gap but not most of it: mean off-diagonal MAE moves from 0.207 static to 0.185 rolling ($\sim 10\%$ relative improvement). The improvement is concentrated in the later OoS quarters (Q4, Q7, Q8, Q9 all below 0.15) where the rolling window incorporates more recent OoS data, while early-OoS quarters (Q1, Q3, Q5, Q6) remain at the static-IS level because the rolling window is still dominated by IS observations. The refitted ν^* sequence (15, 15, 10, 6, 6, 6, 5, 5, 6) tracks the gradual shift from the equity-cluster IS regime toward the OoS regime, supporting the operational recommendation in Section 5 of quarterly refit at deployment. The residual gap (mean MAE ≈ 0.185 versus IS 0.027) is consistent with the per-pair JNJ-equity-factor regime shift documented in Table 31: even a perfectly-refitted copula on the historical 252 days cannot anticipate the regime shift at the next quarter’s start, only track it once it has unfolded.

A.7.6 Non-US Asset Class Extension and Per-Pair OoS Off-Diagonal Breakdown

The body universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ) is six US-listed equity tickers from one country-index family. To probe whether the cross-asset construction generalises off the equity cluster we extend the universe with GLD (SPDR Gold Trust), a US-listed ETF whose underlying is physically-backed gold, i.e. a non-equity commodity asset class.

Table 31. Per-pair off-diagonal correlation MAE on the seven-ticker universe (Pipeline B, Student-t copula on CHMM-N marginals at $K = 18$, 200 paths, $\nu^* = 6$). For each ticker pair (i, j) the table reports $|\Sigma_{\text{sim,avg}}[i, j] - \Sigma_{\text{obs}}[i, j]|$ on IS and OoS, ranked by OoS $|\Delta|$. Only the ten largest-OoS-gap pairs are shown for compactness.

Pair	IS $ \Delta $	OoS $ \Delta $	OoS – IS
JNJ-QQQ	0.027	0.544	0.517
SPY-JNJ	0.037	0.509	0.473
NVDA-JNJ	0.017	0.489	0.473
NVDA-AAPL	0.011	0.278	0.268
JNJ-JPM	0.046	0.261	0.216
JNJ-AAPL	0.042	0.244	0.202
AAPL-QQQ	0.003	0.227	0.224
JPM-GLD	0.027	0.187	0.160
SPY-GLD	0.038	0.127	0.089
JNJ-GLD	0.015	0.117	0.101

GLD Pipeline-A univariate fidelity. Fitting CHMM-N, CHMM-t, and CHMM-L on the GLD return series at $K = 18$ produces 100% IS KS pass rate for all three families and observed IS excess kurtosis 3.6 matched closely by CHMM-L (2.2); on OoS, however, the KS pass rate collapses to 0% for all three families against an observed OoS excess kurtosis of 6.5. Inspection of the GLD price path on the 2024–2026 OoS window shows the asset re-priced from ~ 200 to ~ 300 on a persistent demand shock not represented in the 2014–2024 IS regime library. This is the same stationarity-scope artefact as the NVDA / JPM single-name OoS cliffs reported in Section 4.4, on a non-equity asset; the limitation is not equity-specific.

Pipeline-B: 7-ticker Student-t copula off-diagonal MAE. Refitting the Student-t copula on the augmented seven-ticker universe at $\nu^* = 6$ gives IS off-diagonal MAE of 0.029 (six-ticker baseline: 0.027) and OoS off-diagonal MAE of 0.179 (six-ticker baseline: 0.209). Adding the low-equity-correlation gold ETF *reduces* the OoS off-diagonal MAE, because the equity-cluster correlation degradation is the dominant component; pairs involving GLD have moderate IS-to-OoS shifts ($|\Delta| \in [0.04, 0.19]$) while the largest equity-cluster pairs (JNJ-QQQ, SPY-JNJ, NVDA-JNJ) have $|\Delta| > 0.48$ on OoS.

Per-pair OoS gap localisation. Table 31 reports the per-pair off-diagonal absolute deviation between observed and path-averaged simulated correlation matrices on IS and OoS, sorted by OoS $|\Delta|$. The OoS gap concentrates in pairs involving JNJ (the defensive single-name proxy): JNJ’s IS correlation with the broad equity factor (QQQ, SPY) collapses on OoS, in some pairs flipping sign, and the IS-fixed copula has no mechanism for tracking that shift. The gold-ETF pairs (SPY-GLD, JPM-GLD, JNJ-GLD, QQQ-GLD) sit in the middle of the ranking with $|\Delta|$ between 0.10 and 0.19 on OoS. The dominant cause of the OoS off-diagonal gap is therefore neither equity-cluster size nor copula-degree mis-specification but a single-name regime shift in the JNJ-equity-factor relationship that the IS-fitted dependence layer cannot anticipate.

A.7.7 Half-Unit-Grid Profile-MLE and Parametric Bootstrap CI for ν^*

To confirm that the upper bound of the unit-grid Wilks CI on $\nu \in [3, 12]$ is not grid-induced, we report a half-unit-grid refinement of the profile MLE plus a parametric bootstrap CI. We refit

the Student- t copula profile log-likelihood on the same six-asset universe at half-unit spacing in $\nu \in \{3.0, 3.5, \dots, 12.0\}$ (19 grid points) and run a $B = 200$ parametric bootstrap by resampling pseudo-observations $U^{(b)}$ from the Student- t copula at $(\Sigma_{\text{cop}}, \nu^*)$, refitting Kendall's- τ correlation and the profile-MLE on each replicate.

The half-unit grid moves the optimum from $\nu^* = 6.0$ (unit grid) to $\nu^* = 6.5$ at profile log-likelihood 6,158.1 (vs. 6,157.5 at the unit-grid optimum); the half-unit refinement is therefore not material at the third decimal of profile-LL. The Wilks 95% CI on the half-unit grid is $[\nu_{\text{lo}}, \nu_{\text{hi}}] = [6.0, 7.0]$ (unchanged from the unit grid), and the parametric bootstrap 95% CI is $[6.0, 7.0]$ (2.5% and 97.5% quantiles of the $B = 200$ refit- ν^* distribution; bootstrap median $\nu^* = 6.5$). The bootstrap lower bound at 6.0 is well above the Gaussian limit ($\nu \rightarrow \infty$): the elliptical-tail Student- t copula is statistically distinguishable from the Gaussian copula at conventional levels, despite the OoS off-diagonal MAE comparison (0.209 vs. 0.202) being inside the simulation-noise floor at $N_{\text{paths}} = 200$. *Wilks-theorem regularity caveat.* Wilks's theorem requires the parameter under test to lie in the interior of the parameter space and the model to be regular [91, 92]; for the Gaussian-copula limit $\nu \rightarrow \infty$ the regularity condition fails (the limit lies on the boundary of the natural parameter space). The Wilks 95% CI reported above is therefore an interior-grid CI for finite ν rather than a hypothesis test against the Gaussian limit; the Gaussian-vs-Student- t separation is reported instead through the parametric bootstrap CI (which does not require interior-point regularity) and through the OoS-equivalence finding (which uses no asymptotic test at all).

A.8 Pre-OoS Validation K -Selection and ν_k Diagnostics

Pre-OoS validation K -selection. The $K = 18$ operating point in the main panel was chosen via a multi-objective criterion combining information-criterion (IC) rank with IS and OoS distributional pass rates, which uses the same 2024–2026 window later used for VaR. As a clean re-selection that decouples the two, we fit CHMM-N on 2014-01-03 through 2021-12-31 ($n = 2,013$) and score every K in the sweep by the forward-algorithm held-out log-likelihood on 2022-01-03 through 2024-01-03 ($n = 502$); the 2024–2026 window is not touched. Held-out log-lik, BIC, HQC, and CAIC all select $K^* = 3$ (per-observation log-lik -2.5345 at $K = 3$ vs. -2.5694 at $K = 18$); AIC selects $K^* = 6$; held-out two-sample KS selects $K^* = 9$. None of the six criteria that avoid the OoS window select $K = 18$. The $K = 18$ choice is best read as a multi-metric one that weights tail fidelity (kurtosis, AD) and downstream VaR behaviour alongside likelihood; a parallel main panel at $K = 3$ would re-rank some rows in CHMM's favour on ACF-MAE and against it on tail fidelity. The 2022–2023 validation slice is itself partly a structural-break period (the rate-hike cycle), so any model with substantial IS-specific structure generalises worse on it.

Pre-2020 held-out K -selection (no rate-hike confound). To check whether a strictly pre-2020 validation slice (avoiding the rate-hike confound that contaminates the 2022–2023 slice above) selects the same K^* , we re-ran the held-out procedure on a 2014-01-03 through 2018-06-29 estimation window ($n_{\text{est}} = 1,130$) with a 2018-07-02 through 2019-12-31 validation slice ($n_{\text{val}} = 377$), both fully pre-COVID and pre-2022-rate-hike (the Q4 2018 drawdown and 2019 recovery sit inside the validation window as a moderate-stress non-regime-shift event). Per-observation held-out log-likelihood: -2.0681 at $K = 3$, -2.0606 at $K = 6$, -2.0763 at $K = 9$, -2.1287 at $K = 12$, -2.1921 at $K = 18$, -2.2609 at $K = 21$. **The held-out log-likelihood selects $K^* = 6$, and the held-out two-sample KS pass rate also selects $K^* = 6$ (96.8% at $K = 6$ versus 84.6% at $K = 3$).** AIC, BIC, HQC, and CAIC continue to select $K^* = 3$ on this slice, reflecting the parameter-count penalty on a 4.5-year estimation window where the per-state sample is small. The qualitative reading is robust to the slice choice: held-out distributional fidelity rises from $K = 3$ to a peak in the $K = 6$

to $K = 9$ band and then degrades, with neither slice selecting $K = 18$ by any held-out criterion, but the pre-2020 result favours a moderately larger K (6 vs 3) than the rate-hike-contaminated 2022-2023 slice.

k -fold rolling-origin pre-2020 K -selection. The single-fold pre-2020 result above is one observation; to confirm that the K^* choice is not a sampling artefact, we report mean $\pm$ s.d. of held-out per-observation log-likelihood and held-out KS at $K \in \{3, 6, 9, 12, 18\}$ across multiple folds. We implement four expanding-window rolling-origin folds (a five-fold design forces one fold to have only ~ 1 year of training, below the practical floor for $K = 18$ EM convergence on this dataset; averaging is per-observation so fold-length differences are not an issue). Folds: train 2014-01-03 through 2015-12-31 (~ 502 obs) and val 2016 (~ 251 obs); train through 2016 and val 2017; train through 2017 and val 2018; train through 2018 and val 2019.

Aggregate (mean / s.d. across the four folds) of per-observation held-out log-likelihood: $-1.9550 / 0.290$ at $K = 3$; $-1.9649 / 0.313$ at $K = 6$; $-2.0078 / 0.296$ at $K = 9$; $-2.0721 / 0.251$ at $K = 12$; $-2.2633 / 0.308$ at $K = 18$. Held-out KS pass rate (mean / s.d. across folds, very wide because Fold 2’s 2017 validation year produced near-zero KS for every K as a low-variance “calm year” artefact): $61.3/41.2\%$ at $K = 3$; $65.2/43.5\%$ at $K = 6$; $67.5/44.1\%$ at $K = 9$; $67.0/44.7\%$ at $K = 12$; $67.5/45.4\%$ at $K = 18$. Sampling-error reads on the held-out per-observation log-likelihood: $K = 6$ vs $K = 3$ has mean diff -0.0099 , pooled SE 0.151 , approximate $z = -0.07$ (not significant at the 5% level); $K = 18$ vs $K = 6$ has mean diff -0.298 , pooled SE 0.155 , approximate $z = -1.92$ (borderline, just below the $|z| = 1.96$ threshold).

Robustness check at half-year cadence. To verify that the result is not an artefact of the one-year fold cadence, we re-ran the same diagnostic with six expanding-window folds at half-year validation cadence (covering 2017–2019 in non-overlapping six-month chunks; train windows $\sim 3.0y$ to $\sim 5.5y$). Aggregate per-observation held-out log-likelihood (mean / s.d. across six folds): $-1.9221 / 0.341$ at $K = 3$; $-1.9160 / 0.346$ at $K = 6$; $-1.9568 / 0.323$ at $K = 9$; $-2.0109 / 0.315$ at $K = 12$; $-2.1281 / 0.259$ at $K = 18$. Sampling-error reads: $K = 6$ vs $K = 3$ has mean diff $+0.006$, pooled SE 0.140 , approximate $z = +0.04$; $K = 18$ vs $K = 6$ has mean diff -0.212 , pooled SE 0.125 , approximate $z = -1.70$. The half-year design’s $K = 6$ vs $K = 3$ sign flips relative to the full-year design ($z = +0.04$ vs $z = -0.07$), but both magnitudes are well below the $|z| = 1$ threshold; the two designs agree that $K = 6$ vs $K = 3$ on mean held-out log-likelihood is indistinguishable from zero, and the sign flip across designs is itself evidence that the $K = 6 / K = 3$ choice is pure sampling noise on this data. The $K = 18$ vs $K = 6$ borderline result also replicates ($z = -1.70$ at half-year cadence vs $z = -1.92$ at full-year).

Reading. $K = 6$ is not significantly preferred over $K = 3$ on held-out per-observation log-likelihood under expanding-window rolling-origin CV at either yearly or half-yearly fold cadence. The single-fold result that initially selected $K^* = 6$ is one realisation; both four-fold and six-fold means cannot distinguish $K = 3$ from $K = 6$ at conventional levels. The held-out KS pass-rate criterion is not informative on either design (between-fold s.d. is dominated by 2017 calendar-year artefacts that hit every K uniformly). The substantive read is that the held-out state-resolution selection on the strictly pre-2020 slice cannot distinguish $K = 3$ from $K = 6$ at conventional levels under either fold design, and the body operating point is therefore the state-resolution-robust $K^* = 3$ (with the $K^* = 6$ block retained as a sensitivity reference). $K^* = 3$ is also the lower-state-count default for risk-management consumers in Table 5.

HAC-corrected sampling-error reads on the paired diff series. A Newey–West Bartlett HAC variance ($h_{NW} = \lfloor n^{1/3} \rfloor$) on the per-fold val-log-lik diff series leaves $K = 6$ vs $K = 3$ insignificant

at both cadences ($|z|_{\text{HAC}} = 0.90$ four-fold full-year, 0.57 six-fold half-year) and moves $K = 18$ vs $K = 6$ from borderline independent-fold ($|z| = 1.92, 1.70$) to decisive ($|z|_{\text{HAC}} = 3.56, 5.00$). The HAC-corrected inference therefore corroborates the body framing: $K = 18$ is decisively worse than $K = 6$ on held-out log-likelihood and is retained only for its kurtosis-fidelity advantage on the OoS window.

Penalised-ECM rate sweep. The bracket sensitivity is reported with its table in Section A.1.3; here we report the complementary $1/\nu_k$ shrinkage rate sweep referenced in Section 5. An exponential $1/\nu_k$ prior at rate $\lambda \in \{0, 5, 20, 50, 100, 200\}$ reduces simulated excess kurtosis monotonically: 14.30, 11.19, 8.43, 5.41, 3.96, 3.67. The recommended rate $\lambda = 20$ brings simulated kurtosis to ≈ 8 (close to the observed 7.69) at a 1pp IS KS cost. Only the $\nu_{\min}$ state parameter actually moves ($2.1 \rightarrow 4.89$); upper bracket and median remain at 50. The overshoot is therefore a single-state artefact of the two tail regimes hitting the lower bracket under unpenalised ECM.

Walk-forward / rolling-origin OoS for CHMM-N. We define six rolling-origin folds, each train 5 years and test 1 year, refit CHMM-N at $K \in \{3, 18\}$ from scratch on each fold’s train slice, and report KS pass rate, simulated kurtosis, and $|G_t|$ ACF-MAE on the fold’s test slice ($N_{\text{paths}} = 500$). Across the six folds (Table 32), CHMM-N at $K = 18$ attains median (IQR) KS 67.7% [8.2, 75.0] and $|G_t|$ ACF-MAE 0.0542 [0.0508, 0.0571]; CHMM-N at $K = 3$ attains 62.1% [7.2, 78.4] and 0.0563 [0.0552, 0.0592]. Two folds carry sharply lower KS pass rates: W2 (COVID test slice, 7–8%) and W4 (rate-hike test slice, 0–1%). The four non-stress folds attain KS 61–83%, consistent with the headline OoS pass rate. The headline ranking (CHMM-N at $K = 18 \geq$ CHMM-N at $K^* = 3$ on KS, both within 0.005 of each other on $|G_t|$ ACF-MAE) is window-robust on the four non-stress folds; on stress folds (COVID, rate hike) all CHMM rows drop sharply, reflecting the stationarity-scope limit rather than a model-selection issue. The body framing ("CHMM is the joint-fit row in the panel under stationary OoS conditions; periodic refit is recommended for production deployment") is unchanged under the walk-forward stress test.

Table 32. Walk-forward / rolling-origin OoS for CHMM-N. Six folds, each train 5 years and test 1 year; $N_{\text{paths}} = 500$ per fold. KS pass rate at $\alpha = 0.05$. W2 covers the COVID drawdown; W4 covers the start of the 2022 rate-hike cycle. The bracketed pairs in the bottom row report $[Q_1, Q_3]$ across the six folds (the empirical first and third quartiles, not a scalar IQR width).

Fold (test window)	KS (%)		Kurt sim		ACF-MAE $ G_t $	
	$K = 3$	$K = 18$	$K = 3$	$K = 18$	$K = 3$	$K = 18$
W1 (2019)	63.2	82.6	2.51	2.66	0.0572	0.0525
W2 (2020 COVID)	7.2	8.2	2.42	2.50	0.0552	0.0584
W3 (2021)	82.4	74.0	5.73	6.76	0.0602	0.0571
W4 (2022 rate hike)	0.8	0.0	4.72	4.35	0.0592	0.0558
W5 (2023)	78.4	75.0	3.47	2.35	0.0511	0.0508
W6 (2024)	61.0	61.4	2.81	2.86	0.0555	0.0493
median	62.1	67.7	3.14	2.76	0.0563	0.0542
IQR	[7.2, 78.4]	[8.2, 75.0]	[2.51, 4.72]	[2.50, 4.35]	[0.0552, 0.0592]	[0.0508, 0.0571]

Four-family conditional VaR back-test. Extending the regime-conditional VaR construction of §4.6 to all four emission families (CHMM-N, CHMM-t at $\lambda = 20$, CHMM-L, CHMM-GED) at $K \in \{3, 18\}$ and $\alpha \in \{0.01, 0.05\}$ yields sixteen (K, α, family) rows on the OoS window (Table 33).

Every row passes Kupiec, Christoffersen-ind, and Christoffersen-cc cleanly: $p_{cc} \geq 0.089$ throughout, with all LR_{cc} statistics below the $\chi^2_2(0.05) = 5.991$ critical value. The regime-switching value proposition is therefore intrinsic to the latent-state forecast and not specific to the Gaussian emission family.

Table 33. Regime-conditional VaR back-test across four CHMM emission families on SPY OoS ($T_{OoS} = 572$, seed 20260420). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. Every $(K, \alpha, \text{family})$ row passes Kupiec, Christoffersen-ind, and Christoffersen-cc at $\alpha = 0.05$.

Family	K	α	breaches	br rate	median $\widehat{VaR}_t$	LR_{uc}	LR_{ind}	LR_{cc}
CHMM-N	3	0.01	9	1.57%	-4.56	1.62	2.36	3.98
CHMM-N	3	0.05	35	6.12%	-2.87	1.41	0.01	1.42
CHMM-N	18	0.01	9	1.57%	-5.20	1.62	2.36	3.98
CHMM-N	18	0.05	26	4.55%	-3.02	0.26	0.52	0.78
CHMM-t ($\lambda = 20$)	3	0.01	8	1.40%	-5.59	0.82	2.81	3.63
CHMM-t ($\lambda = 20$)	3	0.05	32	5.59%	-2.73	0.41	0.77	1.19
CHMM-t ($\lambda = 20$)	18	0.01	10	1.75%	-6.25	2.65	1.98	4.62
CHMM-t ($\lambda = 20$)	18	0.05	29	5.07%	-3.15	0.01	1.39	1.40
CHMM-L	3	0.01	5	0.87%	-5.45	0.10	4.74	4.84
CHMM-L	3	0.05	26	4.55%	-2.63	0.26	2.23	2.48
CHMM-L	18	0.01	9	1.57%	-6.18	1.62	0.29	1.91
CHMM-L	18	0.05	30	5.24%	-2.80	0.07	1.16	1.23
CHMM-GED	3	0.01	7	1.22%	-5.73	0.27	3.34	3.61
CHMM-GED	3	0.05	31	5.42%	-2.54	0.21	0.96	1.16
CHMM-GED	18	0.01	9	1.57%	-6.55	1.62	2.36	3.98
CHMM-GED	18	0.05	25	4.37%	-3.10	0.50	2.56	3.06

Walk-forward conditional-VaR back-test. Extending the regime-conditional VaR construction to the six rolling-origin folds of Table 32, with CHMM-N refit per fold from scratch on the train slice and forward-filtered through the test slice under fold-IS-fixed parameters, gives twenty-four (fold, K, α) rows in Table 34. The conditional Christoffersen-cc passes at $\alpha = 0.05$ on ten of twelve fold- K combinations (W1, W3, W4, W5, W6 at both $K \in \{3, 18\}$); the two failures concentrate on W2 (COVID, $p_{cc} \in \{0.011, 0.002\}$ at $K \in \{3, 18\}$). At $\alpha = 0.01$ the construction passes on nine of twelve combinations: W2 fails at both K ($p_{cc} < 10^{-3}$) and W4 at $K = 18$ fails at $p_{cc} = 0.022$ on a LR_{ind} tail-event clustering rather than a LR_{uc} coverage miss ($LR_{ind} = 7.49$ at $LR_{uc} = 0.11$). The pass-rate is therefore 19/24 aggregate at the 5% level and 22/24 if we exclude the two stress folds that the univariate walk-forward already flagged as out-of-distribution by KS in Table 32. The reading is that the conditional construction is window-conditional: under regime shifts of the W2 / W4 magnitude (a regime introduction the IS distribution does not span; Section 5, stationarity-scope paragraph), the conditional VaR inherits the same out-of-scope behaviour as the unconditional generators. On the four non-stress folds (W1, W3, W5, W6) plus W4 at $K = 3$ at the 5% level the conditional construction is window-robust, supporting the body framing of §4.6 extending beyond the headline single-window result.

Per-ticker $\hat{\lambda}^*$ sweep. A per-ticker sweep $\lambda \in \{0, 5, 10, 20, 50, 100\}$ on each of the six body tickers (KS-degradation tolerance $\leq 1.5pp$ from $\lambda = 0$) gives per-ticker optima $\lambda_{SPY}^* = 10$, $\lambda_{NVDA}^* = 20$, $\lambda_{JNJ}^* = 20$, $\lambda_{JPM}^* = 10$, $\lambda_{AAPL}^* = 10$, $\lambda_{QQQ}^* = 0$: $\lambda = 20$ is a reasonable uniform default but per-ticker tuning is recommended whenever a ticker's residual kurtosis at $\lambda = 0$ is within ~ 1 unit of observed.

Table 34. Walk-forward regime-conditional VaR back-test for CHMM-N on six rolling-origin folds (train 5y / test 1y; seed 20260420). At each test day t , conditional $\text{VaR}_t(\alpha)$ is the α -quantile of the K -component Gaussian mixture predictive density under fold-IS-fixed parameters, evaluated on the full-timeline filter $\Pr(s_{t+1} | \mathcal{F}_t = \text{train} \cup \text{test}[1:t-1])$. Critical values: $\chi_1^2(0.05) = 3.841$, $\chi_2^2(0.05) = 5.991$. Bold p_{cc} rows fail at the 5% level.

Fold	K	α	breaches	br rate	median $\widehat{\text{VaR}}_t$	LR _{uc}	LR _{ind}	LR _{cc}	p_{cc}
W1	3	0.01	2	0.80%	-4.01	0.11	0.03	0.15	0.93
W1	3	0.05	15	5.98%	-2.43	0.48	1.18	1.65	0.44
W1	18	0.01	0	0.00%	-5.03	5.05	0.00	5.05	0.08
W1	18	0.05	12	4.78%	-2.66	0.03	0.38	0.40	0.82
W2	3	0.01	16	6.35%	-6.54	32.93	0.87	33.80	0.00
W2	3	0.05	23	9.13%	-4.59	7.34	1.71	9.05	0.01
W2	18	0.01	14	5.56%	-6.67	25.59	1.56	27.15	0.00
W2	18	0.05	25	9.92%	-4.54	10.11	2.56	12.68	0.00
W3	3	0.01	6	2.39%	-4.48	3.53	0.30	3.82	0.15
W3	3	0.05	20	7.97%	-2.47	3.98	0.30	4.28	0.12
W3	18	0.01	2	0.80%	-5.57	0.11	0.03	0.15	0.93
W3	18	0.05	15	5.98%	-3.18	0.48	1.18	1.65	0.44
W4	3	0.01	5	2.00%	-7.98	1.96	0.21	2.16	0.34
W4	3	0.05	18	7.20%	-4.16	2.26	1.99	4.24	0.12
W4	18	0.01	2	0.80%	-9.62	0.11	7.49	7.60	0.02
W4	18	0.05	10	4.00%	-6.63	0.56	3.80	4.36	0.11
W5	3	0.01	0	0.00%	-6.06	5.01	0.00	5.01	0.08
W5	3	0.05	8	3.21%	-3.90	1.91	0.53	2.44	0.30
W5	18	0.01	1	0.40%	-7.10	1.16	0.01	1.17	0.56
W5	18	0.05	7	2.81%	-3.85	2.96	0.41	3.37	0.19
W6	3	0.01	2	0.80%	-4.68	0.11	0.03	0.15	0.93
W6	3	0.05	13	5.18%	-2.85	0.02	0.15	0.17	0.92
W6	18	0.01	1	0.40%	-5.83	1.19	0.01	1.20	0.55
W6	18	0.05	7	2.79%	-2.98	3.06	0.40	3.46	0.18

Cross-asset summary. The full per-construction Pipeline B panel is in Section A.7; for reference, the headline ordering is summarised in Table 35.

Table 35. Cross-asset dependence summary (Pipeline B; CHMM-N marginals at $K = 18$; 200 paths; seed 20260422). Median per-asset KS pass rate (%) and off-diagonal MAE of the simulated correlation matrix vs. observed. Bold marks the column winner. The Student-t copula at $\nu^* = 6$ is the strongest dependence layer on the two IS columns; on OoS the Gaussian copula attains marginally lower off-diagonal MAE (0.202 vs. 0.209) while the truncated C-vine sits between the elliptical copulas and the SIM baseline. Full per-asset detail is in Section A.7.

	Median IS KS	Median OoS KS	Off-diag MAE IS	Off-diag MAE OoS
Single Index Model	77.0	87.0	0.076	0.252
Gaussian copula	97.8	87.5	0.030	0.202
Student-t copula ($\nu^* = 6$)	98.8	85.8	0.027	0.209
Truncated C-vine (root SPY)	97.2	86.0	0.068	0.235